MATH 3700 – Probability and Statistics Course Notes

Summer 2026

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Chapter 0: Introduction to Statistics

What is statistics? Statistics is the science of ____________.

- **Data** is _________________. This information can come from observations, counts, measurements, or responses. Typically data takes the form of observed measurements (e.g. height, temperature) or descriptions (e.g. dog, cat, blue, female).
- Any task involving collecting, processing, displaying, and interpreting data, especially for the purpose of making decisions, is part of the discipline of statistics.

Big picture: statistics is how we use data to say things about ________________________.

Sometimes we do this by displaying data, so that we can get a good visual idea of what is going on.

Generally, if you are taking raw information (data) and trying to make real world sense of it, you're doing statistics.

Unknowns and Uncertainty

Very often we are trying to say something about a quantity that is not only unknown, but is impossible or nearly impossible to know for sure. For instance, how many trees are there in Rocky Mountain National Park? No one knows! But maybe we can get an approximate idea.

________________________ is rampant! We use statistics to answer questions about things that are unknown (such as the average height of trees in Rocky Mountain National Park), but just as importantly, we use statistics to get an idea of how sure we are of our answers. Sometimes it is useful to know that you don't know!

This is what the "margin of error" is in opinion polls. A big "margin of error" means "we don't really know."

Statistics types

We can divide statistics into two categories: ____.

_________________ statistics involve describing a data set:

- We could make a graph of it.
- We can describe its average and how spread out it is.
- We could tell about any interesting features.

However, in descriptive statistics, we limit ourselves to describing the data itself. We do not generalize facts about the data set to a larger group.

If we generalize from a sample to a larger group, we are performing _________________ statistics.

The larger group is called a __________________. These sorts of generalizations are called inferences, and the investigator is said to "draw inference" on the population, using the **sample**. For example, if we want to learn something about the average height of trees in RMNP, we can take the average of a sample of 100 trees.

Inferential statistics jargon

A **population** of items is the entire collection of items of interest to the investigator in his or her study. This is the large group we want to know about. We often want to know a single number that describes a population called a **parameter**.

A **sample** is a subset of the population that we actually collect information from. We can calculate **statistics** from samples to estimate parameters.

We have to be careful when generalizing from a sample to a population. The descriptive characteristics of your sample may not accurately reflect the characteristics of the population you are studying.

Two very important questions in statistical inference are:

1. Is my sample large enough?

2. Is my sample representative of the population of interest?

Survey of Statistical Methods

Goal of Statistics: The underlying goal of statistical methods is to summarize and gain information from data. We will explore two important roles of statistics as they are typically used:

1. **Exploratory Data Analysis (EDA)**: EDA deals with graphical and numerical summaries of data to acquire information about the center, spread, shape, and any unusual values in those data.
2. **Statistical Inference**: Statistical inference concerns the estimation of unknown population values (such as means or variances) typically with confidence intervals, and the formal testing of statistical hypotheses. We will do this later.

Exploratory Data Analysis (EDA)

Goal of EDA: To reduce a "large" set of data to a few numerical and graphical "measures" which provide a rough description of the entire data.

Before looking at methods of EDA, definitions of some important terms are given.

- **Observational Units**: The objects on which data are collected. These might be trees in a forest, areas of land, or people, among others. If the observational units are people, they are generally referred to as **subjects**.
- **Variable**: A characteristic of an observational unit that can be expressed as a number or category (ex: height, age, gender, growth rate, distance run/walked in one day, etc.)
- **Values**: Specific numbers or categories for the individual observational units.

There are two basic types of variables:

1. ______________________: These take numerical values for which arithmetic comparisons make sense.

2. ______________________: These record into which of several categories an experimental unit falls. We will often need to change categorical variables to factor manually in **R** since they will typically be read in as "characters".

Definition: The pattern of variation of a variable is its **distribution**. A distribution records numerical values and how often they occur.

- There are two basic ways to describe the distribution of some data:
  1. Graphical Descriptions
  2. Numerical Descriptions

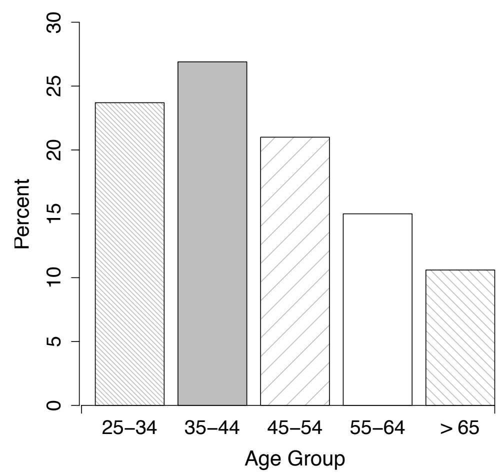
Graphical Descriptions of Data

There are many graphical descriptions of data used in practice; four of the more commonly used displays are considered here.

1. **Bar Chart**: Bar charts are used for categorical variables only. To illustrate the use of a bar chart, consider the following example.

   Example: Suppose we are interested in the association between age and level of college education among adult Americans. The table below represents the percentage of adults in different age categories with 4 years of a college education, among 151,316 sampled adults. [Data from the Census Bureau, 1987]. Also shown is a bar chart which graphically represents these data.

Age Group	% with 4 yrs. of college
25-34	23.7
35-44	26.9
45-54	21.0
55-64	15.0
>64	10.6

   What can we say from these data?

   Notes:

   (a) The y-axis is labeled by the frequency (or relative frequency) of occurrence of the specific categories. The x-axis is labeled by the categories of the variable of interest.

   (b) If the categories are not naturally ordered, it is recommended that they be ordered from largest to smallest frequency.

   (c) The **R** code for creating the barplots shown above can be found on the course Canvas page in the script **Notes 0 Script.R**.

2. **Histogram**: A histogram is a plot for quantitative variables only, which breaks the range of data values of a variable into intervals and displays only the frequency (or relative frequency (%)) of the observations that fall in that interval.

   - When are histograms most useful?

Method

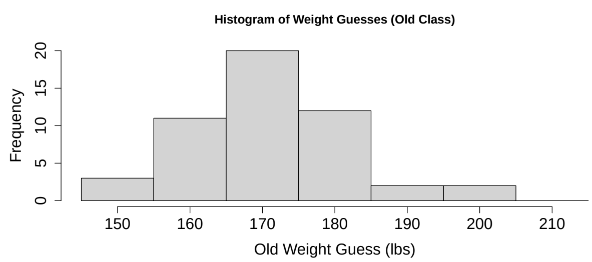
(a) Break the data into intervals of equal width.

(b) Count the number (or compute the percentage) of observations in each interval.

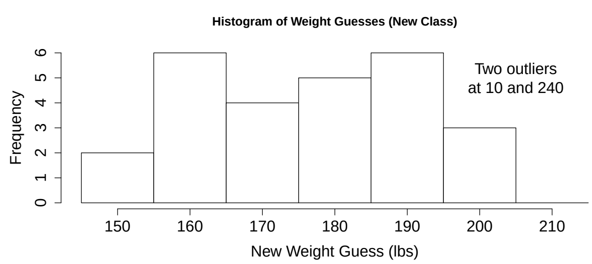
(c) Draw the histogram, as illustrated below.

Example 1: A Professor asks his students to guess his weight in two different semesters. Consider the following guesses of his weight by a previous class, given in the table below, and plotted in the corresponding histogram. Note that the intervals in the histogram have equal width and are ordered.

Interval	146-155	156-165	166-175	176-185	186-195	196-205	206-215	Other
Frequency (Semester 1)	3	11	20	12	2	2	0	0
Relative Frequency	0.060	0.220	0.400	0.240	0.040	0.040	0	0
Frequency (Semester 2)	2	6	4	4	5	3	0	2 (10&240)
Relative Frequency	0.077	0.231	0.154	0.154	0.192	0.115	0	0.077



What do these histograms tell us about the distribution of guesses of his weight?



Are there any differences between the guesses for the two classes?

Distribution Description: Histograms are helpful when examining the following features of a quantitative variable's distribution:

(a) Center: The median is the point at which half the observations are above, and half below.
The mean is the average of the observed values.
The mode is the interval or value of highest frequency.

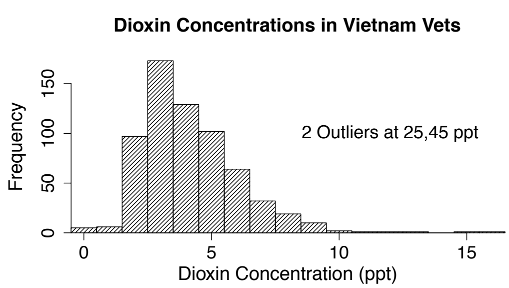
(b) Spread: Common measures of spread will be discussed soon, but include the IQR and the standard deviation.

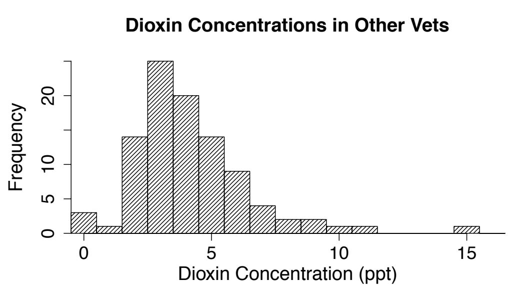
(c) Shape: Some common distribution shapes are indicated below.

(d) Number of Peaks: Most distributions are unimodal (single peaked). When might a distribution of values be bimodal?

(e) Deviations from Overall Shape (Unusual Features): Both stemplots and histograms are good for identifying outlying values or aspects of the distribution which seem to deviate from the overall pattern of values.

Example: An investigation into the effects of Agent Orange on troops in Vietnam was done. Dioxin concentrations for 646 Vietnam veterans and 97 other veterans from the same time period are given in the two histograms below.





Does there appear to be any difference in dioxin concentrations in the Vietnam veterans and the other veterans in this study? It should be noted that there are two outliers not shown for the Vietnam vets. These two vets had dioxin concentrations of 25 and 45 ppt respectively. How would you describe these dioxin concentrations?

Numerical Descriptions for Distributions

The two primary numerical features of a set of data used to describe the distribution of the data are the center and variation (or spread) of the data.

Measures of Center

(Sample) Mean: the average of the data. For a finite sample of data values $x_1, \dots, x_n$, the sample mean is given by:
$$\overline{x} = \frac{1}{n}(x_1 + \cdots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}\sum x_i.$$

Example: Consider the student maximum speed data given by students in my Fall 2019 class. The speeds are given below:

```
 85 188 180 100 100  56 110 100  90  90  90 142  97 105 114  85 105
130  87 110  86 125 150  70 140  85  95 135 100 300  80 110  95 110
100 110 120 110  85 100 111 145
```

1. What is the sample mean? Use technology! **R** or Excel. `=AVERAGE()` in Excel.

2. What is this mean estimating?

(Sample) Median: middle or center of the distribution found by obtaining a number such that 1/2 the values are above, and 1/2 below. `=MEDIAN()` in Excel.

To find the sample median, $\widetilde{m}$:

1. Order the data (as shown above) from smallest to largest:

Value	56	70	80	$\cdots$	100	100	105	$\cdots$	150	180	188
Order Stat	$x_{(1)}$	$x_{(2)}$	$x_{(3)}$	$\cdots$	$x_{(20)}$	$x_{(21)}$	$x_{(22)}$	$\cdots$	$x_{(39)}$	$x_{(40)}$	$x_{(41)}$

2. Two cases: n is even or n is odd (n = sample size)
   - If n is odd, the median is $\widetilde{m} = x_{(\frac{n+1}{2})}$, the middle point in the ordered list.
   - If n is even, the median is $\widetilde{m} = \dfrac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2}$, the average of the two middle points.

Example: What is the median speed for all students?

```
 56  70  80  85  85  85  85  86  87  90  90  90  95  95  97 100 100
100 100 100 100 105 105 110 110 110 110 110 110 111 114 120 125 130
135 140 142 145 150 180 188
```

Example: (n odd), Max Speed data ($n = 41$)

- Median $= \widetilde{m} = x_{(42/2)} = x_{(21)} =$ 100 mph.
- Note: Half the data are above; half are below.

Example: (n even). Suppose we decide to add another speed of 300 that is incorrectly reported in kph instead of mph. Now $n = 42$.

- Median $= \widetilde{m} = (x_{(21)} + x_{(22)})/2 = (100 + 105)/2 =$ 102.5 mph.
- Note: The median is not always one of the data values.

When should we use the median? the mean?

Definition: An **outlier** is an individual observation that falls outside the overall pattern of the data (e.g.: the 300 mph in the speed data).

(Sample) Mode: the value that occurs most often in the data set. `=MODE.MULT()` in Excel.

What is the mode speed in the maximum speeds data set?

$\boxed{\textbf{Measures of Spread (Variation)}}$

The 3 most common measures are:

1. Range
2. Interquartile Range (IQR) (associated with the median)
3. Standard Deviation (associated with the mean)

Range = maximum − minimum value in the data. `=MAX()- MIN()` in Excel.

- The range is simple to compute, but is heavily influenced by (sensitive to) outliers.
- Range for the speeds data:

IQR: The p^{th} **percentile**, denoted $\tilde{\pi}_p$, is the value such that $p\%$ of the observations fall below it. What percentile is the median? The mean?

The **quartiles** can be found by breaking the data into 4 parts in such a way that:

- $\tilde{q}_1 = \tilde{\pi}_{0.25} = 25^{th}$ percentile
- $\tilde{q}_2 = \tilde{\pi}_{0.50} = 50^{th}$ percentile (also the median $\widetilde{m}$)
- $\tilde{q}_3 = \tilde{\pi}_{0.75} = 75^{th}$ percentile

Then the IQR is defined as: $\boxed{IQR = \tilde{q}_3 - \tilde{q}_1}$, the range between the 25^{th} and 75^{th} percentiles. So the middle 50% of the data lie over a distance given by the IQR. In this way, outliers do not play a role in computing the IQR measure of variability.

Example of How to Find Quartiles: Consider the following data representing the number of visits to a web site per month, in the first 15 months of the page's existence. The data are ordered:

$$\underline{10 \quad 12 \quad 15 \quad 15 \quad 16 \quad 16 \quad 17} \quad \boxed{20} \quad \underline{20 \quad 21 \quad 24 \quad 25 \quad 27 \quad 32 \quad 45} \qquad (n = 15)$$

$\Longrightarrow$ IQR $= \tilde{q}_3 - \tilde{q}_1 = 25 - 15 =$ 10 visits (using one simple algorithm).

- Note: If the median is between 2 middle values (n even), the lower value is included in finding $\tilde{q}_1$ and the upper value is included in finding $\tilde{q}_3$. If the median is the middle value (n odd), as in the example, it is not included in computing $\tilde{q}_1$ or $\tilde{q}_3$.
- To find the IQR in Excel, the `=QUARTILE.EXC(data, quartile_number)` function can be used. The quartile number should be 1 (for the first quartile), 2 (for the median), or 3 (for the third quartile).

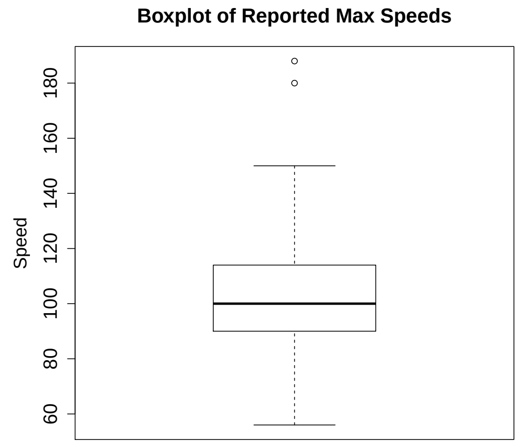
In computing the IQR, we discussed the computation of the quartiles of a distribution, values which split the distribution into four quarters. The five numbers (Min, $\tilde{q}_1, \widetilde{m}, \tilde{q}_3$, Max) provide the "break points" for the data separated into 4 quarters. This group of numbers for a given variable is known as the **5-number summary** of the distribution.

The 5-number summary of the speed variable using Excel is: $(56, 90, 100, 117, 188)$.
If using **R**, the values may be slightly different.

Boxplots

The information in the 5-number summary is often displayed in a **boxplot**, as shown to the right.

- Can see the center, spread, overall range, and degree of symmetry.



Discerning shape using boxplots:

Boxplots have Two Primary Uses:

1. **Comparing two or more distributions**

   Example: Consider the speed data from the previous example now split between females and males. First, the 5-number summaries are:

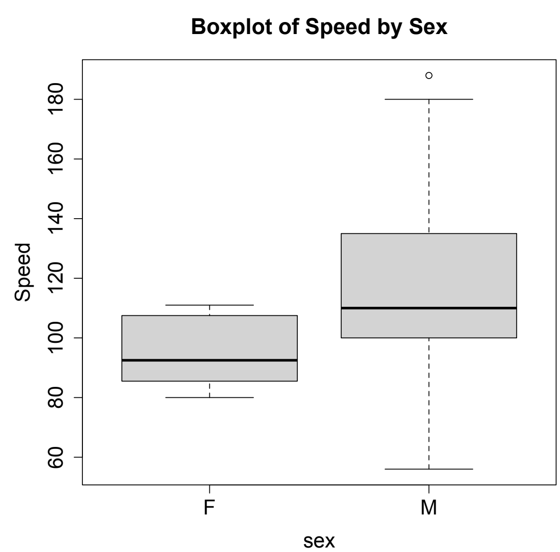
5-Number Summary for Females

Min	Q_1	Median	Q_3	Max
80.00	85.75	92.50	106.25	111.00

5-Number Summary for Males

Min	Q_1	Median	Q_3	Max
56	100	110	135	188

- Side-to-side boxplots for these two speed distributions are drawn to the right.
- What comparisons can be made?



2. **Detecting & Displaying Outliers**

   The IQR is often used to detect outliers in data according to the following **Rule of thumb**:

   (a) Values at least 1.5 x IQR greater than $\tilde{q}_3$ or 1.5 x IQR less than $\tilde{q}_1$ are said to be mild outliers. These lower and upper limits: $(\tilde{q}_1 - 1.5 * IQR, \tilde{q}_3 + 1.5 * IQR)$ are called the lower and upper **inner fences** of the distribution.

   (b) Values at least 3.0 x IQR greater than $\tilde{q}_3$ or 3.0 x IQR less than $\tilde{q}_1$ are said to be extreme outliers. These lower and upper limits: $(\tilde{q}_1 - 3.0 * IQR, \tilde{q}_3 + 3.0 * IQR)$ are called the lower and upper **outer fences** of the distribution.

Example: Reconsider the original 41 speeds (not separated by sex). Which speeds are considered outliers? $1.5 * IQR = 1.5 * (24) = \underline{36}$. So by the $1.5 * IQR$ approach, any value lying outside the interval $(\tilde{q}_1 - 1.5 * IQR, \tilde{q}_3 + 1.5 * IQR) = (90 - 36, 114 + 36) = (54, 150)$ is considered an outlier. Hence, 180 and 188 are outliers. When creating boxplots using software, most programs automatically find these fences for outliers and plot the outliers as points. The lines from the box are then extended to the most extreme data values which are NOT outliers. Boxplots displaying these outliers are sometimes called **modified boxplots**.

Variance and Standard Deviation

- For data $x_1, \dots, x_n$, the standard deviation is based on looking at all differences or deviations from the sample mean: $x_i - \overline{x}$, $i = 1, \dots, n$. What's wrong with just summing these deviations as a measure of spread?

- Square these deviations, sum them, and divide by $\underline{n-1}$ to get the **sample variance** s^2:
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

- The square root of s^2 is the **sample standard deviation**: $s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2}$.

Why do we divide by $(n-1)$ instead of n? Answer: Since the deviations always sum to zero, if we know all but one, we can determine the last one, so that only $n-1$ deviations are free to vary.

Degrees of freedom: the number of deviations free to vary. (For the standard deviation, the degrees of freedom (df) $= n - 1$).

Example to Calculate s: Data: $36, 29, 13, 24, 18$. Suppose these data are the weights of dogs in lbs. $\overline{x} = 24, n = 5$. To compute s directly:

The `=STDEV.S()` or `=STDEV()` functions in Excel can be used to compute the standard deviation.

Standard Deviation Interpretation

The standard deviation can be interpreted (a bit loosely) as

- The sample standard deviation has the same units as the original data.
- Important: The standard deviation s cannot be negative! When is the SD $= 0$?

Which measure of variation (IQR or s) is more resistant (or robust) to outliers?
Why?

Quick Aside – Population Parameters

While $\overline{x}$, s, and n denote the sample mean, sample standard deviation, and sample size, respectively, we have separate symbols that are used when we are describing a population.

The **population** size is sometimes denoted by N.

The **population mean** is $\mu = \dfrac{\sum_{i=1}^{N} x_i}{N}$.

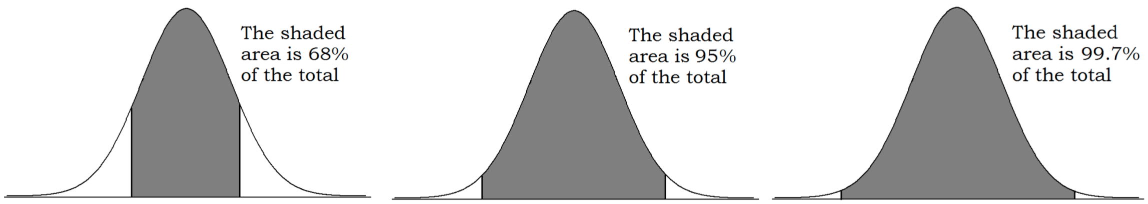
The **population standard deviation** is $\sigma = \sqrt{\dfrac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$.

We usually do not calculate these since we rarely know all of the population information.

Empirical Rule

If a distribution is fairly bell-shaped (or mound-shaped) and roughly symmetric, then:

1. Approximately 68% of the data fall within 1 standard deviation of the mean. (i.e.: $(\mu - \sigma, \mu + \sigma)$ contains $\approx$ 68% of the data).
2. Approximately 95% of the data fall within 2 standard deviations of the mean. (i.e.: $(\mu - 2\sigma, \mu + 2\sigma)$ contains $\approx$ 95% of the data).
3. Nearly all (99.7%) of the data fall within 3 standard deviations of the mean.



Example: Suppose we know that the mean height of adult females in the U.S. is bell-shaded with a mean of 64 inches and a corresponding standard deviation of 2.75 inches. Using the empirical rule, we can say that about 99.7% of all U.S. females are between what two heights?

Probability

Where does probability come into play? Ultimately, we want to know something about **parameters**, which describe population. We use **statistics**, which describe samples, to estimate parameters, but statistics are nearly almost always wrong. We can model the uncertainty in these statistics using probability theory, which we will discuss in Chapters 1 through 5.

Chapter 1: Probability

Two primary way to think about probability:

1. Frequency - probabilities assigned to outcomes of an experiment based on relative frequency of occurrence. Also known as empirical probability. (Ex: Flip coin many times.)
2. Theoretical - all outcomes have the same probability of occurrence in a finite sample space. We think about the outcomes and use them to calculate probabilities. Also known as classical.

Properties of Probability (Section 1.1)

An **experiment** is any process whose outcome is unknown before the process is completed. To characterize the **uncertainty** in these outcomes, we will assign them probabilities to represent how likely they are to occur.

Definition: The **sample space** for an experiment is the set of all possible **outcomes** of the experiment. Often denoted S.

- This can usually be specified before the experiment is performed.

Example: Observe the major of a randomly chosen student. What is S?

Example: Inspect chips from a manufacturing process until the first defective is found. What is S?

Example: Observe the time (in sec) it takes for an **R** script to run. What is S?

Note: Samples spaces are one of three types:

1. **Finite**: There are a finite number of outcomes in the sample space.
2. **Infinite discrete** or **Countably infinite**: There are an infinite number of outcomes in the sample space, but they can be listed discretely such that they are in a 1-1 mapping with the integers.
3. **Infinite**: Infinite sample spaces are typically intervals of real values, and contain an uncountably infinite set of values.

Note: We will focus on **finite** sample spaces in this chapter and take the classical probability approach where we assume that each of the finite number n possible outcomes has the same probability of occurrence, namely $1/n$.

Example: Toss a fair coin & roll a 6-sided die. What is S?

Definition: An event, E, is any subset of the sample space. $E \subset S$.

Example: Let $E = \{H2, H4, H6, T2, T4, T6\}$ be an event in S - might be interested in the event that we roll an even #.

- So E = event that we roll an even #.
- Let F = event that we roll a four or a five, so that $F = \{H4, H5, T4, T5\}$.
- In general, an event E occurs if, when the experiment is performed, the outcome is an element of E.

Set Operations Let E and F be events in S.

1. **Union**: $E \cup F$ - all elements in E OR F or both (E or F occurs).
   Ex: $E \cup F =$

2. **Intersection**: $E \cap F$ - all elements in both E AND F (E and F occur).
   Ex: $E \cap F = EF =$

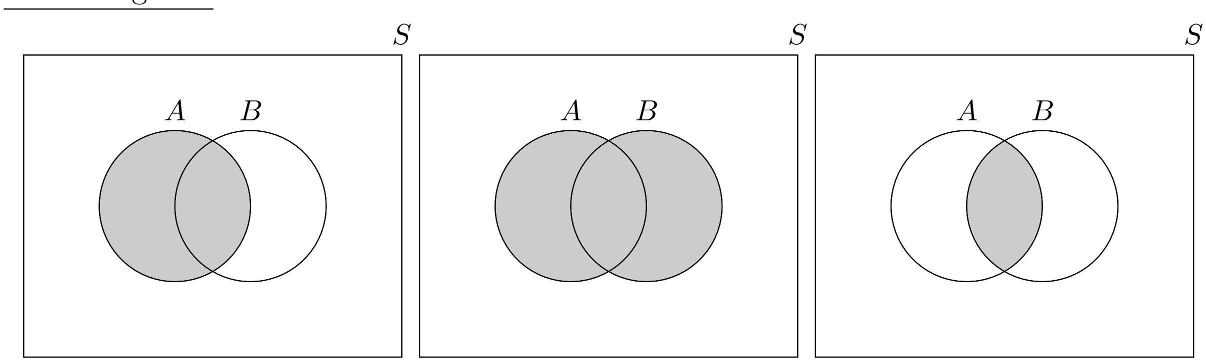
3. **Complement**: E' or E^c - all elements of S that are NOT in E (E does not occur).
   Ex: $E' =$

4. **Subset**: $E \subset F$ - we say that E "is contained in" F if all outcomes in E are also in F.
   (Ex: Roll die. $E = \{1, 3, 5\}$, $F = \{1, 2, 3, 4, 5\}$. Clearly, $E \subset F$.)

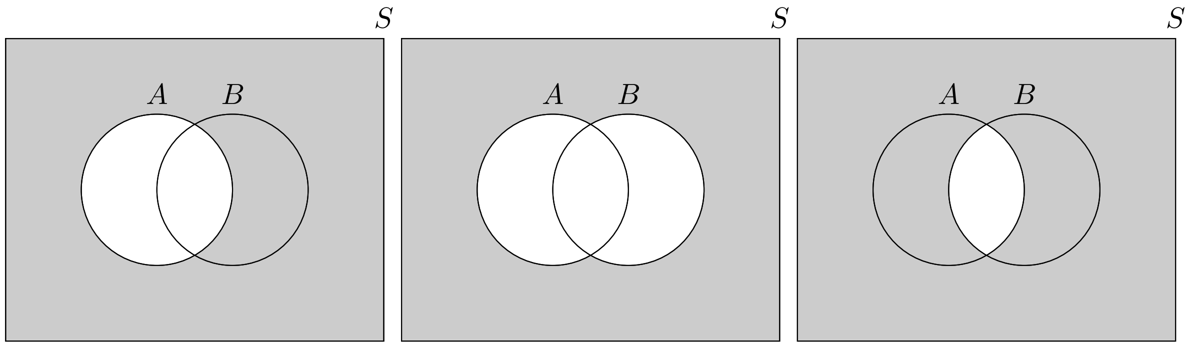
   Notes: (i) $E \subset S$, $F \subset S$.
   (ii) If $E \subset F$ and $F \subset E$, then $E = F$
   (iii) If $E \subset F$ and $F \subset G$, then $E \subset G$ (Transitive Property).

5. **Empty Set (or Null Event)**: $\emptyset$ - the impossible event (also called the empty set), contains no outcomes.
   Notes: (i) $\emptyset \subset E$ for any event E.
   (ii) $\emptyset' = S$, $S' = \emptyset$ (the complement of everything is nothing).

Venn Diagrams:



A $\quad$ $A \cup B$ $\quad$ $A \cap B$



A^c $\qquad$ $(A \cup B)^c$ $\qquad$ $(A \cap B)^c$

Definition: Two events A & B are **mutually exclusive** (or **disjoint**) if $A \cap B = \emptyset$. (i.e.: if they have no points in common).

- So A & B ______________________________
- In the die/coin example, let $G = \{H6, T6\}$, the event that we roll a "6". Since $E' = \{H1, H3, H5, T1, T3, T5\}$, then $E' \cap G = \emptyset$ (we cannot roll an odd # and a "6"), and we say that E' and G are mutually exclusive.

Definition: Two events A & B are **exhaustive** if $A \cup B = S$. (i.e.: together they make up the entire sample space).

- In the die/coin example, let $H = \{H2, H4, H6, T2, T4, T6\}$, the event that we roll a an even. Since $E \cup H = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\} = S$, E and H are exhaustive.

Definition: Two events A & B **partition** a sample space if $A \cup B = S$ and $A \cap B = \emptyset$. (i.e.: together they make up the entire sample space and they are mutually exclusive).

Set Operations Results:

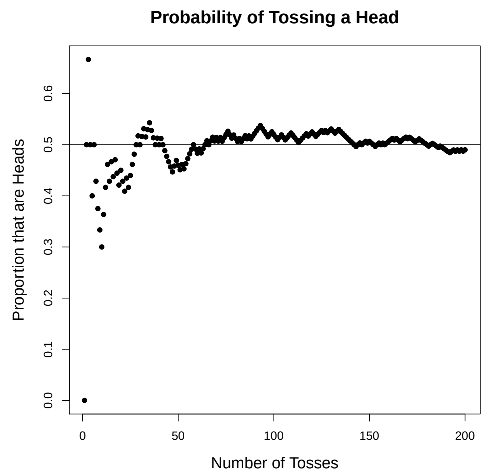
1. Commutativity: $A \cup B =$____________, $A \cap B =$ ____________.
2. Associativity: $(A \cup B) \cup C =$_________________, $(A \cap B) \cap C =$_________________.
3. Distributive Laws: $A \cup (B \cap C) =$___________________________,

   $A \cap (B \cup C) =$___________________________.
4. DeMorgan's Laws: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$.

$$\text{Generalized Demorgan's Laws}: \left(\bigcup_{i=1}^{n} A_i\right)' = \bigcap_{i=1}^{n} A_i', \quad \left(\bigcap_{i=1}^{n} A_i\right)' = \bigcup_{i=1}^{n} A_i'.$$

- Can we verify DeMorgan's Laws with Venn Diagrams?

Axioms of Probability: This section deals with quantifying likelihoods. We don't know if an event E will occur and some events are more likely than others. We want to assign #'s and probabilities of events, to reflect which events are more likely & which are less likely.



- Denote the probability of E as $P(E)$ (or $Pr(E)$).
- Consider the experiment of flipping a coin. The "frequency approach" of defining the probability of getting heads:
  $n =$ number of times coin has been flipped
  $E =$ event of getting heads
  $\mathcal{N}(E) = N(E) = \#$ of outcomes in E

  $\boxed{P(E) \approx \quad\quad\quad\quad}$
- What do we expect to happen to this probability as n gets large?

Note: Under the classical approach, all outcomes are equally likely. Consider some event E in a finite sample space S. Letting $N(E)$ be the number of outcomes in E, the probability of the event E is then defined as:

$$P(E) = \frac{\#\text{ of outcomes in } E}{\#\text{ of outcomes in } S} = \frac{N(E)}{N(S)}$$

Back to the coin/die Example: With $E = \{H2, H4, H6, T2, T4, T6\}$ & $F = \{H4, H5, T4, T5\}$ defined as before, we assign probabilities to these events as:

$$P(E) = \frac{N(E)}{N(S)} = \qquad\qquad\qquad P(F) =$$

Example: Suppose we roll a fair dice twice, recording the value each time. Note that the sample space for this experiment can be written as below, where "ab" indicates an "a" on the 1st roll and a "b" on the 2nd roll.

(a) What is the probability that we roll doubles?

$$S = \left\{ \begin{array}{cccccc} 11 & 12 & 13 & 14 & 15 & 16 \\ 21 & 22 & 23 & 24 & 25 & 26 \\ 31 & 32 & 33 & 34 & 35 & 36 \\ 41 & 42 & 43 & 44 & 45 & 46 \\ 51 & 52 & 53 & 54 & 55 & 56 \\ 61 & 62 & 63 & 64 & 65 & 66 \end{array} \right\}$$

(b) What is the probability of an odd sum?

(c) What is the probability that the difference in the two dice is no larger than 2?

Example: In a computer installation, 300 programs are written each week, 120 in C++, 70 in Java, and 110 in Python. Sixty percent of the programs written in C++ compile on the first run, 80% of the Java programs compile on the first run, and 70% of the Python programs compile on the first run. This information is summarized in the table on the next page.

Language	**Compiles on First Run**	**Does not Compile on First Run**	Total
C++	72	48	120
Java	56	14	70
Python	77	33	110
Total	205	95	300

Define: $E =$ the event that a randomly selected program compiles on the first run,
$F =$ the event that a randomly selected program is not written in Python and does not compile on first run.

Then the probabilities of events E and F are:

Definition: Given a sample space S, a probability function $P(\cdot)$ is any function that satisfies the following 3 axioms:

1. $P(A) \geq 0$ for any event A.
2. $P(S) = 1$.
3. For any infinite sequence of pairwise mutually exclusive events $A_1, A_2, ...$ (i.e.: $A_i \cap A_j = \emptyset$ for all $i \neq j$): $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$.

- The above 3 conditions are called the **Axioms of Probability** or **Kolmogorov's Axioms**. (All prob. theory follows from these 3 axioms).

Checking probability distributions:

Example: Roll fair die. $S = \{1, 2, 3, 4, 5, 6\}$. $P(i) = 1/6, i = 1, ... 6$. Is this a valid probability function?

- Could have assigned: $P(1) = 1/2,\ P(2) = 1/4,\ P(3) = 1/8,\ P(4) = 1/16,\ P(5) = 1/32,\ P(6) = ?$

- Or suppose $P(i) = ci$. What value of c makes this legitimate?

<u>Example</u>: Let E and F be events such that $P(E) = 0.8$, $P(F) = 0.6$, $P(EF) = 0.3$. Is this legitimate?

$\boxed{\textbf{Theorems}}$

<u>Theorem 1.1-1</u>: $P(A) = 1 - P(A')$.

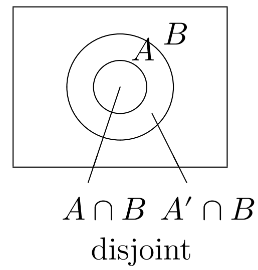
<u>Theorem 1.1-2</u>: $P(\emptyset) = 0$.

<u>Proof</u>: Let $E_1 = S$, $E_2 = \emptyset$, $E_3 = \emptyset$, ... Since these events are mutually exclusive, and $S = \bigcup_{i=1}^{\infty} E_i$, then:

$$\begin{aligned} 1 \overset{\text{Axiom 2}}{=\!=\!=} \; & P(S) = P\left(\bigcup_{i=1}^{\infty} E_i\right) \overset{\text{Axiom 3}}{=\!=\!=} \sum_{i=1}^{\infty} P(E_i) = P(E_1) + \sum_{i=2}^{\infty} P(E_i) \\ \Longrightarrow \quad & P(S) + \sum_{i=2}^{\infty} P(\emptyset) \overset{\text{Axiom 2}}{=\!=\!=} 1 + \sum_{i=2}^{\infty} P(\emptyset) \\ & \Longrightarrow 0 = \sum_{i=2}^{\infty} P(\emptyset) \Rightarrow \underline{P(\emptyset) = 0}. \end{aligned}$$

Theorem 1.1-3: If $A \subset B$, then $P(A) \leq P(B)$.

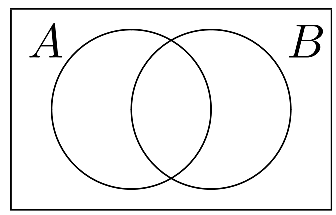
Proof:



Theorem 1.1-4: For each event A, $P(A) \leq 1$.

Theorem 1.1-5: For any two events A, B: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof:



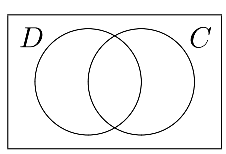
Example: In a survey of 500 people in a local area, a pet food manufacturer discovered that 50% owned a dog, 40% owned a cat, and 10% owned both.

Let D = event a person owns dog, C = event a person owns cat, so:

$P(D) =$

$P(C) =$

$P(D \cap C) =$



(a) What is the probability a randomly chosen person owns either a dog or a cat?

(b) What is the probability a person owns neither?

- Now, suppose R = event a person owns reptile, and: 15% own reptiles, 5% own dogs & reptiles, 2% own cats & reptiles, and 1% own all three.

(c) What is the probability that a randomly chosen person owns at least 1 of the 3 animal types?

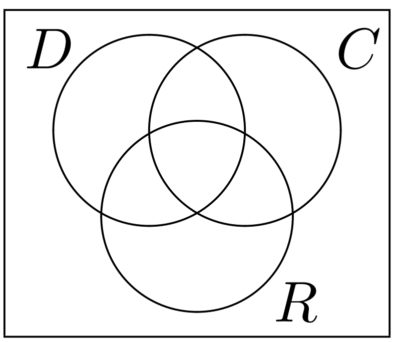
Inclusion/Exclusion Principle: For n events $A_1, \ldots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{\substack{i=1 \\ i<j}}^{n}\sum_{j=1}^{n} P(A_i \cap A_j) + \sum_{i=1}^{n}\sum_{\substack{j=1 \\ i<j<k}}^{n}\sum_{k=1}^{n} P(A_i \cap A_j \cap A_k)$$
$$- \ldots + (-1)^{n+1} P(A_1 \cap A_2 \cap \cdots \cap A_n). \quad \text{(Proof tedious, omitted).}$$

Theorem 1.1-6: For 3 events A_1, A_2, A_3:

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

Back to Example: $P(R \cup C \cup D) = P(R) + P(C) + P(D) - P(R \cap C) - P(R \cap D) - P(C \cap D) + P(R \cap C \cap D)$



$=$

$\boxed{\textbf{Methods of Enumeration (Section 1.2)}}$

Ex: Suppose we have 5 people: $\underbrace{A, B, C}_{\text{Women}}, \underbrace{D, E}_{\text{Men}}.$

- In how many ways can we choose a committee of size 3 from this group?

- ____________ ways.
- How many have 2W, 1M?
- Suppose instead we have 100 people (60W, 40M) and choose at random a committee of 3. How many ways? Guess: ____________ ways.
- Too many outcomes to list, need different way of counting.

Basic Principle of Counting (Multiplication Rule)

Consider an experiment with the following two rules:

1. The experiment is performed in r parts ($r \geq 2$).
2. The i^{th} part of the experiment has n_i possible outcomes, and each part of the experiment is performed separately from each other part.

Then there are a total of $n_1 \cdot n_2 \cdots n_r = \prod_{i=1}^{r} n_i$ possible outcomes for the experiment.

Example: In designing a study of the effectiveness of migraine medicines, 3 factors were considered:

$$\left.\begin{array}{l} \text{(i) Medicine (A,B,C,D, Placebo)} \\ \text{(ii) Dosage Level (Low, Medium, High)} \\ \text{(iii) Dosage Frequency (1,2,3,4 times/day)} \end{array}\right\} r = 3 \text{ parts.}$$

In how many possible ways can a migraine patient be given medicine?

Example: Suppose in this class, I want a subgroup of 1 CS major, 1 Engineering major, and 1 other major. How many subgroups are possible?

$$\left.\begin{array}{ll} n_1 = \#\text{ CS majors} & = \_\_\_\_\_\_ \\ n_2 = \#\text{ Engineering majors} & = \_\_\_\_\_\_ \\ n_3 = \#\text{ Other majors} & = \_\_\_\_\_\_ \end{array}\right\} n_1 \cdot n_2 \cdot n_3 = \_\_\_\_\_\_ \text{ ways.}$$

Permutations & Combinations

Permutations - deal with the # of ordered subsets of size r from n distinct objects.

Combinations - deal with the # of unordered subsets of size r from n distinct objects.

Example: How many possible orderings (positionings) are there for 9 sprinters in the 100-meter sprint on a 9-lane track?

- In general, the # of permutations (orderings) of n distinct objects is: $n(n-1)\cdot\dots\cdot 2\cdot 1 =$

Example: How many different letter arrangements can be formed from the word ARTICHOKES?

Example: How many different letter arrangements can be formed from the word STATISTICS? Why is this different?

- For any one arrangement of the letters (say ITSSCATTIS), we can:

- So, for any "word", there are 3!3!2! possible arrangements of the letters for a total of $\frac{10!}{3!3!2!}$ different letter arrangements.
- In general, the # of permutations of n objects, of which n_1 are alike, n_2 are alike, ..., n_r are alike, is: $\frac{n!}{n_1!n_2!\cdots n_r!}$. This is the number of **distinguishable permutations**.

Sampling without Replacement: Suppose we have n objects, and select $r \leq n$ of these objects, one at a time, without replacement.

1. How many <u>ordered</u> subsets of size r? (Permutations)

$$
\begin{array}{cccc}
 & \multicolumn{3}{c}{\overbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \cdots \qquad\qquad\qquad}^{r\text{ of them}}} \\
 & \uparrow & \uparrow & \uparrow \\
= & \begin{array}{c}\text{\# of ways to} \\ \text{choose 1st object}\end{array} & \begin{array}{c}\text{\# of ways to} \\ \text{choose 2nd object} \\ \text{since there are} \\ n-1 \text{ remaining}\end{array} & \begin{array}{c}\text{\# of ways to} \\ \text{choose rth object}\end{array} \\
 & & & \\
= & \multicolumn{3}{l}{n(n-1)\cdots(n-(r-1))}
\end{array}
$$

2. How many <u>unordered</u> subsets of size r? (Combinations)

   - For each <u>unordered</u> subset, there are $r!$ ordered subsets.

$$
\left[\begin{array}{ll} \underline{\text{Ex}}: & \{A, B, C\} \text{ unordered} : ABC, ACB, BAC, BCA, CAB, CBA \text{ ordered.} \\ & \text{Size } r = 3 \Rightarrow 3! = 3 \cdot 2 \cdot 1 = 6 \text{ ordered subsets.} \end{array}\right]
$$

   - So the # of unordered subsets is: $\dfrac{n!/(n-r)!}{r!} = \dfrac{n!}{r!(n-r)!} = \boxed{\begin{pmatrix} n \\ r \end{pmatrix}}$, the # of possible combinations of n objects, taken r at a time.

<u>Example</u>: Recall the committee of 3 to be chosen from 60W, 40M, at random.

(a) How many total committees are possible?

(b) How many committees with 2W and 1M are possible?

(c) What is the *probability* of getting 2 women and 1 man if we select 3 people at random from these 100?

(d) Suppose the order of selection is important:

$$\left.\begin{array}{ll} 1st - & \text{chair} \\ 2nd - & \text{vice-chair} \\ 3rd - & \text{member} \end{array}\right\} \text{Order of choice}$$

How many <u>ordered</u> committees of size 3 are possible?

<u>Note</u>: This number is $3! = 6$ times larger than the unordered case, because there are 3! orderings for a group of size 3.

(e) How many ways can we have 2W officers, 1M member?

Chair? Vice? Member?

Notes on $\binom{n}{r} = \frac{n!}{r!(n-r)!}$:

1. $\binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{0!n!} = 1$
2. $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)!(n-(n-r))!} = \binom{n}{n-r}$.
3. $$\underbrace{\binom{n}{r}}_{\substack{\text{total \#} \\ \text{groups of} \\ \text{size } r}} = \underbrace{\binom{n-1}{r-1}}_{\substack{\text{\# ways to have} \\ \text{certain object in} \\ \text{group of size } r}} + \underbrace{\binom{n-1}{r}}_{\substack{\text{\# ways not to have} \\ \text{that object in} \\ \text{group of size } r}} \qquad (1 \le r \le n)$$
4. $\binom{n}{r}$ is called the <u>**binomial coefficient**</u>.

<u>Sampling with Replacement, Ordered</u>: Suppose we have n objects, and select $r < n$ of these, one at a time, <u>replacing</u> the object after each selection.

n ways to choose 1st object,

n ways to choose 2nd object, etc., for r selections

$\Rightarrow n^r$ possible subsets (ordered).

Example: Roll die 3 times. Total # sample points = $6^3 = \underline{216}$ $(n = 6, r = 3)$.

How many ways can we get exactly 2 sixes?

Sampling with Replacement, Unordered: This situation is much less common, so we will not discuss it. It is included in the following table, however.

Table for counting:

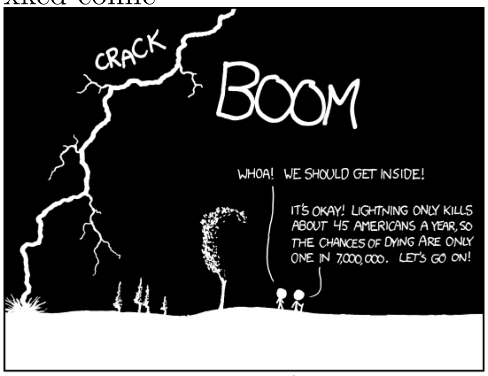
	With replacement	Without replacement
ordered	n^r	$n!/(n-r)!$
unordered	$\binom{n+r-1}{r}$	$\binom{n}{r}$

Example: Suppose we have a random # generator, which generates 4-digit #'s. Each digit is selected at random from $(0, \ldots, 9)$, one at a time. What proportion of randomly generated #'s contain no alike digits?

- Sampling with or without replacement?
- Ordered or unordered?
- Total # of 4-digit numbers?
- For all digits to be different, there are:
- So proportion with no alike digits =

Conditional Probability (Sections 1.3)

xkcd comic



THE ANNUAL DEATH RATE AMONG PEOPLE WHO KNOW THAT STATISTIC IS ONE IN SIX.

As suggested in the comic to the left, probabilities can change based on the condition that another event has occurred. For example, what is the probability that a randomly chosen man is over 6 feet tall? About 0.14. What is that same probability given that you are selecting from an NBA basketball team? Almost 1. The probability changes given the fact that you are selecting from a basketball team.

Example: Medical Testing. A disease affects 0.1% of the population. The medical test for this disease can be described as 95% accurate. If you test positive, what is the chance you have the disease?

Example: Suppose we roll a fair dice twice, recording the value each time. Note that the sample space for this experiment can be written as below, where "ab" indicates an "a" on the 1st roll and a "b" on the 2nd roll.

(a) What is the probability that we roll two sixes?

$$S = \left\{ \begin{array}{cccccc} 11 & 12 & 13 & 14 & 15 & 16 \\ 21 & 22 & 23 & 24 & 25 & 26 \\ 31 & 32 & 33 & 34 & 35 & 36 \\ 41 & 42 & 43 & 44 & 45 & 46 \\ 51 & 52 & 53 & 54 & 55 & 56 \\ 61 & 62 & 63 & 64 & 65 & 66 \end{array} \right\}$$

(b) What is the probability that we roll two sixes if we know the first die roll was a 6?

(c) What is the probability that we roll two sixes if we know at least one of the die rolls was a 6?

Definition: If A & B are any events with $P(B) > 0$, then the **conditional probability of A given that B occurred** is:

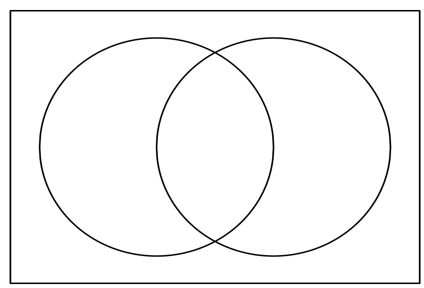
$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{N(A \cap B)}{N(B)}} \quad \text{"probability of } A \text{ given } B\text{".}$$

Example: Roll fair die 3 times ($6^3 = 216$ total outcomes in S).
Let $A =$ event that sum is 4 or less, so $A =$
Let $B =$ event that first roll is even. What is $P(A|B)$?

- B can be viewed as a restricted sample space (i.e.: we know the outcome is in B, now what is prob. it is also in A?).
- What is $P(A|B')$? $=$

  (since 3 outcomes in A have first roll being odd.)

Example: Randomly choose a student at a certain school. Suppose 60% of the students are female, and 20% of the females are English majors.



(a) What is the probability you choose a female English major?

(b) Suppose 15% of the students are English majors. Given that we choose an English major, what is the probability she is female?

Example: Suppose in manufacturing 20 soccer balls, 4 have defective stitching.

(a) If we randomly select 2 balls without replacement, what is the probability that neither have defective stitching?

(b) If we randomly select 4 balls without replacement, what is probability that none have defective stitching?

General Result: **Multiplication Rule**: For any events $A_1, \dots, A_n$:

$$P(A_1 A_2 \cdots A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 A_2) \cdots P(A_n|A_1 A_2 \cdots A_{n-1}).$$

Proof: RHS =

Independent Events (Section 1.4)

Sometimes, for events A, B, $P(A|B) = P(A)$. If this occurs, then A and B are said to be **independent**. (i.e.: knowing whether B occurred has no effect on whether A occurs).

Note:

Definition: A & B are independent events if $P(A \cap B) = P(A)P(B)$.

- Independence comes about in 2 ways:
  1. *Build* models which incorporate independence.
  2. *Discover* independence in a given model.

Example: Suppose we roll a fair die twice. Let:

A_6 = event we roll 6 on 1st roll $\Rightarrow P(A_6) = 1/6$.

B_6 = event we roll 6 on 2nd roll $\Rightarrow P(B_6) = 1/6$.

What is the probability that we get 6 on both rolls? $P(A_6 \cap B_6) = 1/36$.
Notice that $P(A_6 \cap B_6) = P(A_6)P(B_6) \Rightarrow$ A_6, B_6 independent.

- Makes sense since the first roll should have no effect on 2nd roll & vice versa. This is an example of building a model which incorporates independence.

Example: Consider the following table:

	Utah Resident	Not Utah Resident	Total
CS Major	85	15	100
Not CS Major	680	120	800
Total	765	135	900

Let C = event a randomly chosen SUU student is a CS major,

U = event a randomly chosen SUU student is a Utah resident.

Then $P(C) =$ ______________ & $P(U) =$ ________________. Are C and U independent?

$P(C \cap U) =$ _________________. $P(C)P(U) =$ _________________ $\stackrel{?}{=} P(CU)$.

- Example of discovering independence in a model.

Warning: We should never automatically assume independence. Always show it unless the problem states the events are independent.

Relationship Between Mutually Exclusive & Indep. Events

- Suppose A, B are mutually exclusive ($A \cap B = \emptyset$), where $P(A), P(B) > 0$.

- Now, suppose A, B are independent with $P(A) > 0$, $P(B) > 0$.

- Both arguments show:

Theorem 1.4-1: If A and B are independent, then:

(a) A and B' are independent.

(b) A' and B are independent.

(c) A' and B' are independent.

Proof (a) :

Theorem 1.4-2: Events A, B, and C are mutually independent if and only if the following two conditions hold:

1. A, B, and C are pairwise independent; that is
   $P(A \cap B) = P(A)P(B)$, $P(A \cap C) = P(A)P(C)$, and $P(B \cap C) = P(B)P(C)$.
2. $P(A \cap B \cap C) = P(A)P(B)P(C)$.

Does pairwise (2-way) independence $\Rightarrow$ mutual (all-way) independence?

Counterexample: S = {all 2-digit #'s using 1,2,3} = {11, 12, 13, 21, 22, 23, 31, 32, 33}

$$\begin{aligned} \text{Let } E_1 &= \{\text{odd \#'s}\} = \{11, 13, 21, 23, 31, 33\}, \ P(E_1) = 2/3 \\ E_2 &= \{\text{\#'s} < 20\} = \{11, 12, 13\}, \qquad\qquad P(E_2) = 1/3 \\ E_3 &= \{\text{\#'s w/same digits}\} = \{11, 22, 33\}, P(E_3) = 1/3. \end{aligned}$$

$$\left.\begin{aligned} &E_1E_2 = \{11, 13\}, \quad P(E_1E_2) = 2/9 = (2/3)(1/3) = P(E_1)P(E_2). \checkmark \\ &E_1E_3 = \{11, 33\}, \quad P(E_1E_3) = 2/9 = (2/3)(1/3) = P(E_1)P(E_3). \checkmark \\ &E_2E_3 = \{11\}, \qquad P(E_2E_3) = 1/9 = (1/3)(1/3) = P(E_2)P(E_3). \checkmark \end{aligned}\right\} \text{Pairwise indep.}$$

Are E_1, E_2, E_3 (mutually) independent? $E_1E_2E_3 = \{11\}$, &
$P(E_1E_2E_3) = 1/9 \neq (2/3)(1/3)(1/3) = 2/27 = P(E_1)P(E_2)P(E_3) \Rightarrow$ not mutually independent.

Example: Students Wednesday and Edin skip class independently from one another. Let:

W = event that Wednesday attends class, $P(W) = 0.60$.

E = event that Edin attends class, $P(E) = 0.80$.

What is the probability that Edin is in class if we know at least one is in class?

$\boxed{\text{Bayes' Theorem (Section 1.5)}}$

<u>Useful Result</u>: For any events A, B (where $0 < P(B) \leq 1$):

$$\begin{aligned}\underline{P(A)} &= P(AB \cup AB') = P(AB) + P(AB') \text{ (Axiom of Probability \#3)} \\ &= \underline{P(A|B)P(B) + P(A|B')P(B')} \text{ (definition of conditional probability).}\end{aligned}$$

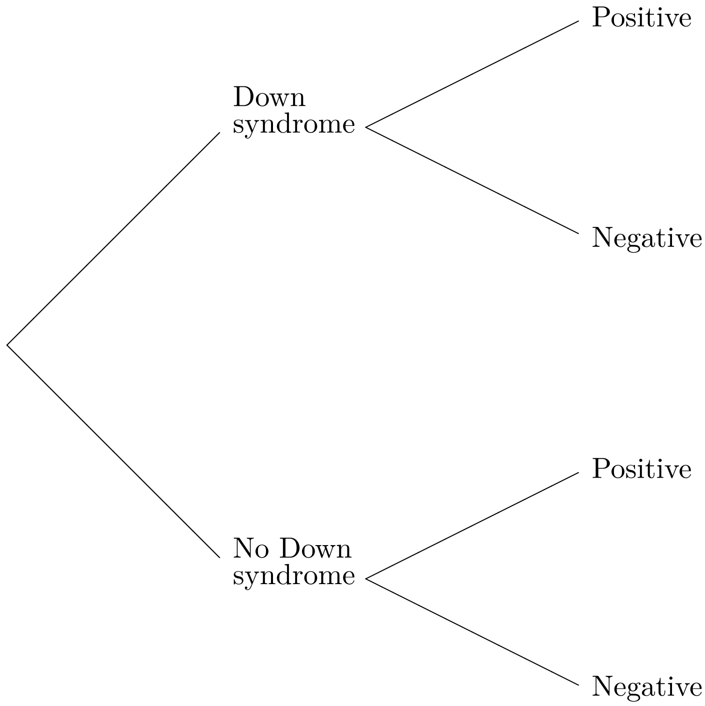
<u>Example</u>: (False positives): For pregnant women over age 35, doctors often recommend a test for Down syndrome early in the second trimester. This is due to a higher risk of having a child with Down syndrome. One test screens for the presence of alpha fetoprotein (AFP) in the mother's blood. This test is somewhat controversial because of its high **<u>false positive</u>** rate.

- When this test is given to a mother of age 36, the probability that:
  - the test is positive when there is a baby with Down syndrome is 0.999,
  - the test is positive when there is not a baby with Down syndrome is 0.05,
  - the test is negative when there is a Down syndrome baby is
  - the test is negative when there is not a Down syndrome baby is

Problem: some women are told they have a Down syndrome baby when they don't (false positive).

- Suppose 1 of every 400 babies born to mothers over the age of 35 have Down syndrome.
- Given that the AFP test is positive, what is the probability a randomly chosen mother of this age has a Down syndrome baby?
- Let:
- Know:
- Want:
- So there is a ______ chance of having a Down syndrome baby when the mother tests positive!
- What happened here? With only 1 in 400 having Down syndrome, in every 400 babies, we expect: <u>1</u> to have Down syndrome & nearly <u>20</u> to test positive w/o Down syndrome (5%).
- Your chances of being in the false positive group are <u>much</u> higher.

- Another way to solve this problem is with a tree diagram:



General Useful Result: Let A be any event in S and let $B_1, \ldots, B_m$ form a partition (disjoint) of S, with $P(B_i) > 0$. Then $A = (AB_1) \cup (AB_2) \cup \cdots \cup (AB_m)$, a disjoint union since B_i's are disjoint.

$$\text{So: } \underline{P(A)} = P\left(\bigcup_{i=1}^{m} AB_i\right) = \sum_{i=1}^{m} P(AB_i) \ \text{(AP3)}$$
$$= \sum_{i=1}^{m} P(A|B_i)P(B_i) \ \text{(Mult. Rule)}.$$

Bayes Theorem: Let $B_1, \ldots, B_m$ partition S, so that $P(B_k) > 0$ for $k = 1, \ldots, m$ and let A be any event such that $P(A) > 0$. Then for each $k = 1, \ldots, m$:

$$P(B_k|A) = \frac{P(A \cap B_k)}{P(A)} = \frac{P(A|B_k)P(B_k)}{\sum_{i=1}^{m} P(A|B_i)P(B_i)}.$$

Example: In a certain factory, machines A,B,C produce springs.

- Machines A, B, C produce 2%, 1%, 3% defective springs respectively.
- Of the total production of springs in the factory, Machines A,B,C produce 35%, 25%, and 40% respectively.

(a) Given that a randomly chosen spring is defective, what is the probability it was produced by Machine C?

Let A, B, C = events the spring was produced by Machines A,B,C respectively.
D = event the spring is defective.

- Given:
- We want:
- Note that A, B, C partition the sample space S of all possible machines from which the spring could be produced. By Bayes' Theorem:

- Note: $P(C) = 0.40 < P(C|D)$. Hence, knowing the spring is defective increases the probability the spring was produced by machine C.

(b) What is $P(A|D) + P(B|D) + P(C|D)$?

Chapter 2: Discrete Distributions

Often, we are interested in a numeric summary of outcomes of an experiment rather than ______________________________________. In a past experiment we had 100 people: (60 women, 40 men) and we looked at choosing a committee of 3.

Let $X = \#$ women chosen for the committee.

- X is an example of a random variable (X is uncertain).

Definition 2.1-1: A **random variable** is a function from the sample space S to $\mathbb{R}$ (the real numbers).

Example: Toss fair coin twice. $S = \{HH, HT, TH, TT\}$.

s	$X(s)$

Let $X = \#$ heads observed. $X : S \to \{0, 1, 2\} \subset \mathbb{R}$.

- What is the probability of the event:
  $\{X = 1\} = \{HT, TH\}$?
  - Concise "random variable" notation is $P(X = 1)$.

$P(X = 0) =$, $P(X = 1) =$, $P(X = 2) =$, $P(X = 7) =$

Example: 100 people (60 women, 40 men) $\Rightarrow$ choose committee of 3 at random.

Let $X = \#$ women chosen. What values can X have?

$P(X = i) =$

i	$P(X = i)$

Example: Suppose we have a committee of 9 Democrats, 8 Republicans, & 3 Independents. We want to form a subcommittee of 3 at random from this group.

Let $X =$ (# of Democrats selected) - (# of Repub's selected).
What values can X have?

$P(X = 3) =$

$\underline{P(X=1)}$ =

$\underline{P(X=0)}$ = $P((1D,1R,1I)\cup(0D,0R,3I)) = \cdots = \frac{217}{1140}$,
$P(X=-1), P(X=-2), P(X=-3) = \ldots.$

- The collection of probabilities for all values of the random variable X is known as the **probability distribution** (or **probability mass function (pmf)**) for the random variable X. In this example, we might express the probability distribution in the form shown to the right.

$$P(X=x) = \left\{\begin{array}{ll} 7/95 & \text{if } x=3 \\ 9/95 & \text{if } x=2 \\ 21/76 & \text{if } x=1 \\ 217/1140 & \text{if } x=0 \\ 23/95 & \text{if } x=-1 \\ 7/95 & \text{if } x=-2 \\ 14/285 & \text{if } x=-3 \end{array}\right\}.$$

Discrete Random Variables

Definition: A random variable X is **discrete** if it can only take on a countable # of values i.e.: $X = x_1, \ldots, x_n$ (finite) or $X = x_1, x_2, \ldots$ (infinite, but countable).

Definition: The **probability mass function (pmf)**, f, of the random variable X is the function $f : \mathbb{R} \to [0,1]$ defined by $f(a) = P(X=a)$.

Coin tossing Example: $P(X=x) = \left\{\begin{array}{ll} 1/4 & \text{if } x=0 \\ 1/2 & \text{if } x=1 \\ 1/4 & \text{if } x=2 \end{array}\right\}.$

$f(0) =$, $f(1) =$, $f(2) =$

Definition 2.1-2: A function $f : \mathbb{R} \to [0,1]$ which satisfies the following 3 conditions is a **pmf**:

1. $f(x) > 0$, $x \in S$ (and $f(x) = 0$ for $x \notin S$).
2. $\displaystyle\sum_{x\in S} f(x) = 1$. $\longrightarrow$ KEY FACT.
3. $P(X \in A) = \sum_{x\in A} f(x)$, where $A \subset S$.

Example: Suppose we count the # of visitors X to visit my home page in 1 day, and suppose
$f(x) = P(X = x) = c \cdot \dfrac{3^x}{x!}$, $x = 0, 1, 2, \ldots.$ What value of c makes this a pmf?

- We have, $P(X = x) > 0$ if $x = 0, 1, \ldots$ & $c > 0$; $P(X = x) = 0$ otherwise.
- What else do we need?

So $f(x) = P(X = x) =$

- The probability mass function (pmf) of X can be written succinctly in the form above, or alternatively, one could list the probabilities in a table such as the following:

x	0	1	2	3	4	$\cdots$
$P(X = x)$						$\cdots$

$P(X = 0) =$

$P(X > 1) =$

Notes (Reminders):

- In all of the examples considered thus far, the sample space has been finite or countable. When S is finite or countable, X is said to be **discrete**.
- When asked to find the **probability mass function** (or **probability distribution**) of X, you must give $P(X = x)$ for all possible values x of X.

Definition: The **cumulative distribution function (cdf)** of a random variable X is given by: $F(x) = P(X \leq x)$ for all $x \in \mathbb{R}$.

Example: Let $X = \#$ of heads in 2 fair coin flips. What is the probability mass function (pmf)? What is the cumulative distribution function (cdf)?

$P(X < 0) =$ $\qquad$ $P(X \leq 0) =$ $\qquad$ $P(X < 1) =$

$P(X \leq 1) =$ $\qquad$ $P(X < 2) =$ $\qquad$ $P(X \leq 2) =$

$F(6)$? $\qquad$ $F(3/2)$? $\qquad$ $F(-372)$?

Properties of a cdf: Let $F(\cdot)$ be a cdf.

1. $F(\cdot)$ is a nondecreasing function (if $a < b$, then $F(a) \leq F(b)$).
2. $\lim_{x\to\infty} F(x) = 1$.
3. $\lim_{x\to-\infty} F(x) = 0$.

Mathematical Expectation (Section 2.2)

We have seen that a random variable X can take on values $x_1, x_2, ...$ with probabilities $f(x_1), f(x_2), ...$ If we perform an experiment, we only observe one value of X. What value do we expect to observe?

Definition: The **expected value** or **expectation** of X, written $E[X]$ (or $E(X)$ or EX), is defined as:

- What are we doing?

Example: Flip coin 3 times, and let $X = \#$ of heads observed.

x	0	1	2	3
$f(x) = P(X = x)$	1/8	3/8	3/8	1/8

- How many do you expect?

Note: $E[X]$ is not necessarily a value that we can actually observe; it is what we "expect", or the average # of heads observed if we repeated the experiment many times.

Example: It costs \$2 to play a certain lottery game. Suppose you have a 1/275 chance of winning \$500 (plus the original \$2) and thus lose your \$2 with prob. 274/275. How much do you expect to win?

Expectation of Functions of Rand. Variables

- We now know how to find $E[X]$ for a discrete random variable X. How do we find $E[u(X)]$ for some function u of X?

Example: Suppose X is a random variable with pmf: What is $E[X^4]$?

x	$P(X = x)$	$y = x^4$
-1	.20	
0	.30	
1	.40	
2	.10	

Definition 2.2-1: Suppose X is a discrete random variable with pmf $f(x)$ and if $u(\cdot)$ is a real-valued function, then:

$$\boxed{E[u(X)] = \sum_{x \in S} u(x) f(x)}.$$

Example: Suppose it costs \$300/week to run a particular Airbnb rental. Let X = # of nights rented per week, and suppose X has pmf:

x	$f(x)$
0	.10
1	.20
2	.25
3	.15
4	.10
5	.05
6	.05
7	.10

- If a one-night stay in this Airbnb costs \$150, what is the expected weekly profit?

Theorem 2.2-1: For constants a, b: $E[au_1(X) + bu_2(X)] = aE[u_1(X)] + bE[u_2(X)]$.

- In the last example, $a = 150$, $b = -300$ and $u_1(X) = X$ and $u_2(X) = 1$.

Proof :

Example: Three buses take concertgoers to the Kettle House Amphitheater to watch the Goo Goo Dolls. There are 36, 40 and 44 concertgoers on the three buses. When they arrive, one concertgoer is selected to introduce the band. What is the expected value of the number of concert goers on the bus of the selected individual?

Special Mathematical Expectations (Section 2.3)

- The mean or expected value of X, $E[X]$, gives info about the "center" of the X-values. Consider the random variables: X, Y, where:

$$P(X=9) = P(X=10) = P(X=11) = 1/3,$$
$$P(Y=1) = P(Y=10) = P(Y=19) = 1/3.$$

$$\left.\begin{array}{l} E[X] = (9)(1/3) + (10)(1/3) + (11)(1/3) = 10 \\ E[Y] = (1)(1/3) + (10)(1/3) + (19)(1/3) = 10 \end{array}\right\} \text{ same mean.}$$

- What is different about X and Y?

- Need a way of measuring spread in a random variable.

Definition: If X is a random variable with mean $\mu = E[X]$, then the **variance of X**, $\mathrm{Var}(X)$, is defined as:

$$\mathrm{Var}(X) = E[(X-\mu)^2] \quad \left(\begin{array}{l} \text{expected } \underline{\text{squared}} \text{ distance from} \\ X \text{ to the } \overline{\text{mean } \mu} \end{array} \right)$$

- This is called the computing formula for the variance - easiest for computing.

Example: Recall the Airbnb example, where $X = \#$ of night rented per week, and X had pmf given to the right:

x	$f(x)$
0	.10
1	.20
2	.25
3	.15
4	.10
5	.05
6	.05
7	.10

We found $E[X] = 2.8$ nights. What is the variance?

Useful Identity: For any constants a, b: $\text{Var}(aX + b) = a^2\text{Var}(X)$.

Back to Example: In the Airbnb example, we were interested in weekly profit: $R = 150X - 300$, and found $E[R] = \$120$. What is $\text{Var}(R)$?

- The square root of the variance of X is called the **standard deviation** of X, and is denoted by: $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

- Units are the same as X. So with $E[X] = 2.8$, renting 4 nights is: 1.2 more than the mean:

- Also, $\text{SD}(R) = \sqrt{\text{Var}(R)} =$

Variance Property: $\text{Var}(X) \geq 0$ always. Why? - Since $(x - \mu)^2 \geq 0$ and since expected values of nonnegative quantities are nonnegative.

- When does $\text{Var}(X) = 0$?

Definition: For any random variable X, the value $E[X^n] = \sum_x x^n f(x)$ is called the **n^{th} moment of X**.

- So:
  $E[X]$ = first **moment** of X, or **mean** of X, often written as μ.
  $E[X^2] = 2^{nd}$ moment of X, etc.
  $E[(X - \mu)^n] = n^{th}$ **central moment** of X, where $\mu = E[X]$.
  $E\left[\left(\frac{X-\mu}{\sigma}\right)^n\right] = E[(X - \mu)^n]/\sigma^n = n^{th}$ **standardized moment** of X, where $\sigma = \text{SD}(X)$.
  $E[(X - \mu)^3]/\sigma^3$ = **skewness** of X.
  $E[(X - \mu)^4]/\sigma^4$ = **kurtosis** of X.

- The 2nd central moment of X is the variance.

Definition 2.3-1: Let X be a discrete random variable with pmf $f(x)$ and sample space S. If there is a positive number h such that $E(e^{tX}) = \sum_{x \in S} e^{tX} f(x)$ exists and is finite for $-h < t < h$, then the function defined by $\boxed{M(t) = E(e^{tX})}$ is called the **moment-generating function** (mgf) of X.

- Note that the mgf is a function of t, not a function of X.
- How does the mgf work in generating moments?

To find the rth moment for a random variable X, we can use $\boxed{E[X^r] = M^{(r)}(0)}$.

Since $E[X] = M'(0)$ and $E[X^2] = M''(0)$, we can find the mean using $\mu = E[X] = M'(0)$ and the variance using $\sigma^2 = E[X^2] - E[X]^2 = M''(0) - M'(0)^2$.

Bernoulli & Binomial Distributions (Section 2.4)

Consider an experiment with two possible outcomes: success or failure. If it's a success, let $X = 1$; if a failure, let $X = 0$. The probability mass function (pmf) of X is:

$$\left\{ \begin{array}{ll} P(X=1) = & p \\ P(X=0) = & 1-p=q \end{array} \right\}, \text{ where } p = \text{prob. of success, } 0 \leq p \leq 1.$$

Definition: Any experiment of this type is known as a **Bernoulli trial**, and X is a **Bernoulli random variable**. *More Info Here.*

Note: The pmf of X can be written: $P(X = x) =$, $x = 0, 1$.

When this occurs, we write: $X \sim \text{Bernoulli}(p)$.

What if we have more than one trial?

Example: Suppose that whenever I play ultimate frisbee, I have a 70% chance of at some point getting hit in the face. Let $X = 1$ if I get hit in the face (success(?)) and $X = 0$ if I don't (failure(?)). Assume that this probability holds no matter how many times I play (i.e. I never get any better) and getting hit in the face one day is independent of getting hit another day.

Suppose I play frisbee one day. What is the probability that I get hit in the face?

Suppose I play twice. What is the probability that I get hit both times? What is the probability that I get hit exactly once?

Suppose I play thrice. What is the probability that all I get hit all three times? What is the probability that I get hit exactly once?

What about playing four times?

$P(X = x) =$

Definition: Let $X = \#$ of successes in n independent Bernoulli trials. Then X is a **Binomial random variable**, and has pmf given by:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x},\ x = 0, 1, \ldots, n.$$

We write $X \sim \text{Bin}(n, p)$ or $X \sim b(n, p)$.

Example: Suppose with 4 children, each child independently is left-handed with probability $p = 1/10$. Let $X = \#$ of children (out of 4) who are left-handed. Pmf of X?

$$P(X = x) = \frac{4!}{x!(4-x)!}(1/10)^x (9/10)^{4-x} = \begin{matrix} (9/10)^4 \\ 4(1/10)(9/10)^3 \\ \vdots \\ \vdots \end{matrix} = \left\{ \begin{array}{ll} .6561, & x = 0 \\ .2916, & x = 1 \\ .0486, & x = 2 \\ .0036, & x = 3 \\ .0001, & x = 4 \end{array} \right\}.$$

(a) What is the probability that at least 1 of the 4 is left-handed?

(b) What is $E(X)$?

Properties of a Binomial Process:

1. There are a fixed number of independent trials, n.
2. There are only two possible outcomes from each trial (success or failure).
3. The probability of success, p, is the same for each trial.
4. The probability of failure, q, is the same for each trial.
5. We are counting the number of successes in the n trials.

Calculator Tips:

Example: Choose 100 cars randomly & count the # of defects on each car. Is this binomial?

Example: Suppose 8% of cars at a certain dealer has at least one minor defect. If you inspect 100 cars, what is the probability at least 10 cars will have a minor defect?

Expected Value and Standard Deviation of Binomial Random Variables

We just saw that for a binomial random variable, X,

$$\boxed{E(X) = n \cdot p}$$

Another way to think about the expected value:

How about the variance and standard deviation?

$\text{Var}(X) = \sum x^2 f(x) - \mu^2.$

Find the variance for the left-handed example with probability distribution $\left\{ \begin{array}{ll} .6561, & x=0 \\ .2916, & x=1 \\ .0486, & x=2 \\ .0036, & x=3 \\ .0001, & x=4 \end{array} \right\}.$

So,

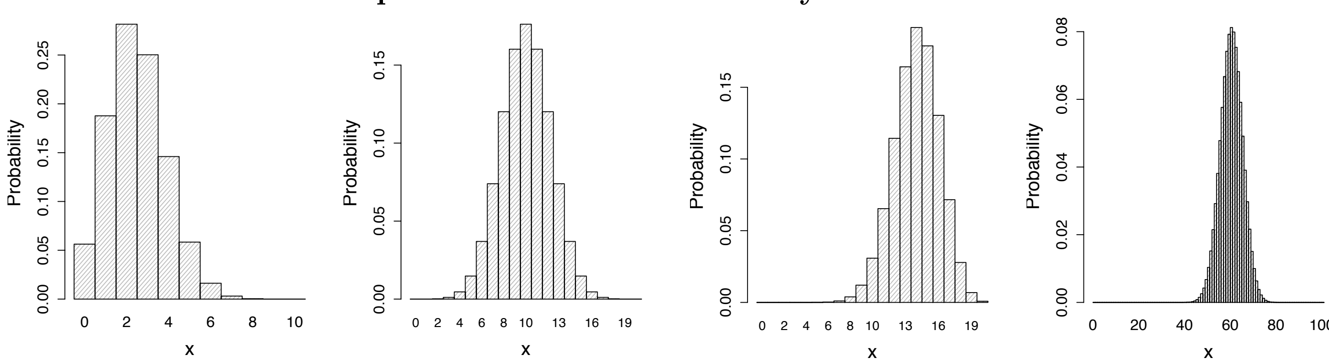
$$\boxed{\text{Var}(X) = \sigma^2 = \qquad\qquad\qquad}$$

and

$$\boxed{\text{SD}(X) = \sigma = \qquad\qquad\qquad}$$

Below are some examples of Binomial(n, p) pmfs. Can you guess which distributions they are?

Examples of Binomial Probability Mass Functions



<u>Definition</u>: The cumulative distribution function (cdf) for a random variable X where $X \sim \text{Bin}(n, p)$, is given by:

$P(X \leq x) =$

<u>Definition</u>: The $q\%$ quantile of the binomial random variable $X \sim \text{Bin}(n, p)$ is the minimum value k such that:

$$P(X \leq k) \geq q/100.$$

- As an example, the 95% quantile is the smallest value k such that $P(X \leq k) \geq 0.95$. I.e.: It's the smallest value k necessary to encounter 95% of the distribution.

Calculating binomial probabilities in **Excel**: Suppose I take 25 free throws independently one after another and let X be the number of free throws made; $X \sim \text{Bin}(n = 25, p = 0.4)$. Consider the following questions.

1. Find the probability that I make exactly 10 free throws.

$$\begin{aligned} P(X=10) &= \binom{25}{10}(0.4)^{10}(0.6)^{15} = 3268760(0.0001048576)(0.000470185) = \underline{0.1612} \\ &= \boxed{\textbf{BINOM.DIST(10, 25, 0.4, FALSE)}} = \underline{0.1612} \end{aligned}$$

2. Find the probability that I make between 10 and 15 free throws inclusive.

$$\begin{aligned} P(10 \le X \le 15) &= P(X=10) + P(X=11) + \cdots + P(X=15) \\ &= \binom{25}{10}(.40)^{10}(.60)^{15} + \cdots + \binom{25}{15}(.40)^{15}(.60)^{10} \\ &= 0.1612 + 0.1465 + 0.1140 + 0.0760 + 0.0434 + 0.0212 \\ &\boxed{\textbf{BINOM.DIST(15, 25, 0.4, FALSE)} - \textbf{BINOM.DIST(9, 25, 0.4, FALSE)}} \\ &= \underline{0.5622} \end{aligned}$$

3. Find the probability that I make no more than 15 free throws.

4. Given that you know I made at least 10 free throws, find the probability that I made at least 12 free throws.

5. Up to how many free throws can I make with at least 95% certainty? (i.e.: if the probability of making more than k free throws is less than 5%, what is the smallest value of k possible?)

The Geometric and Hypergeometric Distributions (Section 2.5)

Geometric Random Variable

In this scenario, instead of counting the number of successes in n Bernoulli trials like we did for the Binomial, we will instead be concerned with the number of trials needed to get one success.

So when $X = \#$ Bernoulli trials required for the first success, X is geometric with prob. p.

What should the pmf of X be in this scenario?

- So: $P(X = x) =$
- And $E(X) =$ and $\text{Var}(X) =$
- $M(t) =$

Properties of a Geometric:

1. There are a sequence of independent Bernoulli trials.
2. There are only two possible outcomes from each trial (success or failure).
3. The probability of success, p, is the same for each trial.
4. The probability of failure, q, is the same for each trial.
5. We are counting the number of trials needed to get the first success.

Calculator Tips

Example: An industrial process manufactures components with a 10% defective rate. Each day the process is subjected to a quality control inspection; products are inspected sequentially until the first defective is obtained.

(a) What is the probability that the first defective will occur in one of the first three components inspected?

(b) What is the probability that more than three components will be inspected before a defective is found?

(c) If three products have been inspected without finding a defective, what is the probability that the next three inspections will be free of defectives?

(d) What is the number of inspections expected until the first defective?

Hypergeometric Random Variable

Suppose you are selecting from two types of items where order doesn't matter. Suppose we have N items to choose from, there are N_1 of Type 1 and $N_2 = N - N_1$ of Type 2. If we select n items and let $X = \#$ of items of Type 1 that are selected, then X is a hypergeometric random variable.

$P(X = x) =$

Properties of a Hyptergeometric:

1. We are selecting n items from N where order of selection does not matter and each item can only be selected once.
2. The probability of an item being selected changes when another has been selected.
3. We are counting the number of type 1 items that are being selected.

Example: Suppose we have 30 ovens, where 5 are defective, & we select 2 at random. What is $P(X = x)$, $x = 0, 1, 2$, where $X = \#$ of defective ovens?

More properties:

$$E[X] = \underbrace{n}_{\#\text{ items selected}} \cdot \underbrace{\frac{N_1}{N}}_{\substack{\text{proportion of type 1}\\ \text{items in population}}}$$

- Through messy calculations:

$$\boxed{\text{Var}(X) = n\left(\frac{N_1}{N}\right)\left(\frac{N_2}{N}\right)\left(\frac{N-n}{N-1}\right)}$$

- If N is large relative to n, then $\text{Var}(X) \approx np(1-p) = \text{Var}(\text{Bin}(n,p))$ where $p = N_1/N$.

In the Example: $E[X] = 2 \cdot (5/30) = 1/3$ ovens. So we expect to find 1/3 defective ovens if we test 2 at random.

It should be noted that there is no closed form of the moment generating function for a hypergeometric distribution.

$\boxed{\textbf{The Negative Binomial Distribution (Section 2.6)}}$

Example: Suppose when shooting free throws, Shaq has a 40% chance of making a free throw and the shots are independent of each other. What is the probability that Shaq makes his third free throw on the sixth attempt?

Suppose we have a sequence of independent Bernoulli trials, where $p =$ the probability of a success. Let $X = \#$ trials required for the r^{th} success. Then $\underline{X \sim \text{NegBin}(r, p)}$ with pmf given by:

$$P(X = x) = \underbrace{\begin{pmatrix} x-1 \\ r-1 \end{pmatrix}} \quad \cdot \quad \underbrace{p^r} \quad \cdot \quad \underbrace{(1-p)^{x-r}} \;,\; x = r, r+1, ...$$

Example: Suppose $r = 3$.
Possible outcomes?

$r = 3$	X	Prob

Note: The negative binomial distribution comes up often in the following type of medical trial: Suppose we keep giving a drug to subjects until we get a certain # of responses (successes) we want (we don't know the sample size in advance).

- Might give patients a new drug until we get 5 responses (successes). Here $r = 5$, p=probability the patient responds, and the study is done to gain info about p.

$$\begin{aligned}
\boxed{\text{E[X]}} &= \sum_{x=r}^{\infty} x \begin{pmatrix} x-1 \\ r-1 \end{pmatrix} p^r (1-p)^{x-r} = \sum_{x=r}^{\infty} \frac{x!}{(r-1)!(x-r)!} \left(\frac{1}{p}\right) p^{r+1} (1-p)^{x-r} \\
&= \frac{r}{p} \sum_{x=r}^{\infty} \frac{x!}{r!(x-r)!} p^{r+1} (1-p)^{x-r} = \frac{r}{p} \sum_{x=r}^{\infty} \begin{pmatrix} x \\ r \end{pmatrix} p^{r+1} (1-p)^{x-r} \\
&= \frac{r}{p} \sum_{y=r+1}^{\infty} \begin{pmatrix} y-1 \\ r \end{pmatrix} p^{r+1} (1-p)^{y-1-r} \quad \text{(letting } y = x+1 \ (x = y-1))
\end{aligned}$$

- With more tedious algebra, it can be shown that:

$$\boxed{\text{Var}(X) = \frac{r(1-p)}{p^2}} = \frac{r}{p}\frac{1-p}{p}.$$

- The moment generating function

$$\boxed{M(t) = \frac{(pe^t)^r}{[1-(1-p)e^t]^r}}$$

Properties of a Negative Binomial:

1. There are a sequence of independent Bernoulli trials.
2. There are only two possible outcomes from each trial (success or failure).
3. The probability of success, p, is the same for each trial.
4. The probability of failure, q, is the same for each trial.
5. We are counting the number of trials needed to get the rth success.

Example: Suppose we attempt a new surgery on lab rats until 10th one dies, where $p = .20$ is the probability an individual rat dies. How many rats do we expect to test?

- This makes sense (need to test 5 on average to have 1 die).

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{10(0.8)}{(.2)^2} = 320 \Rightarrow \text{SD}(X) \approx 18 \text{ rats.}$$

- So, anywhere between 32 & 68 rats is within 1 standard deviation of the mean.
- What is the probability that it takes between 30 and 40 rats inclusive until the 10th one dies?

- What is the probability that no more than 50 rats are needed in this experiment?

Poisson Distribution (Section 2.7)

Definition: A random variable X is said to be a Poisson random variable with parameter $\lambda > 0$, written: $X \sim \text{Poisson}(\lambda)$, if it has pmf:

$$f(x) = P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!},\ x = 0, 1, 2, \ldots.$$

Notes: (i) X can be any positive integer or zero.

(ii) $\sum_{x=0}^{\infty} f(x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda}e^{\lambda} = \underline{1}$, so this is a legitimate pmf.

Example: Suppose $X \sim \text{Poisson}(\lambda = 4)$. Then:

$$P(X = 0) = \frac{e^{-4} \cdot 4^0}{0!} = .0183,\quad P(X = 1) = \frac{e^{-4} \cdot 4^1}{1!} = .0733,$$
$$f(2) = .1465,\ f(3) = .1954,\ f(4) = .1954,\ f(5) = .1563,\ \ldots.$$

Properties of a Poisson:

1.

2. The probability of the event occurring is the same for each interval.
3. The number of occurrences in one interval is independent of the number of occurrences in other intervals.
4. We are counting the number of events that occur.

Examples of Poisson Random Variables

1. The # of texts received on your cell phone in some time unit.
2. The # of "hits" at a web site on a given day.
3. The # of misprints on a page of a book.
4. The # of men killed by horse kicks in the Prussian army in 1 year.

Why are these Poisson?

- If X = # of texts, what distribution does X have? (assuming either 1 or 0 texts per second).
- In this way, Poisson random variables are really approximations to Binomial random variables, where $\lambda = np$.

Calculator and Excel Tips:

Example: Suppose the probability a person in a store buys kombucha is $p = 0.05$. In a sample of $n = 30$ people, what is the probability that no more than 1 buys kombucha?
Let $X = \#$ people who buy kombucha. *Assumptions?*

- Distribution of X?

  $P(X \leq 1) =$

- Since n is "large" and p is small,

Moment-Generating Function:

$M(t) =$

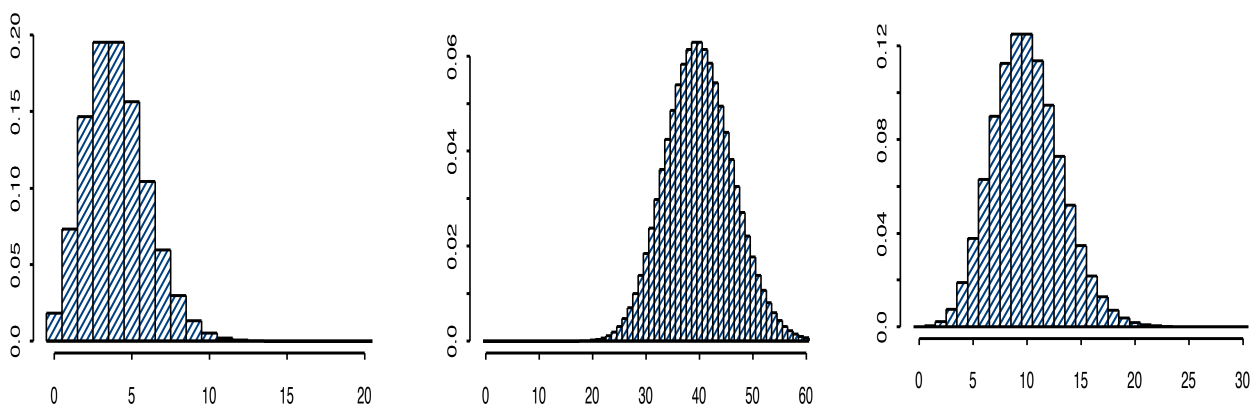
Mean & Variance of a Poisson: Let $X \sim$ Poisson (λ).

$E[X] \;\; =$

$$
\begin{aligned}
E[X^2] &= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda}\lambda^x}{x!} = \lambda \sum_{x=1}^{\infty} \frac{x e^{-\lambda}\lambda^{x-1}}{(x-1)!} = \lambda \sum_{y=0}^{\infty} \frac{(y+1)e^{-\lambda}\lambda^y}{y!} \\
&= \lambda \left[\sum_{y=0}^{\infty} y \frac{e^{-\lambda}\lambda^y}{y!} + \sum_{y=0}^{\infty} \frac{e^{-\lambda}\lambda^y}{y!} \right]
\end{aligned}
$$

$\Rightarrow \mathrm{Var}(X) =$

- So the mean and variance are the <u>same</u> for Poisson random variables.
- What do Poisson pmf's look like? Below are four examples of Poisson pmf's.



Important fact:

If $X_1 \sim \text{Poisson}(\lambda_1)$, $X_2 \sim \text{Poisson}(\lambda_2)$, ... , $X_n \sim \text{Poisson}(\lambda_n)$, then
$Y = X_1 + X_2 + \cdots + X_n \sim \text{Poisson}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$.

<u>Example</u>: The number of tornados that occur in Des Moines in a month has a Poisson distribution with a mean of 1 between June and August and has a Poisson distribution with a mean of 0.5 between September and May. The occurrence of tornados in a given month is independent of the occurrences in any other month. Calculate the probability that there are at most two tornados that occur between July and November.

Give the Distribution for the Following:

1. A casino patron will continue to make \$5 bets on red in roulette until she has won one of these bets. On any bet, she either wins \$5 with probability 18/38 or loses \$5 with probability 20/38. What is the distribution of X = the number of bets necessary to win a bet?

2. A large number of insects are expected to be attracted to a certain variety of rose plant. A commercial insecticide is advertised as being 99% effective. Suppose 2000 insects infest a rose garden where the insecticide has been applied, and let X = the number of surviving insects. What type of random variable is X?

3. Suppose I wish to select a committee of size 3 from this class, which has 17 men and 7 women. Let X = the number of men selected. If the selection is random, what type of random variable is X?

4. On any given weekday, a 911 dispatcher receives an average of 14 calls. Let X be the number of phone calls received by the dispatcher on a Tuesday. What type of random variable is X?

Summary

For a random variable X with associated probability mass function $f(x)$,

- For $f(x)$ to be valid, $0 \le P(x) \le 1$ and $\sum f(x) = 1$.
- $E(X) = \mu = \sum x f(x)$.
- $\mathrm{Var}(X) = E\left((x-\mu)^2\right) = \sigma^2 = \sum (x-\mu)^2 f(x) = \sum x^2 f(x) - \mu^2$.
- $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{\sum x^2 f(x) - \mu^2}$.

Table 1: **Summary Sheet**

Distribution	**Summary**	**Formulas**
Binomial Distribution	A binomial experiment satisfies these conditions: 1. The experiment has a fixed number of n of independent trials. 2. There are only two possible outcomes for each trial. Each outcome can be classified as a success or a failure. 3. The probability of success p is the same for each trial. 4. We are counting the number of successes x in the n trials.	n = the number of trials x = the number of successes in n trials p = the probability of success in a single trial q = probability of failure in a single trial $= 1-p$ $f(x) = \binom{n}{x} p^x q^{n-x} = \dfrac{n!}{(n-x)!x!} p^x q^{n-x}$, $0 \leq x \leq n$. $E(X) = \mu = np$ $\text{Var}(X) = \sigma^2 = npq$, $\text{SD}(X) = \sigma = \sqrt{npq}$ $M(t) = (1-p+pe^t)^n$ $P(X = x) = \text{binompdf}(n, p, x)$ $= \text{BINOM.DIST}(x, n, p, FALSE)$ $P(X \leq x) = \text{binomcdf}(n, p, x)$ $= \text{BINOM.DIST}(x, n, p, TRUE)$
Geometric Distribution	Conditions: 1. A trial is repeated until the first success occurs. 2. The repeated trials are independent of each other. 3. The probability of success p is the same for each trial. 4. The random variable x represents the number of the trials until the first success occurs.	x = the number of trials in which the first success occurs p = probability of success in a single trial q = probability of failure in a single trial $f(x) = p(1-p)^{x-1}$, $x \geq 1$. $E(X) = 1/p$ $\text{Var}(X) = (1-p)/p^2$, $\text{SD}(X) = \sqrt{1-p}/p$ $M(t) = \frac{pe^t}{1-(1-p)e^t}$ $P(X = x) = \text{geometpdf}(n, p, x)$ $= \text{NEGBINOM.DIST}(x-1, 1, p, FALSE)$ $P(X \leq x) = \text{geometcdf}(n, p, x)$ $= \text{NEGBINOM.DIST}(x-1, 1, p, TRUE)$
Hypergeometric Distribution	Conditions: 1. A sample of size n is randomly selected without replacement from a population of N items where order of selection doesn't matter. 2. In the population, N_1 items can be classified as type 1, and $N_2 = N - N_1$ items can be classified as type 2. 3. We are counting the number of type 1 items x that are being selected.	x = the number of Type 1 items that are selected from the N items. $f(x) = \dfrac{\binom{N_1}{x} \cdot \binom{N_2}{n-x}}{\binom{N}{n}}$, $0 \leq x \leq r$. $E(X) = n \cdot N_1/N$ $\text{Var}(X) = n \left(\frac{N_1}{N}\right) \left(\frac{N_2}{N}\right) \left(\frac{N-n}{N-1}\right)$ $\text{SD}(X) = \sqrt{n \left(\frac{N_1}{N}\right) \left(\frac{N_2}{N}\right) \left(\frac{N-n}{N-1}\right)}$ $P(X = x) = \text{HYPGEOM.DIST}(x, n, N_1, N, FALSE)$ $P(X \leq x) = \text{HYPGEOM.DIST}(x, n, N_1, N, TRUE)$
Negative Binomial Distribution	Conditions: 1. A trial is repeated until the rth success occurs. 2. The repeated trials are independent of each other. 3. The probability of success p is the same for each trial. 4. The random variable x represents the number of the trials until the rth success occurs.	x = the number of trials needed for the rth success p = probability of success in a single trial q = probability of failure in a single trial $f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x \geq r$. $E(X) = r/p$ $\text{Var}(X) = r(1-p)/p^2$, $\text{SD}(X) = \sqrt{r(1-p)}/p$ $M(t) = \frac{(pe^t)^r}{[1-(1-p)e^t]^r}$ $P(X = x) = \text{NEGBINOM.DIST}(x-r, r, p, FALSE)$ $P(X \leq x) = \text{NEGBINOM.DIST}(x-r, r, p, TRUE)$
Poisson Distribution	Conditions: 1. The experiment consists of counting the number of times x an event occurs over a specified interval of time, area, or volume. 2. The probability of the event occurring is the same for each interval. 3. The number of occurrences in one interval is independent of the number of occurrences in other intervals.	x = the number of occurrences in the given interval λ = the mean number of occurrences in a given interval unit $f(x) = \dfrac{e^{-\lambda} \lambda^x}{x!}$. $E(X) = \lambda$ $\text{Var}(X) = \lambda$, $\text{SD}(X) = \sqrt{\lambda}$ $M(t) = e^{\lambda(e^t - 1)}$ $P(X = x) = \text{poissonpdf}(\lambda, x)$ $= \text{POISSON.DIST}(x, \lambda, FALSE)$ $P(X \leq x) = \text{poissoncdf}(\lambda, x)$ $= \text{POISSON.DIST}(x, \lambda, TRUE)$

Chapter 3: Continuous Distributions

In Chapter 2 we learned that a random variable is **discrete** if

Not all random variables are discrete. Some can

E.g.,

Random Variables of the Continuous Type (Section 3.1)

Here's the formal definition: A random variable X is said to be **continuous** if there exists a *nonnegative* function $f(x)$, defined on $\mathbb{R}$ so that, for any subset of real numbers, $B \subseteq \mathbb{R}$,

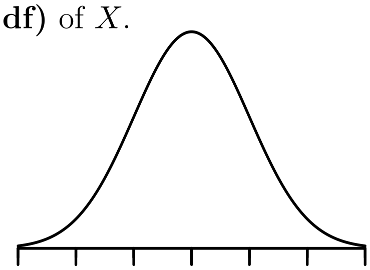
$$P(X \in B) = \int_B f(x)dx.$$

The function $f(x)$ is called the **probability density function (pdf)** of X.

Notes:

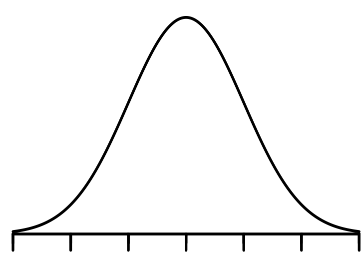
1. What does B look like? - typically an interval ($B = [a, b]$).

   So: $P(X \in B) = P(a \leq X \leq b) = \displaystyle\int_a^b f(x)dx.$



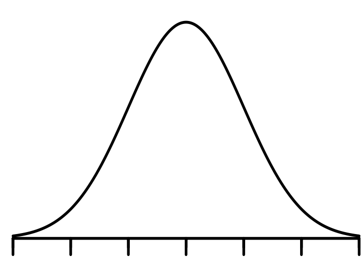
2. $\displaystyle\int_{-\infty}^{\infty} f(x)dx = P(-\infty < X < \infty) = 1$

   (total probability = 1).



3. $P(X = a) = P(a \leq X \leq a) = \displaystyle\int_a^a f(x)dx = 0.$

4. $P(X < a) = P(X \leq a) = F(a) = \displaystyle\int_{-\infty}^{a} f(x)dx.$



5. $\displaystyle\int_a^b f(x)dx = P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$

   $= F(b) - F(a)$ (Fundamental Thm. of Calculus).

6. $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b).$

Properties of a PDF: (i) $f(x) \geq 0 \ \forall x \in S$ (ii) $\displaystyle\int_S f(x)dx = 1$ (Total prob. = 1)

(iii) If the interval $(a, b) \subseteq S$, then $P(a < X < b) = \displaystyle\int_a^b f(x)dx$

Example: Suppose X is a continuous random variable with pdf:

$$f(x) = \begin{cases} c(4x - x^2), & 0 < x < 4 \\ 0 & \text{o/w} \end{cases}$$

(a) What value of c makes this a pdf?

(b) Plot the pdf: $f(x) = \qquad (4x - x^2),\ 0 < x < 4.$

(c) $P(1 \leq X < 3) \ =$

Example: Let X = lifetime of certain computer chip (in years) and suppose X has pdf:

$$f(x) = \left\{ \begin{array}{cc} \frac{2}{9} x e^{-\frac{1}{9}x^2}, & 0 < x < \infty \\ 0 & \text{o/w} \end{array} \right\}.$$

(a) What is the probability the chip lasts more than 3 years?

(b) What is the cdf of X?

$F(x) \ =$

Relationship Between CDF & PDF: If X is a random variable with cdf $F(\cdot)$ and pdf $f(\cdot)$, then:

$$F(x) = \int_{-\infty}^{x} f(t)dt. \quad \left[\text{And } \frac{d}{dx}F(x) = f(x)\right].$$

However, we must be sure $F(x) \to 1$ as $x \to \infty$ and $F(x) \to 0$ as $x \to -\infty$.

Example: $F(x) = 1 - e^{-\frac{1}{9}x^2}$. And $f(x) = \frac{d}{dx}F(x) = -e^{-\frac{1}{9}x^2} \cdot \left(-\frac{2}{9}x\right) = \frac{2}{9}xe^{-\frac{1}{9}x^2}$, $0 < x < \infty$, which is what we had in the example.

Expectation & Variance

For discrete random variables, $E(X) = \sum_x xP(X = x)$.

With continuous random variables, we replace the $\sum$ with a $\int$, so:

$$\boxed{E(X) = \int_x xf(x)dx, \quad \text{when } X \text{ is a continuous RV.}}$$

Ex: $f(x) = \begin{cases} \frac{1}{\theta}e^{-\frac{x}{\theta}}, & x > 0 \\ 0 & \text{o/w} \end{cases}$, $\lambda > 0$.

$E(X) =$

- Most results for expectations with discrete random variables have analogous results for continuous random variables:

1. If X is continuous with pdf $f(x)$, then for any real-valued $u(\cdot)$:

$$E[u(X)] = \int_S u(x)f(x)dx. \text{ (Helpful for mean, variance, and mgf).}$$

2. Var(X) $= E[X^2] - (E[X])^2$ (Computing Formula for variance).
3. If a, b are constants, $E[aX + b] = aE[X] + b$
4. If a, b are constants, $\text{Var}[aX + b] = a^2\text{Var}(X)$.

Example: Suppose X = arsenic concentration (in ppb) at a certain dam below a mining operation, and suppose X has pdf:

$$f(x) = \left\{ \begin{array}{cc} \frac{1}{64}x^{1/2}, & 0 \leq x \leq 16 \\ \frac{1}{16}e^{-\frac{3}{16}x+3}, & x > 16 \end{array} \right\}$$

(a) What is the mean arsenic concentration?

$$E(X) \quad = \quad \int_x xf(x)dx =$$

(b) EPA limit is 18ppb. What is the probability this limit is not exceeded?

$$\begin{aligned} P(X \leq 18) &= \int_0^{16} \frac{1}{64}x^{1/2}dx + \int_{16}^{18} \frac{1}{16}e^{-\frac{3}{16}x+3}dx \\ &= \frac{1}{64} \cdot \frac{2}{3}x^{3/2}\Big|_0^{16} - \frac{1}{3}e^{-\frac{3}{16}x+3}\Big|_{16}^{18} \\ &= \frac{1}{96}(64) - \frac{1}{3}\left(e^{-3/8} - 1\right) = \frac{2}{3} + \frac{1}{3}\left(1 - e^{-3/8}\right) \approx \underline{0.7709}. \end{aligned}$$

OR we could compute: $1 - \int_{18}^{\infty} \frac{1}{16}e^{-\frac{3}{16}x+3}dx.$

(c) What is the median arsenic concentration?

(d) Suppose the measurements were biased due to miscalibration of the equipment used and that the true arsenic concentration is $Y = 0.9X + 3$. What is the mean arsenic concentration?

Uniform Distribution

Definition: A random variable is said to have a uniform distribution on the interval (0,1) if it has pdf:

$$f(x) = \left\{ \begin{array}{cc} 1 & 0 < x < 1 \\ 0 & \text{o/w} \end{array} \right\}$$

Is it a valid pdf? Area under $f(x)$ = Area of the box = $(1)(1) = 1$.

$\left[\text{Or: } \int_0^1 1dx = x|_0^1 = 1 - 0 = 1.\right]$

General Definition: We say that $X \sim \text{Unif}(a, b)$ or $X \sim U(a, b)$ if X has pdf:

$$f(x) = \left\{ \begin{array}{cc} \dfrac{1}{b-a}, & a < x < b \\ 0 & \text{o/w} \end{array} \right\}.$$

For a random variable $X \sim U(-4, 12)$, find:
$P(0 \leq X \leq 2)$

Cumulative Distribution Function (CDF): The cdf for a Uniform(a, b) random variable is:

$F(x) = P(X \leq x) =$

So:

$F(x) \quad =$

- What is the uniform random variable used for?

Mean and Variance of a Uniform(a, b) Random Variable

$$\begin{aligned}
\boxed{E(X)} &= \int_{-\infty}^{\infty} x \frac{1}{b-a} dx = \\
&= \\
E(X^2) &= \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{1}{3} x^3 \Big|_a^b = \frac{b^3 - a^3}{3(b-a)} \\
&= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{b^2 + ab + a^2}{3}. \\
\boxed{\mathrm{Var}(X)} &= E(X^2) - [E(X)]^2 = \frac{b^2 + ab + a^2}{3} - \frac{(b+a)^2}{4} \\
&= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3a^2}{12} = \frac{b^2 - 2ab + a^2}{12} = \boxed{\frac{(b-a)^2}{12}}.
\end{aligned}$$

Example: Suppose you arrive at a bus stop at 10:00am, knowing that the bus will arrive at some time uniformly distributed between 10:00 and 10:30am.

(a) What is the probability you wait no longer than 10 minutes?

(b) If the bus has not come by 10:15am, what is the probability we wait at least an additional 10 minutes?

The Exponential, Gamma, and Chi-Square Distributions (Section 3.2)

The Exponential Distribution

Recall that the Poisson random variable is often used when modeling the number of events that occur within a given time interval. The exponential random variable is used to model the time between events of a Poisson Process.

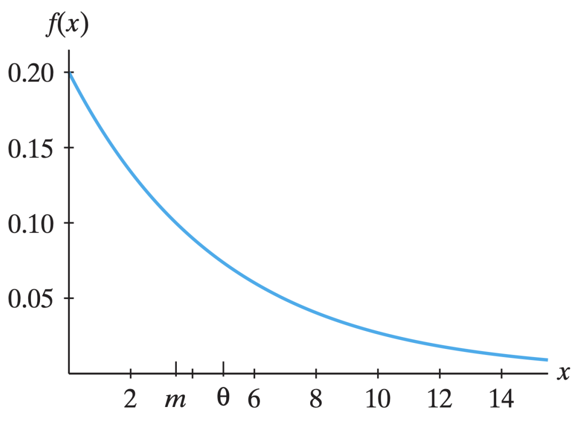
Definition: A random variable X has an **exponential distribution** with parameter θ, if X has probability density function (pdf):

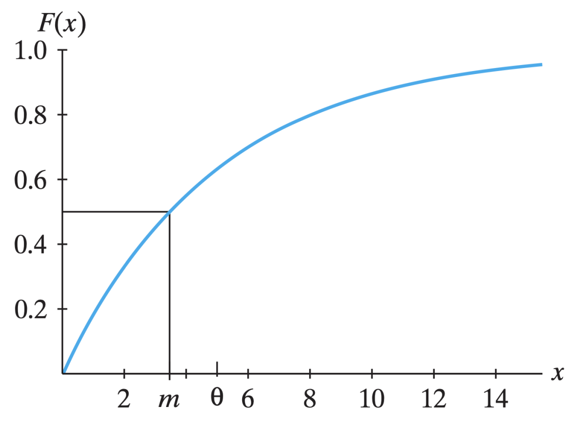
$$f(x) = \frac{1}{\theta} \exp\left(-\frac{x}{\theta}\right), \ x \geq 0, \ 0 < \theta < \infty$$

- We can write $X \sim \text{Exp}(\theta)$
- $\theta = 1/\lambda$ where $\lambda =$ average number of events of the Poisson process in a unit interval.

Cumulative Distribution Function (CDF): The cdf for an Exp(θ) random variable is:

$F(x) =$





Moment-Generating Function (MGF):

$M(t) =$

On page 63, we calculated the expected value for the exponential random variable:

$\boxed{E(X) = \theta}$. It can be shown $\boxed{\text{Var}(X) = \theta^2}$.

Calculator and Excel Tips:

Example: Suppose that recordable injuries at a manufacturing plant are randomly distributed throughout the year, with an average of 1.5 recordable injuries per 30 days.

(a) Calculate the prob. that there are at least three recordable injuries in the next 30 days.

(b) Calculate the prob. that the time until the next recordable injury is less than 30 days.

(c) Calculate the prob. that the time until the there are two recordable injuries is less than 30 days.

The Gamma Distribution

Recall that the Poisson random variable is often used when modeling the number of events that occur within a given time interval. The gamma random variable is used to model the time until the αth event of a Poisson Process.

Definition: A random variable X has an **gamma distribution** with parameters α and θ, if X has probability density function (pdf):

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^{\alpha}} x^{\alpha-1} \exp\left(-\frac{x}{\theta}\right), \; x \geq 0, \; 0 < \theta, \alpha < \infty$$

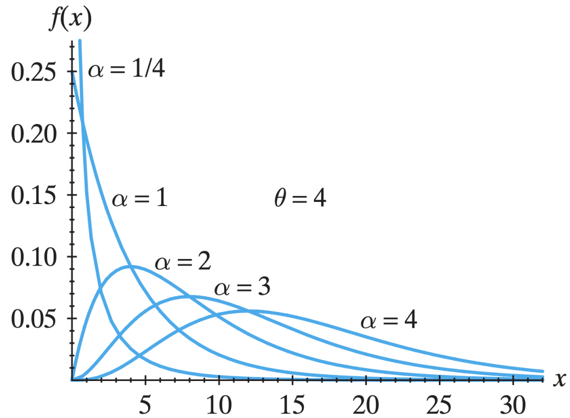
- We can write $X \sim \text{Gam}(\alpha, \theta)$
- $\theta = 1/\lambda$ where $\lambda =$ average number of events of the Poisson process in a unit interval.
- $\Gamma(\alpha)$ is the **gamma function**.

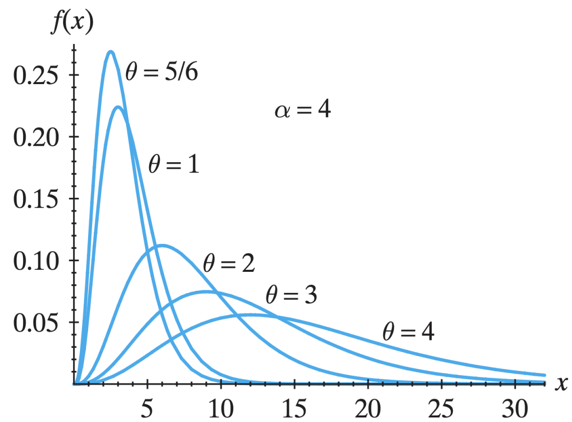
The gamma function:

$$\Gamma(t) = \int_0^{\infty} y^{t-1} e^{-y} dy \text{ for } t > 0$$

You will investigate more about the gamma function in the homework, but you can think of the gamma function as an extension of a factorial. Namely, $\Gamma(n) = (n-1)!$.

Cumulative Distribution Function (CDF): There is no closed form function for the cdf of a gamma distribution, but we can still usually calculate probabilities by integrating by parts.





Gamma Properties:

$$M(t) = \int_0^{\infty} e^{tx} \frac{1}{\Gamma(\alpha)\theta^{\alpha}} x^{\alpha-1} \exp\left(-\frac{x}{\theta}\right) dx = \frac{1}{(1-\theta t)^{\alpha}} \text{ for } t < 1/\theta$$

$E[X] = \mu = \alpha\theta$

Excel Tips:

$\text{Var}(X) = \sigma^2 = \alpha\theta^2$

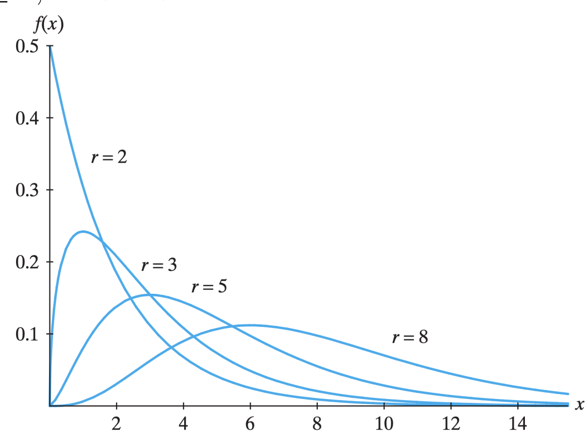
The Chi-Square Distribution

A special case of the gamma distribution that is widely used in inferential statistics is known as the **chi-square distribution** or the **χ^2 distribution**. This distribution is a gamma with $\theta = 2$ and $\alpha = r/2$. All chi-square distributions have degrees of freedom, which we denote with r.

Definition: A random variable X has an **chi-square distribution** with r degrees of freedom if X has probability density function (pdf):

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \ x \geq 0, \ 0 < r < \infty$$

- We can write $X \sim \chi^2(r)$



Chi-square Properties:

$$M(t) = \frac{1}{(1-2t)^{r/2}} \text{ for } t < 1/2$$

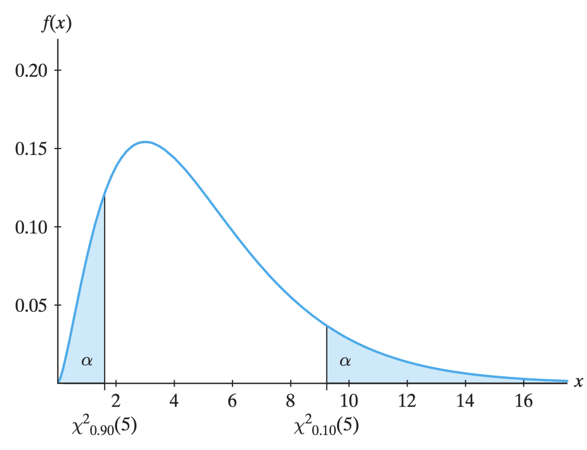
$E[X] = \mu = \alpha\theta =$

$\text{Var}(X) = \sigma^2 = \alpha\theta^2 =$

The chi-square distribution is very important, so tables have been created that can help us find probabilities. Table IV in appendix B has values for the chi-square.

Calculator and Excel Tips:

Example: Let $X \sim \chi^2(5)$. Find the following probs:



$P(X < 7) =$

$P(X > 4) =$

$P(2 < X < 6) =$

Notation

We use special notation for probabilities like these. Namely, we define $\chi^2_\alpha(r)$ as the value such that $P[X > \chi^2_\alpha(r)] = \alpha$. That is, $\chi^2_\alpha(r)$ is the $100(1-\alpha)$th percentile.

So, $\chi^2_{0.1}(5)$ is the number on the chi-square distribution with 5 degrees of freedom such that 10% of the area lies above the number or 90% lies below it. Thus, $\chi^2_{0.1}(5)$ is the 90% percentile and is equal to 9.2364. We can find this in Excel using `=CHISQ.INV(0.90, 5)`.

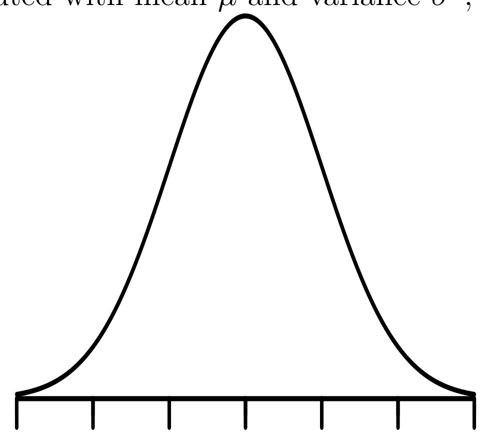
$\boxed{\text{The Normal Distribution (Section 3.3)}}$

Definition: A random variable X has a **normal (Gaussian) distribution** with parameters μ, σ^2 if X has probability density function (pdf):

$$f(x) = \frac{1}{\sqrt{2\pi}\cdot\sigma}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\},\ -\infty < x < \infty,\ -\infty < \mu < \infty,\ \sigma^2 > 0.$$

- In this case, we say that X is normally distributed with mean μ and variance σ^2, and write $\boxed{X \sim N(\mu, \sigma^2)}$.

Normal distributions all have the same bell-shape, as shown to the right. A particular normal pdf is specified by giving its mean μ and standard deviation σ. The mean occurs at the peak of the curve. The standard deviation is the distance between the mean and the first tick mark next to it, which corresponds to the *inflection point* of the distribution.



Desmos Link: https://www.desmos.com/calculator/o0dmsplvb1

Properties of a Normal Distribution

1. Normal densities are always bell-shaped.
2. Normal densities are symmetric about the mean μ.
3.

Why are normal random variables so important?

1. Many random phenomena are approximately normally distributed.
(Examples: Heights of men or women in this class, arsenic concentrations in groundwater, diameter of a tree, annual snowfall, etc.)
2. Normal probabilities can be used to approximate binomial probabilities. This was the original use of the normal distribution, and the result governing this approximation is the **DeMoivre-Laplace Theorem** (1812).
3. Sample means drawn from virtually *any* population are approximately normal! This result is one of the most important results in statistics and is known as the **Central Limit Theorem**.

Are normal densities legitimate pdfs?: By inspection of the normal pdf given on the previous page, for $\sigma^2 > 0$, $f(x) > 0 \; \forall \; x \in (-\infty, \infty)$. So we need only verify that the density function is a *probability* density function, i.e.: we need:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx = 1.$$

Before attacking this integral directly, consider the following result:

Fact : $\displaystyle\int_{-\infty}^{\infty} \exp\left\{-\frac{y^2}{2}\right\} dy = \sqrt{2\pi}.$ (Uses a polar transformation and double integral)

Using this fact then:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx =$$

Mean and Variance of a Normal Random Variable: Computing:

$$\begin{aligned}
\boxed{E(X)} &= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx \\
&= \frac{1}{\sqrt{2\pi} \cdot \sigma} \left[\int_{-\infty}^{\infty} (x-\mu) \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx\right. \\
&\quad \left. + \int_{-\infty}^{\infty} \mu \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx\right] \quad \text{(adding and subtracting } \mu) \\
&= \frac{1}{\sqrt{2\pi} \cdot \sigma} \left[-\sigma^2 \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}\Bigg|_{-\infty}^{\infty} + \mu \int_{-\infty}^{\infty} \exp\left\{-\frac{y^2}{2}\right\} \sigma dy\right] \\
&\quad \left(\text{letting } y = \frac{x-\mu}{\sigma}, \text{ so: } dy = \frac{1}{\sigma} dx \;\Rightarrow\; \sigma dy = dx\right) \\
&= \frac{1}{\sqrt{2\pi} \cdot \sigma} \left[0 + \mu\sigma\sqrt{2\pi}\right] = \boxed{\mu}. \\
E(X^2) &= \mu^2 + \sigma^2 \;\; \text{(via tedious algebra and calculus), so that:} \\
\boxed{\mathrm{Var}(X)} &= E(X^2) - [E(X)]^2 = \mu^2 + \sigma^2 - \mu^2 = \boxed{\sigma^2}.
\end{aligned}$$

Other Results for Normal Distributions

1. If $X \sim N(\mu, \sigma^2)$, then $Y = aX + b \sim$

Hence, any linear function of a normal random variable is normal.

2. <u>**Theorem 3.3-1**</u> If $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X-\mu}{\sigma} \sim$

Z here is called the <u>**standard normal**</u> random variable, and has pdf given by:

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-z^2/2\right\}, \quad \infty < z < \infty.$$

<u>CDF of a Normal Random Variable</u>: Suppose $X \sim N(\mu, \sigma^2)$. The cdf of X is given by:

$$F(x) \;=\; P(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}\cdot\sigma} \exp\left\{-\frac{(t-\mu)^2}{2\sigma^2}\right\} dt. \text{ (by definition)}$$

Problem?

$F(x) \;=\; P(X \leq x) =$

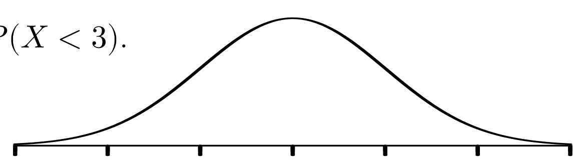
$=$

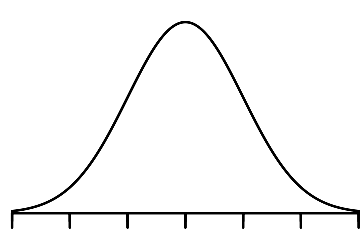
- The <u>standard normal cdf</u> Φ has been computed numerically with the resulting values given in the standard normal tables that are found in the book's appendices and on Canvas.
- The technique of converting the normal probability statement to be in terms of the *standard* normal random variable Z is known as <u>**"standardizing"**</u>.
- The utility of the Standard Normal Table comes from noting that any probability calculation involving a normal random variable can be standardized and thus written in terms of the standard normal random variable Z, and subsequently calculated using this table.

<u>**Calculator and Excel Tips:**</u>

<u>Example</u>: Suppose $X \sim N(\mu = 2, \sigma^2 = 16)$. Find $P(X < 3)$.

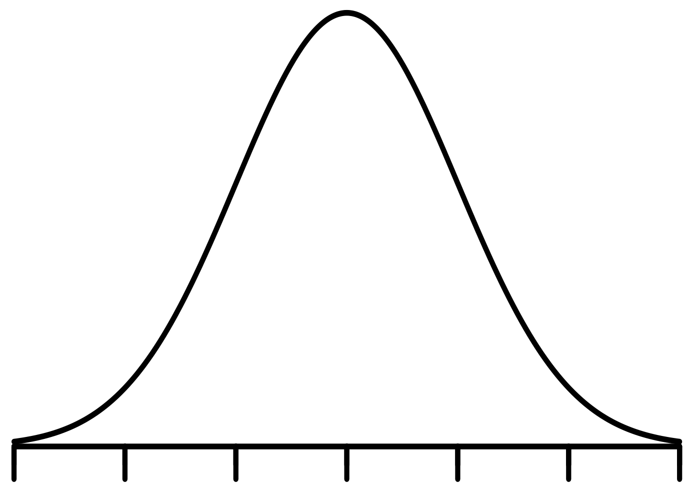
$P(X < 3) \;=$



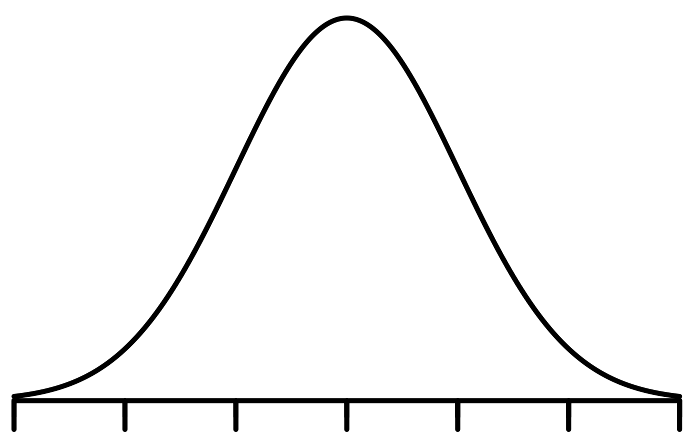


Example: Suppose the annual snowfall in Missoula is normally distributed with mean $\mu = 42$ inches, and standard deviation $\sigma = 8$ inches.

(a) What is the probability that starting this year, it will take over 10 years before a year occurs having a snowfall over 60 inches? What assumptions are you making?



(b) At what snowfall level is the probability of exceeding that snowfall level 0.05? (i.e.: 5% of all snowfall levels fall above what snowfall?)



Other random variables: There are many other random variable distributions. Some we will discuss later and others, such as the the Weibull Distribution, we will not focus on in this class.

Relation between random variables

There is an association between different continuous random variables we discussed. For example, if $Z \sim N(0,1)$ is a standard normal random variable, then we can find the type of variable that Z^2 is:

The technique we used above is known as the **cdf technique** for transforming the distribution of a transformed variable. We will investigate this further in Chapter 5.

Theorem 3.3-2 If the random variable X is $N(\mu, \sigma^2)$, with $\sigma^2 > 0$, then the random variable $V = \dfrac{(X-\mu)^2}{\sigma^2} = Z^2$ is $\chi^2(1)$.

Chapter 4: Bivariate Distributions

In Chapters 2 and 3, all of the distributions we discussed pertained to only one variable. In many applications, we have two (or more) random variables, known as a **bivariate random variable** and we would like to know about the probability of the two variables.

Bivariate Distributions of the Discrete Type (Section 4.1)

Definition 4.1-1: Let X and Y be two random variables defined on a discrete sample space. Let S denote the corresponding two-dimensional space of X and Y, the two discrete random variables. The probability that $X = x$ and $Y = y$ is denoted by $f(x, y) = P(X = x, Y = y)$. The function $f(x, y)$ is called the **joint probability mass function** (joint pmf) of X and Y and has the following properties:

1. $0 \leq f(x, y) \leq 1$.
2. $\displaystyle\sum\sum_{(x,y)\in S} f(x, y) = 1$.
3. $P[(X, Y) \in A] = \displaystyle\sum\sum_{(x,y)\in A} f(x, y)$, where $A \subset S$.

Example: Suppose you are selecting a committee of 3 people from 6 math faculty, 4 CS faculty, and 5 engineering faculty. Let X be the number of math faculty on the committee and Y be the number of CS faculty on the committee. The table below shows most of the $f(x, y)$ values.

		X			
		0	1	2	3
Y	0		12/91	15/91	4/91
	1		24/91	12/91	0
	2	6/91	36/455	0	0
	3	4/455	0	0	0

Find $f(0, 0)$ and $f(0, 1)$.

Definition 4.1-2 Part 1: Let X and Y have the joint probability mass function $f(x,y)$ with sample space S. The probability mass function of X alone, which is called the **marginal probability mass function** of X is defined by

$$f_X(x) = \sum_y f(x,y) = P(X=x), \text{ for } x \in S_X$$

and the marginal pmf of Y is defined by function of Y alone, which is called the marginal probability mass function of Y is defined by

$$f_Y(y) = \sum_x f(x,y) = P(Y=y), \text{ for } y \in S_Y.$$

In our previous example,

$$\begin{aligned}
f_X(x) = \sum_y f(x,y) &= \sum_{y=0}^{3} \frac{\binom{6}{x}\binom{4}{y}\binom{5}{3-x-y}}{\binom{15}{3}} \\
&= \frac{\binom{6}{x}\binom{4}{0}\binom{5}{3-x-0}}{\binom{15}{3}} + \frac{\binom{6}{x}\binom{4}{1}\binom{5}{3-x-1}}{\binom{15}{3}} + \frac{\binom{6}{x}\binom{4}{2}\binom{5}{3-x-2}}{\binom{15}{3}} + \frac{\binom{6}{x}\binom{4}{3}\binom{5}{3-x-3}}{\binom{15}{3}} \\
&= \frac{\binom{6}{x}\left[\binom{4}{0}\binom{5}{3-x-0} + \binom{4}{1}\binom{5}{3-x-1} + \binom{4}{2}\binom{5}{3-x-2} + \binom{4}{3}\binom{5}{3-x-3}\right]}{\binom{15}{3}} \\
&= \qquad\qquad\qquad \text{(using Vandermonde's Identity)}
\end{aligned}$$

Vandermonde's Identity: $\binom{x+y}{n} = \sum_{k=0}^{n} \binom{x}{k}\binom{y}{n-k}$. In our case, $x = 4, y = 5, n = 3-x$.

So, $f_X(0) = \frac{\binom{6}{0}\binom{9}{3-0}}{\binom{15}{3}} =$ $\qquad f_X(1) = \frac{\binom{6}{1}\binom{9}{3-1}}{\binom{15}{3}} =$

$f_X(2) = \frac{\binom{6}{2}\binom{9}{3-2}}{\binom{15}{3}} =$ $\qquad f_X(3) = \frac{\binom{6}{3}\binom{9}{3-3}}{\binom{15}{3}} =$

Similar idea for Y. We now sum out the x:

$$\begin{aligned}
f_Y(y) = \sum_x f(x,y) &= \sum_{x=0}^{3} \frac{\binom{6}{x}\binom{4}{y}\binom{5}{3-x-y}}{\binom{15}{3}} \\
&= \frac{\binom{6}{0}\binom{4}{y}\binom{5}{3-y-0}}{\binom{15}{3}} + \frac{\binom{6}{1}\binom{4}{y}\binom{5}{3-y-1}}{\binom{15}{3}} + \frac{\binom{6}{2}\binom{4}{y}\binom{5}{3-y-2}}{\binom{15}{3}} + \frac{\binom{6}{3}\binom{4}{y}\binom{5}{3-y-3}}{\binom{15}{3}} \\
&= \frac{\binom{4}{y}\left[\binom{6}{0}\binom{5}{3-y-0} + \binom{6}{1}\binom{5}{3-y-1} + \binom{6}{2}\binom{5}{3-y-2} + \binom{6}{3}\binom{5}{3-y-3}\right]}{\binom{15}{3}} \\
&= \frac{\binom{4}{y}\binom{11}{3-y}}{\binom{15}{3}} \text{ (using Vanermonde Identity)}
\end{aligned}$$

So, $f_Y(0) = \frac{\binom{4}{0}\binom{11}{3-0}}{\binom{15}{3}} = 33/91$ $\qquad$ $f_Y(1) = \frac{\binom{4}{1}\binom{11}{3-1}}{\binom{15}{3}} = 44/91$

$f_Y(2) = \frac{\binom{4}{2}\binom{11}{3-2}}{\binom{15}{3}} = 66/455$ $\qquad$ $f_Y(3) = \frac{\binom{4}{3}\binom{11}{3-3}}{\binom{15}{3}} = 4/455$

Independence

Recall that for probabilities, we can check if two events are independent if $P(A \cap B) = P(A \text{ and } B) = P(A)P(B)$. For bivariate variables, the ideas is similar.

Definition 4.1-2 Part 2: The random variables X and Y are **independent** if and only if $P(X = x, Y = y) = P(X = x)P(Y = y)$ or, equivalently, $f(x, y) = f_X(x)f_Y(y)$ for $x \in S_X$ and $y \in S_Y$. Otherwise, the random variables are said to be **dependent**.

Example: Let the joint pmf of X and Y be $f(x, y) = \dfrac{xy}{c}$ for $x = 1, 2, 3$, and $y = 1, 2$. Find the value for c that would make this a legitimate pmf and then determine whether X and Y are independent.

Bivariate Expectation

Just like in the univariate case, we can find expected values in the bivariate case as well. For two random variables X and Y, the **expected value** (or mathematical expectation) of $u(X, Y)$ is $E[u(X, Y)] = \sum\sum_{(x,y)\in S} u(x, y)f(x, y)$.

Example: Returning to the example with joint pmf $f(x,y) = \frac{xy}{}$ for $x = 1, 2, 3$, and $y = 1, 2$, find $E[X]$, $E[Y]$, $E[X+Y]$ and $E[XY]$.

Important Expectations:

- $E[X] = \mu_X = \sum_{x \in S_X} x f_X(x) = \sum\sum_{(x,y) \in S} x f(x,y)$
- $E[Y] = \mu_Y = \sum_{y \in S_Y} y f_Y(y) = \sum\sum_{(x,y) \in S} y f(x,y)$
- $$E[(X-\mu_X)^2] = \sigma_X^2 = \sum_{x \in S_X} (x-\mu_X)^2 f_X(x) = \sum\sum_{(x,y) \in S} (x-\mu_X)^2 f(x,y)$$
  $$= \left[\sum_{x \in S_X} x^2 f_X(x)\right] - \mu_X^2 = \left[\sum\sum_{(x,y) \in S} x^2 f(x,y)\right] - \mu_X^2$$
- $$E[(Y-\mu_Y)^2] = \sigma_Y^2 = \sum_{y \in S_Y} (y-\mu_Y)^2 f_Y(y) = \sum\sum_{(x,y) \in S} (y-\mu_Y)^2 f(x,y)$$
  $$= \left[\sum_{y \in S_Y} y^2 f_Y(y)\right] - \mu_Y^2 = \left[\sum\sum_{(x,y) \in S} y^2 f(x,y)\right] - \mu_Y^2$$

The Correlation Coefficient (Section 4.2)

We know that the variance is a measure of how spread out the values of a random variable are (specifically, the average squared distance the values are from the mean, taking probabilities into account). Now that we are discussing two random variables simultaneously, we can also get measures of how related they are to each other.

The first of these measures is known as the **covariance**. For random variables X and Y (discrete or continuous) with respective means μ_X and μ_Y and respective standard deviations σ_X and σ_Y, the covariance of X and Y is

$$E[(X-\mu_x)(Y-\mu_Y)] = \sigma_{XY} = \text{Cov}(X,Y).$$

We can compute the covariance in a couple difference ways in the discrete case:

- $\text{Cov}(X,Y) = \sum\sum_{(x,y)\in S}(x-\mu_X)(y-\mu_Y)f(x,y).$
- $\text{Cov}(X,Y) = E[XY] - \mu_X\mu_Y = \left[\sum\sum_{(x,y)\in S} xyf(x,y)\right] - \mu_X\mu_Y$

Although the covariance is useful, it is often helpful to standardize it so that it (a) is scaled to be between -1 and 1 and (b) is unitless so it can be easily compared for different pairs of variables. This standardized measure is known as the **correlation coefficient**, which is denoted as ρ or sometimes ρ_{XY}:

$$\rho = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

A few comments about the correlation coefficient:

- If $\rho = 1$, that means the variables X and Y are perfectly positively correlated.
- If $\rho = -1$, that means the variables X and Y are perfectly negatively correlated.
- If $\rho = 0$, that means the variables X and Y are not correlated at all.

So, the farther from zero the correlation coefficient is, either in the positive or negative direction,

If ρ is known, we can it is sometimes useful to know that $E[XY] = \rho\sigma_X\sigma_Y + \mu_X\mu_Y$.

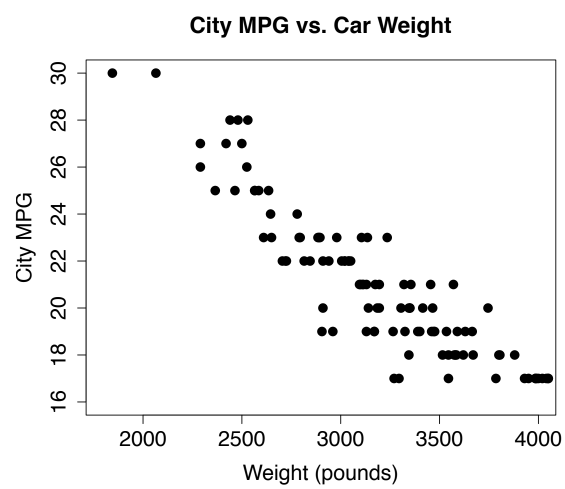
Example: For the example with joint pmf $f(x,y) = \dfrac{xy}{18}$ for $x = 1,2,3$, and $y = 1,2$, find $\text{Cov}(X,Y)$ and ρ.

Correlation in Applied Statistics

In applied statistics, we can calculate a sample correlation coefficient. This is also known as the Pearson correlation coefficient and the formula for calculating it is

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left[\left(\frac{x_i - \bar{x}}{s_x} \right) \cdot \left(\frac{y_i - \bar{y}}{s_y} \right) \right]$$

It is most useful in determining how related to quantitative variables are. Or, if we look at a **scatterplot** of two variables, the correlation coefficient tells us information about both the slope of a line that fits the data best and how close the points lie to that line.



Independence facts related to covariance

- If X and Y are independent, then $E[u(X)v(Y)] = E[u(X)]E[v(Y)]$.
- If X and Y are independent, then $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = 0$.

Proof of first fact:

Conditional Distributions (Section 4.3)

By now, you've noticed that many of the probability topics from Chapter 1 can be applied to bivariate probability distributions. This is true for conditional probability as well. Recall that $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$.

Definition 4.3-1: The **conditional probability mass function** of X, given that $Y = y$ is defined by

$$g(x|y) = \frac{f(x, y)}{f_Y(y)}, \text{ as long as } f_Y(y) > 0.$$

Similarly, the condition pmf of Y, given that $X = x$, is defined by

$$h(y|x) = \frac{f(x, y)}{f_X(x)}, \text{ as long as } f_X(x) > 0.$$

Example: Let's go back to the example with selecting a committee of 3 and let's find the conditional distribution of selecting math faculty given that there are no computer science faculty.

$$f(x, y) = \frac{\binom{6}{x}\binom{4}{y}\binom{5}{3-x-y}}{\binom{15}{3}}, \qquad f_Y(y) = \frac{\binom{4}{y}\binom{11}{3-y}}{\binom{15}{3}}$$

		X				
		0	1	2	3	Total
Y	0	2/91	12/91	15/91	4/91	33/91
	1	8/91	24/91	12/91	0	44/91
	2	6/91	36/455	0	0	66/455
	3	4/455	0	0	0	4/455
	Total	12/65	216/455	27/91	4/91	1

Conditional Expectation

Since conditional distributions are probability distributions in their own right, we can find the **conditional mean** and the **conditional variance**.

- $E[u(X)|Y=y] = \sum_x u(x)g(x|y)$
- $E[v(Y)|X=x] = \sum_y v(y)h(y|x)$
- The conditional mean of X is $\mu_{X|y} = E[X|y] = \sum_x xg(x|y)$
- The conditional mean of Y is $\mu_{Y|x} = E[Y|x] = \sum_y yh(y|x)$
- The conditional variance of X is
  $\sigma^2_{X|y} = \mathrm{Var}(X|y) = E[(X-\mu_{X|y})^2|y] = \sum_x (x-\mu_{X|y})^2 g(x|y) = E[X^2|y] - \mu^2_{X|y}$
- The conditional variance of Y is
  $\sigma^2_{Y|x} = \mathrm{Var}(Y|x) = E[(Y-\mu_{Y|x})^2|x] = \sum_y (y-\mu_{Y|x})^2 h(y|x) = E[Y^2|x] - \mu^2_{Y|x}$

Example: Find the conditional mean and variance for the number of math faculty on the committee given there are no computer science faculty on the committee.

It can also be useful to know the following laws of total probability.

<u>**Theorem 4.3-1: Law of Total Probability for Expectation**</u>: Let X and Y be discrete random variables such that $E(Y)$ exists. Then

$$E[E(Y|X)] = E(Y) \text{ where } E[E(Y|X)] = \sum_x \left[\sum_y y h(y|x) \right] f_X(x).$$

The same is true for X: $E[E(X|Y)] = E(X)$.

<u>**Theorem 4.3-2: Law of Total Probability for Variances**</u>: If X and Y are discrete random variables, then

$$E[\mathrm{Var}(Y|X)] + \mathrm{Var}(E[Y|X]) = \mathrm{Var}(Y)$$

where $\mathrm{Var}(Y|X) = E[Y^2|X] - E[Y|X]^2$ and $\mathrm{Var}(E[Y|X]) = E[E(Y|X)^2] - E[Y]^2$ provided all of the expectations exist.

<u>Example</u>: Verify the law of total probability for expectation holds by finding $E[Y]$ where $f(x, y) = \dfrac{x + y}{12}$ for $x = 1, 2$, and $y = 1, 2$.

Showing the law of total probability for variances holds here. First,

$$\begin{aligned}\text{Var}(Y) &= E[Y^2] - E[Y]^2 \\ &= \sum_{y=1}^{2} y^2 f_Y(y) - \left(\frac{19}{12}\right)^2 = \sum_{y=1}^{2} y^2 \frac{2y+3}{12} - \left(\frac{19}{12}\right)^2 = 1^2 \cdot \frac{5}{12} + 2^2 \cdot \frac{7}{12} - \left(\frac{19}{12}\right)^2 = \boxed{\frac{35}{144}}.\end{aligned}$$

Now, we can find $E[\text{Var}(Y|X)]$:

$$\begin{aligned}\text{Var}(Y|X) &= E[Y^2|X] - E[Y|X]^2 \\ &= \sum_{y=1}^{2} y^2 h(y|x) - \left(\frac{3x+5}{2x+3}\right)^2 = \sum_{y=1}^{2} y^2 \left(\frac{x+y}{2x+3}\right) - \left(\frac{3x+5}{2x+3}\right)^2 \\ &= 1^2 \cdot \left(\frac{x+1}{2x+3}\right) + 2^2 \cdot \left(\frac{x+2}{2x+3}\right) - \left(\frac{3x+5}{2x+3}\right)^2 = \frac{5x+9}{2x+3} - \left(\frac{3x+5}{2x+3}\right)^2\end{aligned}$$

So

$$\begin{aligned}E[\text{Var}(Y|X)] &= \sum_{x=1}^{2} \text{Var}(Y|X) f_X(x) \\ &= \sum_{x=1}^{2} \left[\frac{5x+9}{2x+3} - \left(\frac{3x+5}{2x+3}\right)^2\right] \frac{2x+3}{12} = \left(\frac{14}{5} - \frac{64}{25}\right) \cdot \frac{5}{12} + \left(\frac{19}{7} - \frac{121}{49}\right) \cdot \frac{7}{12} = \boxed{\frac{7}{70}}.\end{aligned}$$

Finally, we need to find $\text{Var}(E[Y|X])$:

$$\begin{aligned}\text{Var}(E[Y|X]) &= E[E(Y|X)^2] - E[E(Y|X)]^2 \\ &= E[E(Y|X)^2] - E[Y]^2 \text{ by the law of total probability for expectation} \\ &= \sum_{x=1}^{2} E[Y|X]^2 f_X(x) - \left(\frac{19}{12}\right)^2 = \sum_{x=1}^{2} \left(\frac{3x+5}{2x+3}\right)^2 \left(\frac{2x+3}{12}\right) - \left(\frac{19}{12}\right)^2 \\ &= \left(\frac{8}{5}\right)^2 \cdot \frac{5}{12} + \left(\frac{11}{7}\right)^2 \cdot \frac{7}{12} - \left(\frac{19}{12}\right)^2 = \boxed{\frac{1}{5040}}\end{aligned}$$

Therefore, we have $\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X]) = \frac{7}{70} + \frac{1}{5040} = \boxed{\frac{35}{144}}$, which matches the calculation at the top of the page!

<u>Example</u>: Let $X \sim \text{Poisson}(\lambda)$ with $\lambda \sim \text{Gam}(2, 1)$. Find $E[X]$ and $\text{Var}(X)$.

Bivariate Distributions of the Continuous Type (Section 4.4)

In this section, we will be extending most of the topics from the discrete case to the continuous case. That means that we will now be working with probability density functions (pdfs) rather than probability mass functions (pmfs). Remember that for a pdf, $f(x) \neq P(X = x)$ since $P(X = x) = 0$ for any x in the continuous case. We can still use pdfs to find probabilities by integrating. Now, however, we will have double integrals just as we had double sums in the previous sections of this chapter. Double integration is one of the few Calculus III topics in this course. Let's discus double integration quickly before moving on.

Aside: Double Integration

For the most part, double integrals work exactly as you would expect. We begin with the inner most integrals and work our way out. When integrating, we treat the variable we are not integrating with respect to as a constant.

Example: Evaluate $\int_0^2 \int_{-1}^4 (x+y)dxdy$

We can also have variables in the bounds:

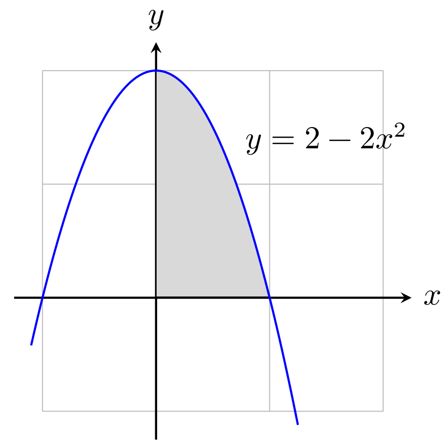
Example: Evaluate $\int_0^2 \int_x^{x^2} (x+y)dydx$

It is not always the case that we can switch the order of integration, although you can under special circumstances:

Example: Evaluate $\int_x^{x^2} \int_0^2 (x+y)dxdy$

We can find the area of a region using double integrals instead of single ones. The key is that the inner integral goes from boundary to boundary and the outer integral goes from extreme to extreme.

Example: Find the area of the region shown here:



Back to the Probability

Definition: Let X and Y be two random variables defined on a continuous sample space. Let S denote the corresponding two-dimensional space of X and Y. The function $f(x, y)$ is called the **joint probability density function** (joint pdf) of X and Y and has the following properties:

1. $f(x, y) \geq 0$, where $f(x, y) = 0$ when (x, y) is not in the support (sample space) S of X and Y.
2. $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)dxdy = \iint_S f(x, y)dxdy = 1.$
3. $P[(X, Y) \in A] = \displaystyle\iint_{(x,y)\in A} f(x, y)$, where $\{(X, Y) \in A\}$ is an event defined in the plane.
   (We can think of that as a region for which we want the area).

The **marginal pdfs** of X and Y are found by integrating out the other variable and are given by:

$$f_X(x) = \int_{S_Y} f(x,y)dy, \text{ for } x \in S_X$$

and

$$f_Y(y) = \int_{S_X} f(x,y)dx, \text{ for } y \in S_Y$$

We can find the **mean, variances, covariance, and correlation** in a similar way as we did in the discrete case:

- $E[X] = \mu_X = \int_{S_X} x f_X(x)dx = \iint_S x f(x,y)dA.$
- $E[Y] = \mu_Y = \int_{S_Y} y f_Y(y)dy = \iint_S y f(x,y)dA$
- $$\begin{aligned} E[(X-\mu_X)^2] = \sigma_X^2 &= \int_{S_X} (x-\mu_X)^2 f_X(x)dx = \iint_S (x-\mu_X)^2 f(x,y)dA \\ &= \left[\int_{S_X} x^2 f_X(x)dx\right] - \mu_X^2 = \left[\iint_S x^2 f(x,y)dA\right] - \mu_X^2 \end{aligned}$$
- $$\begin{aligned} E[(Y-\mu_Y)^2] = \sigma_Y^2 &= \int_{S_Y} (y-\mu_Y)^2 f_Y(y)dy = \iint_S (y-\mu_Y)^2 f(x,y)dA \\ &= \left[\int_{S_Y} y^2 f_Y(y)dy\right] - \mu_Y^2 = \left[\iint_S y^2 f(x,y)dA\right] - \mu_Y^2 \end{aligned}$$
- $$\begin{aligned} E[(X-\mu_X)(Y-\mu_Y)] = \text{Cov}(X,Y) &= \iint_S (x-\mu_X)(y-\mu_Y)f(x,y)dA \\ &= E[XY] - \mu_X\mu_Y = \iint_S xyf(x,y)dA - \mu_X\mu_Y \end{aligned}$$
- The correlation is still $\rho = \dfrac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$

In addition, X and Y are **independent** if and only if $f(x,y) = f_X(x) \cdot f_Y(y)$.
Conditional distributions are analogous to the discrete case as well:

- $g(x|y) = \dfrac{f(x,y)}{f_Y(y)}$ if $f_Y(y) > 0$.
- $h(y|x) = \dfrac{f(x,y)}{f_X(x)}$ if $f_X(x) > 0$.
- $\mu_{X|y} = E[X|y] = \int_{S_X} xg(x|y)dx$
- $\mu_{Y|x} = E[Y|x] = \int_{S_Y} yh(y|x)dy$
- $$\begin{aligned} \sigma_{X|y}^2 &= E[(X-\mu_X)^2|y] \\ &= \int_{S_X} (x-\mu_{X|y})^2 g(x|y)dx \\ &= \left[\int_{S_X} x^2 g(x|y)dx\right] - \mu_{X|y}^2 \end{aligned}$$
- $$\begin{aligned} \sigma_{Y|x}^2 &= E[(Y-\mu_Y)^2|x] \\ &= \int_{S_Y} (y-\mu_{Y|x})^2 h(y|x)dy \\ &= \left[\int_{S_Y} y^2 h(y|x)dy\right] - \mu_{Y|x}^2 \end{aligned}$$

Example: Let $f(x,y) = x/4$ for $0 \le x \le 2$ and $x^2 \le y \le 4$.

(a) Verify $f(x,y)$ is a pdf.

(b) Find $P(0 < X < 1)$.

(c) Find $P(0 < X < 1 \text{ and } 2 < Y < 3)$.

(d) Find μ_Y.

(e) Find $P(2 < Y < 3|X = 1)$.

(f) Find $\mu_{Y|1}$.

(g) Find $P(X > Y)$.

(h) Find $P(2 < Y < 3)$.

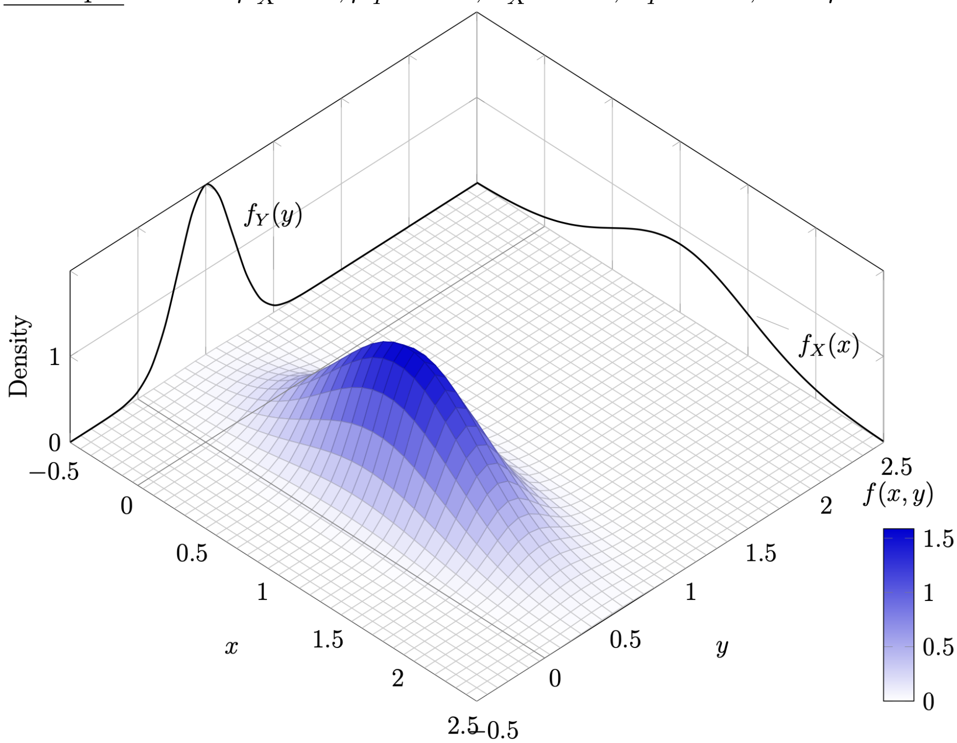
The Bivariate Normal Distribution (Section 4.5)

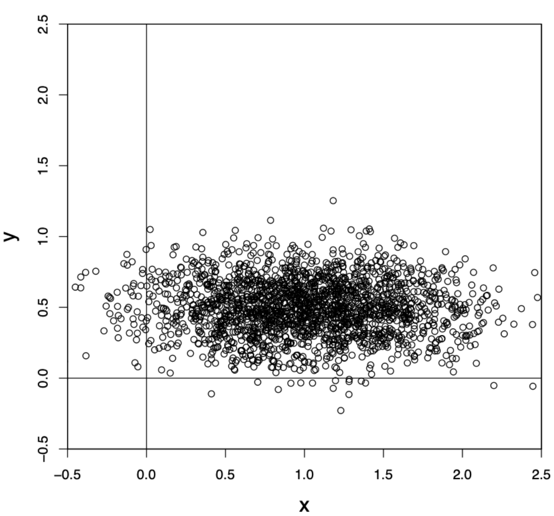
The only "special" bivariate distribution we will discuss in this class is the **bivariate normal distribution**. Two continuous variables X and Y are bivariate normal if their pdf is

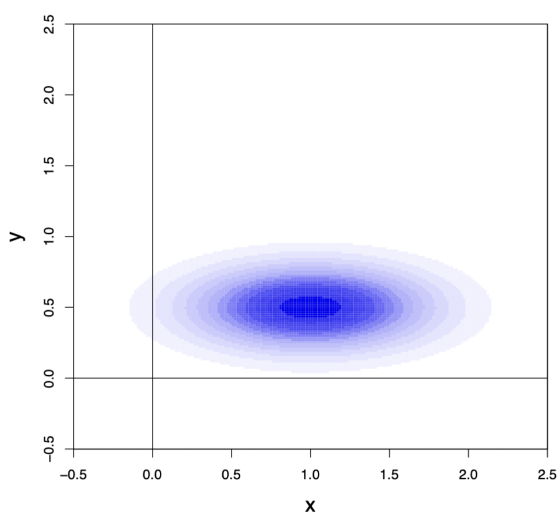
$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}$$

for $-\infty < x, y < \infty$ where ρ is the correlation coefficient between X and Y. We write $X, Y \sim BN(\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2, \rho)$.

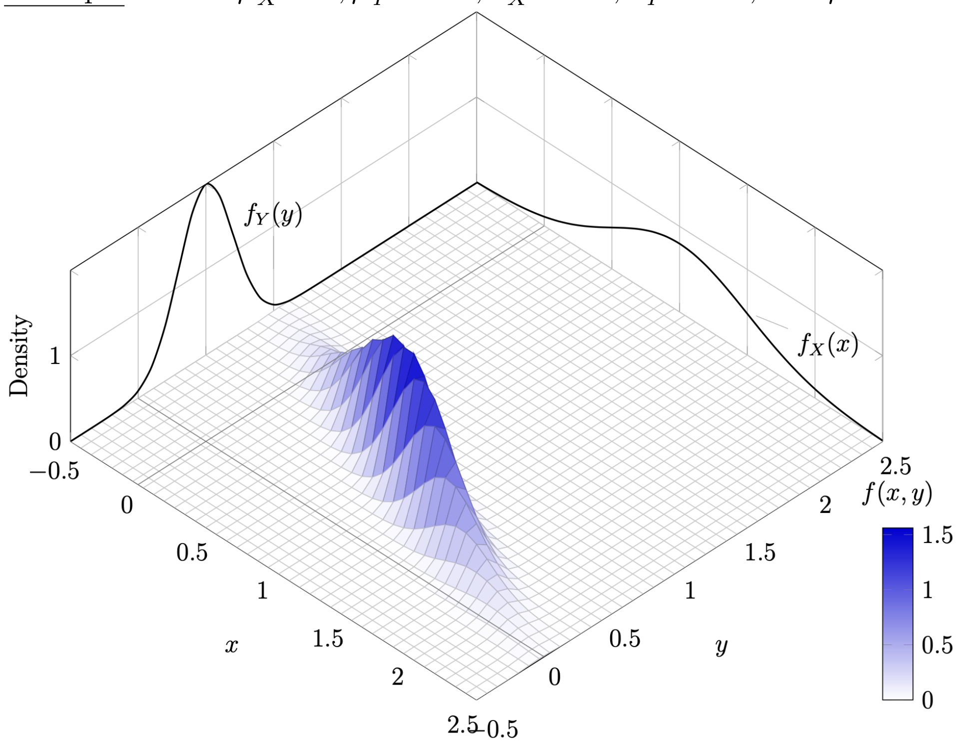
Example: When $\mu_X = 1, \mu_Y = 0.5, \sigma_X = 0.5, \sigma_Y = 0.2$, and $\rho = 0$.

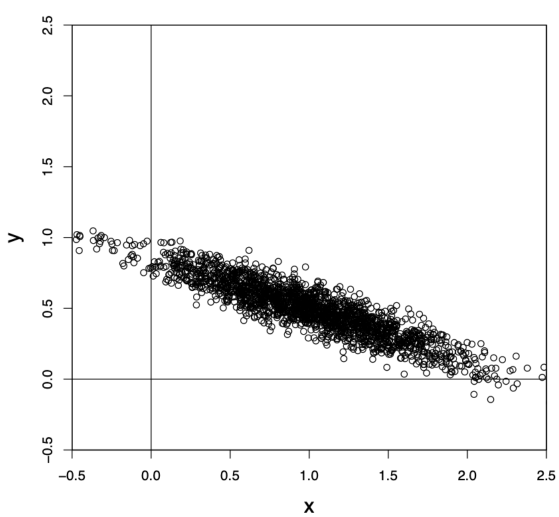


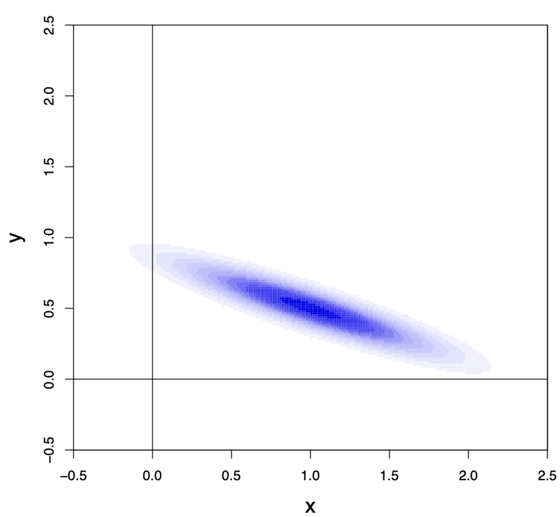




Example: When $\mu_X = 1, \mu_Y = 0.5, \sigma_X = 0.5, \sigma_Y = 0.2$, and $\rho = -0.9$.







As we can see from the images above, the **marginal distributions** for X and Y are normal as well! In particular,

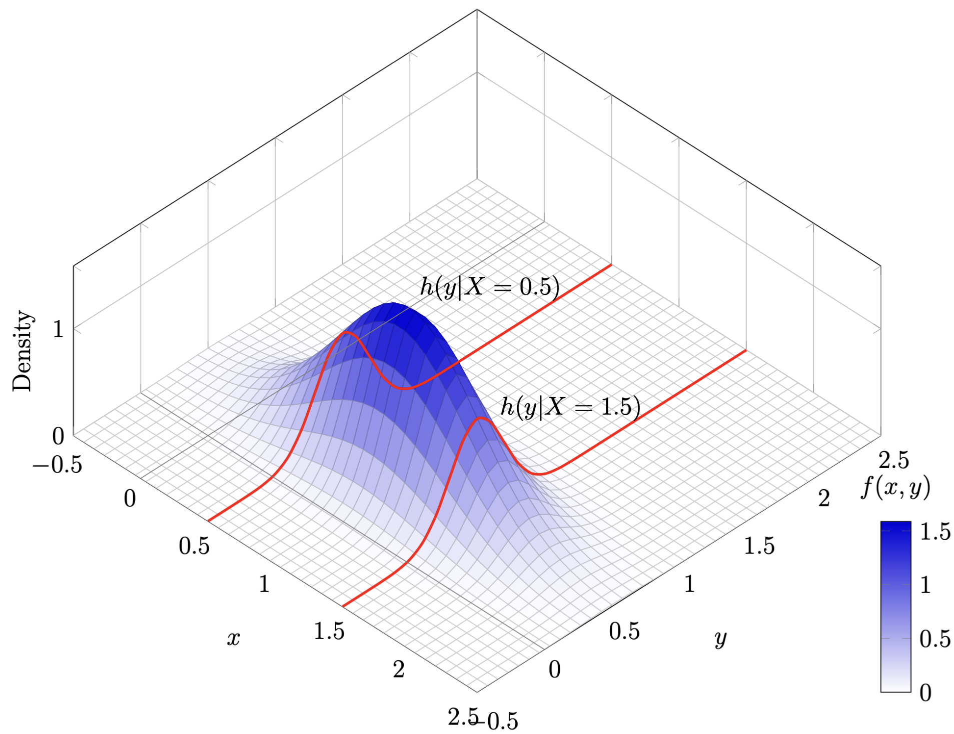
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \text{ for } -\infty < x < \infty$$

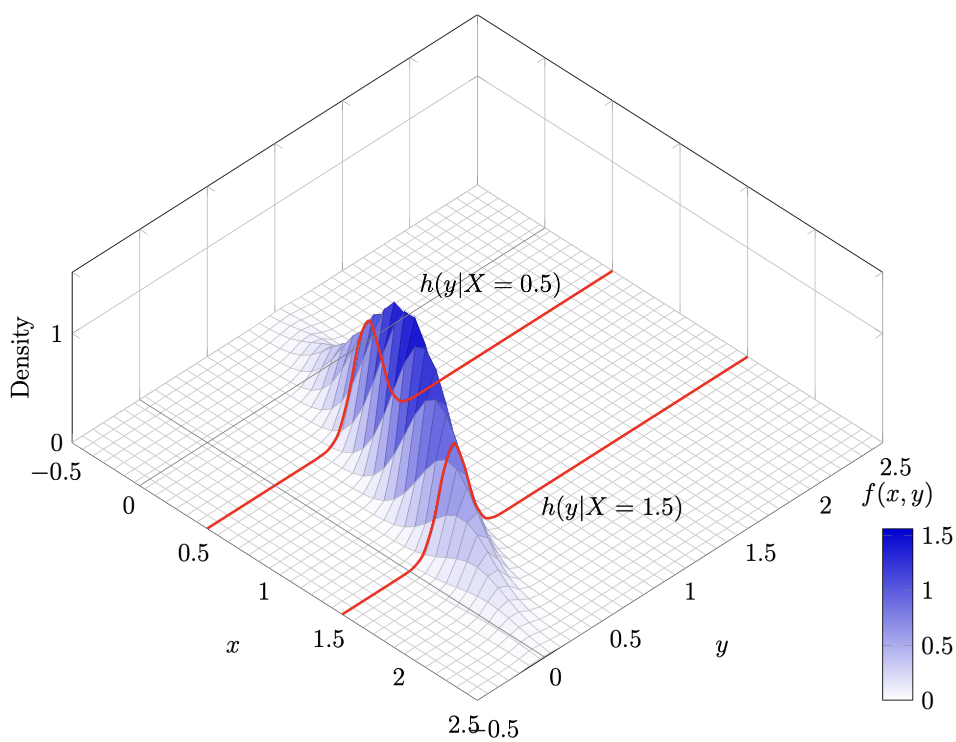
and

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}} \text{ for } -\infty < y < \infty$$

These depend only on the respective means and variances for the variable, not on ρ.

The **conditional distributions** of X given $Y = y$ and of Y given $X = x$ are also normal, but these conditional distributions are a bit more complicated. We are essentially taking a "slice" of the joint distribution, as the images below show.





The **conditional distributions** are:

$$g(x|y) = \frac{1}{\sqrt{2\pi\sigma_X^2(1-\rho^2)}} \exp\left\{-\frac{[x - \mu_X - \rho(\sigma_X/\sigma_Y)(y-\mu_Y)]^2}{2\sigma_X^2(1-\rho^2)}\right\}$$

and

$$h(y|x) = \frac{1}{\sqrt{2\pi\sigma_Y^2(1-\rho^2)}} \exp\left\{-\frac{[y - \mu_Y - \rho(\sigma_Y/\sigma_X)(x-\mu_X)]^2}{2\sigma_Y^2(1-\rho^2)}\right\}$$

Notice the following:

1. $g(x|y) \sim N(\mu_X + \rho(\sigma_X/\sigma_Y)(y-\mu_Y), \sigma_X^2(1-\rho^2))$
2. $h(y|x) \sim N(\mu_Y + \rho(\sigma_Y/\sigma_X)(x-\mu_X), \sigma_Y^2(1-\rho^2))$

That means the conditional means and variances are:

- $\mu_{X|y} = \mu_X + \rho\dfrac{\sigma_X}{\sigma_Y}(y-\mu_Y)$
- $\sigma_{X|y}^2 = \sigma_X^2(1-\rho^2)$
- $\mu_{Y|x} = \mu_Y + \rho\dfrac{\sigma_Y}{\sigma_X}(x-\mu_X)$
- $\sigma_{Y|x}^2 = \sigma_Y^2(1-\rho^2)$

Note on independence: We stated earlier that if two variables are independent, then $\text{Cov}(X,Y) = 0$ and thus $\rho = 0$. But, in general, if $\rho = 0$, that does not mean the variables are independent. However, if X and Y are bivariate normal, then knowing $\rho = 0$ does guarantee the variables are independent.

Theorem 4.1-1: If X and Y have a bivariate normal distribution with correlation coefficient ρ, then X and Y are independent if and only if $\rho = 0$.

Example: Let X and Y have a bivariate normal distribution with $\mu_X = 10$, $\sigma_X = 3$, $\mu_Y = 20$, $\sigma_Y = 5$, and $\rho = 0.65$. Find $P(9 < X < 14 \text{ and } 18.8 < Y < 20.5)$ using integrals (and WolframAlpha).

Example: For the case a few pages ago when $\mu_X = 1, \mu_Y = 0.5, \sigma_X = 0.5, \sigma_Y = 0.2$, and $\rho = -0.9$, first determine $P(0.4 < Y < 0.5)$, then find the exact conditional distributions of Y given $X = 0.5$ and $X = 1.5$, and finally use the conditional distributions to find $P(0.4 < Y < 0.5|X = 0.5)$ and $P(0.4 < Y < 0.5|X = 1.5)$.

Chapter 5: Distributions of Functions of Random Variables

In Chapter 3, it was noted that if $Z \sim N(0,1)$ then $Z^2 \sim \chi^2(1)$. In that example, we found the distribution of a function of our random variable, Z. In this chapter, we expand on that idea in both the univariate and bivariate cases for both discrete and continuous variables.

Functions of One Random Variable (Section 5.1)

We begin in the **univariate and discrete case**. Let X be our orignal variable with known pmf $f(x) = P(X = x)$ with support $x = c_1, c_2, \ldots$ and let $Y = u(X)$: the new variable we want the distribution for. Then the pmf of Y is

$$g(y) = P(Y = y) = P(u(X) = y) = \sum_{x:u(x)=y} f(x) \text{ for } y \in S_y$$

If $u(X)$ is a one-to-one function, then this is rather simple.

1. Solve $Y = u(X)$ for X.
2. Replace all x with $u^{-1}(y)$ in the pmf of X.
3. Apply the function u to the support of X to get the support of Y.

Example: Let X have a Poisson distribution with $\lambda = 1$. Find the pmf for the variable $Y = e^X$. What about for $W = X^2$?

If $u(X)$ is not one-to-one, we need to be more careful:

Example: Let X have pmf $f(x) = \dfrac{4-x}{21}$ for $x = -2, -1, 0, 1, 2, 3, 4$. Find the pmf for $Y = X^2$.

The more interesting (and more complicated) case is the **univariate and continuous case**. There are three primary ways to find the pdf of a variable transformation:

1. The cdf technique.
2. The change-of-variable technique.
3. The moment-generating function technique (discussed in Section 5.4).

The cdf technique

This is the technique we used back in Chapter 3 to show $Z^2 \sim \chi^2(1)$. The steps are listed below for finding the pdf of $Y = u(X)$ when the pdf of X, $f(x)$, is known:

1. Begin by finding the cdf of X: $F(x) = P(X \leq x) = \displaystyle\int_{-\infty}^{x} f(t)dt$.
2. Solve $y = u(x)$ for x to obtain $x = u^{-1}(y) = v(y)$.
3. Plug in $v(y)$ for x in the cdf of X to obtain the cdf of Y: $G(y) = F[v(y)]$.
4. Take the derivative to obtain the pdf of Y: $g(y) = \dfrac{d}{dy}G(y)$.
5. Write the support of Y by applying u to the support of X.

Note: We still have to be careful with functions that are not one-to-one (not a focus for us).

Example: Let $X \sim U(0, 2)$. Find the pdf of $Y = X^3$.

Example: Let the pdf of X be $f(x) = \dfrac{2-x}{2}$ for $0 < x < 2$. Find the pdf of $R = e^{-X}$.

The change-of-variable technique

The cdf technique is great, but only works when taking an integral of a pdf is relatively simple to do. The idea of the change-of-variable technique is to apply the Second Fundamental Theorem of Calculus so we don't have to take an integral just to take a derivative again.

Background: We ultimately want to find
$g(y) = \frac{d}{dy}G(y) = \frac{d}{dy}\left[\int_{-\infty}^{x} f(t)dt\right] = \frac{d}{dy}\left[\int_{-\infty}^{v(y)} f(t)dt\right].$

Apply the chain rule:
$\frac{d}{dy}\left[\int_{-\infty}^{v(y)} f(t)dt\right] = \frac{d}{dv}\left[\int_{-\infty}^{v} f(t)dt\right]\frac{dv}{dy} = f(v)\frac{dv}{dy}.$

So, we can find $g(y)$ by letting

$$g(y) = f\big(v(y)\big)\frac{d}{dy}\left[v(y)\right]$$

where $x = v(y)$. Since a pdf must be positive, we actually need to make sure that derivative is positive, so the pdf of y is

$$\boxed{g(y) = f\big(v(y)\big)\left|\frac{d}{dy}\left[v(y)\right]\right|}$$

Steps:

1. Solve $y = u(x)$ for x to obtain $x = u^{-1}(y) = v(y)$.
2. Take the derivative of $v(y)$:
3. Plug $v(y)$ in for x in the pdf of X and multiply by $|v'(y)|$ to obtain the pdf of Y: $g(y) = f(v(y))|v'(y)|$.
4. Write the support of Y by applying u to the support of X.

Note: We *still* still have to be careful with functions that are not one-to-one.

Example: Let $X \sim U(0, 2)$. Find the pdf of $Y = X^3$.

Example: Let X have a **Cauchy** distribution with $f(x) = \dfrac{1}{\pi(1+x^2)}$ on $-\infty < x < \infty$. Let $Y = \arctan(X)$. Find the pdf of Y using both the cdf and change-of-variable techniques.

Transformations of Two Random Variables (Section 5.2)

Now we will consider the **bivariate and continuous case**. We will only discuss the change-of-variable method for these transformations since working with cdfs is not straightforward with two variables. In addition, we will only consider the case in this class where there is only one inverse, which is analogous to the univariate case when the transformation function was one-to-one.

In the univariate case, we began with $f(x)$ and $Y = u(X)$ and then $X = u^{-1}(Y) = v(Y)$. Then $g(y) = f\big(v(y)\big)|v'(y)|$ and the support of X was transformed through the function u. We need to extend this idea to two variables.

First of all, we will have two original variables, X_1 and X_2 that have a joint distribution $f(x_1, x_2)$. Then we will have two new variables $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$. So, solving for X_1 and X_2 gives $X_1 = v_1(Y_1, Y_2)$ and $X_2 = v_2(Y_1, Y_2)$. But to extend the idea of $|v'(y)|$ we need a two-dimensional derivative. This is known as the **Jacobian**.

$$J = \det \begin{bmatrix} \dfrac{\partial x_1}{\partial y_1} & \dfrac{\partial x_1}{\partial y_2} \\ \\ \dfrac{\partial x_2}{\partial y_1} & \dfrac{\partial x_2}{\partial y_2} \end{bmatrix} = \det \begin{bmatrix} \dfrac{\partial v_1(y_1, y_2)}{\partial y_1} & \dfrac{\partial v_1(y_1, y_2)}{\partial y_2} \\ \\ \dfrac{\partial v_2(y_1, y_2)}{\partial y_1} & \dfrac{\partial v_2(y_1, y_2)}{\partial y_2} \end{bmatrix} = \begin{vmatrix} \dfrac{\partial x_1}{\partial y_1} & \dfrac{\partial x_1}{\partial y_2} \\ \\ \dfrac{\partial x_2}{\partial y_1} & \dfrac{\partial x_2}{\partial y_2} \end{vmatrix} = \frac{\partial x_1}{\partial y_1}\frac{\partial x_2}{\partial y_2} - \frac{\partial x_1}{\partial y_2}\frac{\partial x_2}{\partial y_1}$$

Where each $\dfrac{\partial x_i}{\partial y_j}$ are **partial derivatives**, another Calculus III topic. Then

$$\boxed{g(y_1, y_2) = f\Big(v_1(y_1, y_2), v_2(y_1, y_2)\Big) \cdot |J|}$$

The support of X_1 and X_2 will also have to be changed and we must be very careful when changing it!

Example: Let X_1 and X_2 have the joint pdf $f(x_1, x_2) = 6x_1^2 x_2$ on $0 \le x_1, x_2 \le 1$. Find the pdf of $Y_1 = X_1 - X_2$ and $Y_2 = X_2$.

For bounds, we want to think about every curve that is bounding the current region and change each one individually according to the transformation. That will give you the new curves.

Example: Suppose $X \sim \text{Gam}(\alpha, \theta)$ and $Y \sim \text{Gam}(\beta, \theta)$ and X and Y are independent. If $U = \dfrac{X}{X+Y}$ and $V = X + Y$, find the joint pdf of U and V and then find the marginal pdf of U.

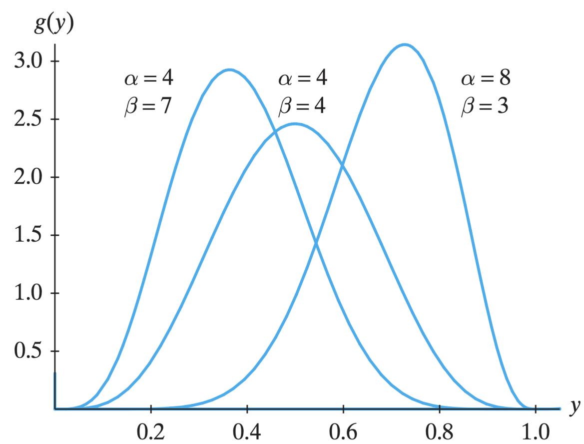
The Beta Distribution

The marginal pdf of U in the last example is also a special one. It is called a **beta distribution**. Specifically, a variable X has a beta distribution if its pdf is

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} \text{ for } 0 < x < 1$$

We write $X \sim \text{Beta}(\alpha, \beta)$. The beta distribution gives lots of flexibility for modeling continuous random variables that are restricted to be between 0 and 1.

Let's find the mean of the beta distribution:



Find note that since $\displaystyle\int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} = 1$,

it is the case that $\displaystyle\int_0^1 x^{\alpha-1}(1-x)^{\beta-1} = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

Example: If $X \sim \text{Beta}(5, 2)$, find $P(0.25 < X < 0.75)$.

The F Distribution

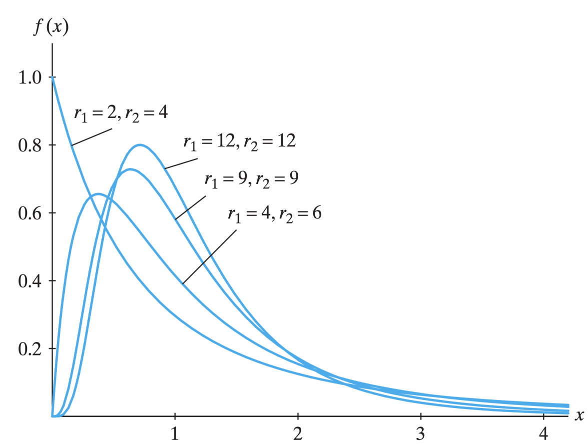
Another very important probability distribution in inferential statistics is known as the F distribution.

Theorem 5.2-1: If $U \sim \chi^2(r_1)$ and $V \sim \chi^2(r_2)$, then $X = \dfrac{U/r_1}{V/r_2}$ follows an F distribution. That is, and F distribution is the ratio of χ^2 random variables divided by their degrees of freedom. The pdf of an F is

$$f(x) = \frac{\Gamma\left(\frac{r_1+r_2}{2}\right)(r_1/r_2)^{r_1/2}x^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)[1+(r_1/r_2)x]^{(r_1+r_2)/2}} \text{ for } x > 0$$

We write $X \sim F(r_1, r_2)$. So every F distribution has two different degrees of freedom: one for the numerator and one for the denominator. We will simply use our calculators and Excel to find probabilities regarding F distributions. It is worth knowing that $E[X] = \dfrac{r_2}{r_2 - 2}$.

Calculator and Excel Tips



Example: For $X \sim F(5, 20)$, find $P(X > 1)$ and $F_{0.05}(5, 20)$.

Several Independent Random Variables (Section 5.3)

In this section, we will focus on when we have multiple independent random variables. We have discussed some important facts about two independent random variables and now we will extend those to the case when we have many. Remember that if events are independent, then $P(A \cap B) = P(A)P(B)$. So, $P(X = x, Y = y) = P(X = x)P(Y = y) = f(x)g(y)$ in the discrete case. That is, $f(x, y) = f_X(x)g_Y(y)$. That is true for continuous variables as well.

If $X_1, \dots, X_n$ are independent random variables with respective pdfs (or pmfs) $f_1(x_1), \dots, f_n(x_n)$, then their joint pdf is

$$f(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} f_i(x_i) = f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n)$$

If all n variables have the same distribution, then $X_1, \dots, X_n$ is said to be a **random sample** of size n from that distribution and the random variables are said to be **independent and identically distrib**

The following properties hold for any independent variables (whether they are iid or not):

- $E[X_1 \cdot X_2 \cdot \dots \cdot X_n] = \prod_{i=1}^{n} E[X_i] = E[X_1] \cdot E[X_2] \cdot \dots \cdot E[X_n]$
- $E\big[(X_1 - \mu_1) \cdot (X_2 - \mu_2) \cdot \dots \cdot (X_n - \mu_n)\big] = 0$

Theorem 5.3-1: Say $X_1, X_2, \dots, X_n$ are n independent random variables and $Y = u_1(X_1)u_2(X_2) \cdot \dots \cdot u_n(X_n)$ (the product of functions of each variable), then $E[Y] = E\big[u_1(X_1)u_2(X_2) \cdot \dots \cdot u_n(X_n)\big] = E\big[u_1(X_1)\big]E\big[u_2(X_2)\big] \cdot \dots \cdot E\big[u_n(X_n)\big] = \prod_{i=1}^{n} E\big[u_i(X_i)\big]$ if each expected value exists.

Theorem 5.3-2: If $X_1, X_2, \dots, X_n$ are n random variables with respective means $\mu_1, \mu_2, \dots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, then the mean of $Y = \sum_{i=1}^{n} a_i X_i = a_1 X_1 + \dots + a_n X_n$, where $a_1, \dots, a_n$ are real constants, is

$$\mu_Y = \sum_{i=1}^{n} a_i \mu_i = a_1 \mu_1 + \dots + a_n \mu_n.$$

If $X_1, X_2, \dots, X_n$ are also independent, then

$$\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2 = a_1^2 \sigma_1^2 + \dots + a_n^2 \sigma_n^2.$$

If the variables are not independent, we can still find the variance of Y, but it is more complicated. We would have to take the covariance into account.

$$\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2 + 2 \sum_{j=2}^{n} \sum_{i=1}^{j-1} a_i a_j \mathrm{Cov}(X_i, X_j).$$

Example: Suppose $X_1, X_2 \overset{iid}{\sim} \text{Exp}(3)$ and $X_3 \sim U(1,3)$ and all variables are independent. Find each of the following:

(a) $f(x_1, x_2, x_3)$

(b) $P(1 < X_1 < 2, X_2 < 3, 1.5 < X_3 < 2.5)$

(c) The mean of $Y = X_2^2 X_3 (2X_1 - 4)$

(d) The mean of $Z = 2X_1 - 4X_2 + X_3$

(e) The variance of Z

The Moment-Generating Function Technique (Section 5.4)

We have discussed the cdf and change-of-variable techniques for finding the distribution of a transformation of variables. We now turn to the last method: the **moment-generating function technique**.

Theorem 5.4-1: If $X_1, X_2, \dots, X_n$ are independent random variables with respective moment-generating functions $M_{X_i}(t)$ for $i = 1, \dots, n$, then the moment-generating function of $Y = \sum_{i=1}^{n} a_i X_i$ is

$$M_Y(t) = \prod_{i=1}^{n} M_{X_i}(a_i t).$$

Proof:

If $X_1, \dots, X_n$ is a random sample from a distribution with mgf $M(t)$, then was is the mgf of $Y = \sum_{i=1}^{n} X_i$? How about of $\overline{X}$?

$\boxed{\textbf{The mgf technique}}$ The way we can use mgfs to find the distribution of new independent variables is to apply Theorem 5.4-1 and determine if the mgf of the new variable matches one of the mgfs that we know.

Example: If $X_1 \sim \text{Exp}(1)$ and $X_2 \sim \text{Gam}(5, 2)$ are independent, then what is the distribution of $Y = X_1 + \frac{1}{2}X_2$?

Example: Use the fact that the mgf for a χ^2 distribution with r degrees of freedom is $M(t) = (1-2t)^{-r/2}$ to find the distribution of $Y = \sum_{i=1}^{n} X_i$ if $X_i \overset{indep}{\sim} \chi^2(r_i)$ for $i = 1, \ldots, n$.

Theorem 5.4-2: Let $X_1, \ldots, X_n$ be independent chi-square random variables with $r_1, \ldots, r_n$ degrees of freedom, respectively. Then $Y = X_1 + X_2 + \cdots + X_n$ is

Corollary 5.4-2: Let $Z_1, \ldots, Z_n$ have independent standard normal distributions, $N(0,1)$. If these random variables are independent, then $W = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ has a distribution that is

Corollary 5.4-3: Let $X_1, \ldots, X_n$ have independent normal distributions, $N(\mu_i, \sigma_i^2)$ for $i = 1, \ldots, n$, respectively. Then the distribution of $W = \sum_{i=1}^{n} \frac{(X_i - \mu_i)^2}{\sigma_i^2}$ is $\chi^2(n)$.

Proof:

This technique works for discrete variables too.

Example: If X_1, X_2, and X_3 are random samples from a Poisson with $\lambda = 4$, what is the distribution of $Y = X_1 + X_2 + X_3$? How about of $W = X_1 - X_2 + 2X_3$?

Random Functions Associated with Normal Distributions (Section 5.5)

In inferential statistics, normal distributions play a huge role and in many applications, we need to assume (hopefully with verification) that our sample comes from a normal distribution. Let's discuss some important characteristics related to this topic.

Theorem 5.5-1: If $X_1, X_2, \dots, X_n$ are n independent normal variables with means $\mu_1, \mu_2, \dots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, respectively, then the linear function $Y = \sum_{i=1}^{n} c_i X_i$ has the normal distribution $N\left(\sum_{i=1}^{n} c_i \mu_i, \sum_{i=1}^{n} c_i^2 \sigma_i^2\right)$.

Proof:

Theorem 5.5-2: Let $X_1, X_2, \dots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu, \sigma)^2$. That is, $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Then the sample mean $\overline{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$ and the sample variance $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ are independent and $\dfrac{(n-1)S^2}{\sigma^2} = \dfrac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{\sigma^2}$ is $\chi^2(n-1)$.

The proof for this can be found in the textbook on page 201-202.

Example: The weight (in grams) of a certain type of construction nail that is most often used in making roofs is normally distributed with a mean of 9.4 grams and a standard deviation of 0.5 grams. Suppose you have a random sample of 12 such nails.

(a) What is the probability that the total weight of the nails in your sample is less than 110 grams?

(b) What is the probability that the sample standard deviation of the weights of your nails is between 0.4 and 0.55 grams?

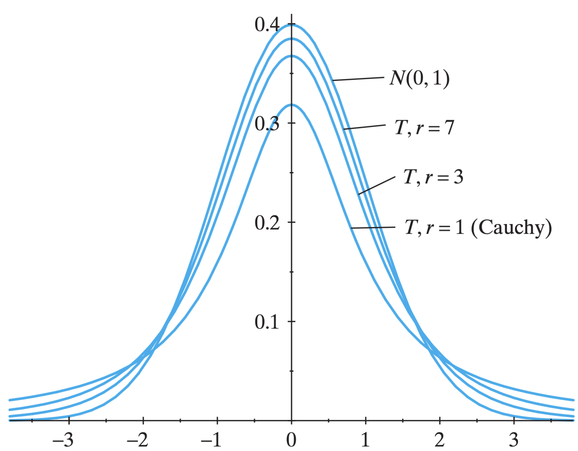
Student's t Distribution

The last named distribution we will consider in this class is called Student's t distribution or just the **distribution**. This is another one that is used extremely often in statistical inference (along with the normal, χ^2, and F distributions).

Theorem 5.5-3: Let $T = \dfrac{Z}{\sqrt{U/r}}$ where Z is a standard normal variable, U is a random variable that is $\chi^2(r)$, and Z and U are independent. Then T has a t distribution with r degrees of freedom and the pdf is

$$f(t) = \frac{\Gamma((r+1)/2)}{\sqrt{\pi r}\Gamma(r/2)} \frac{1}{(1+t^2/r)^{(r+1)/2}} \text{ for } -\infty < t < \infty$$

We write $T \sim t(r)$. The proof for this can be found in the textbook on page 203-204. Important facts about the t:



- They are symmetric about zero and bell-shaped (like the standard normal distribution).
- The t-distribution has wider tails than the normal distribution.
- The degrees of freedom completely specify which t-distribution to use.
- As the degrees of freedom increase, the t gets closer to the standard normal distribution.
- There is no mgf for the t.
- $E[T] = 0$ (when $r > 1$) and $\text{Var}(T) = r/(r-2)$ (when $r > 2$).

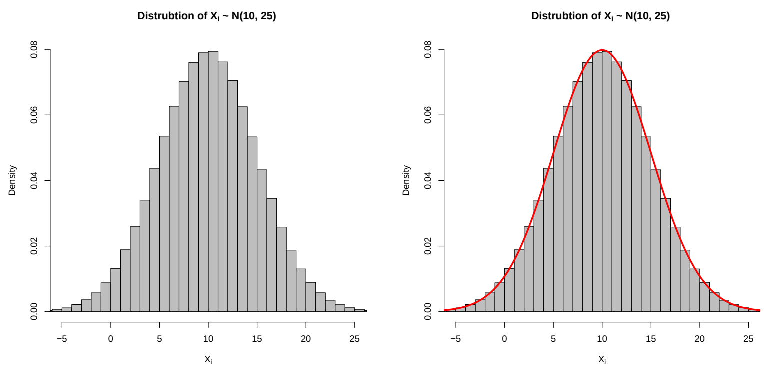
Calculator and Excel Tips:

Example: Find $P(0.5 < T < 1.5)$ and $t_{0.01}$ when $r = 10$.

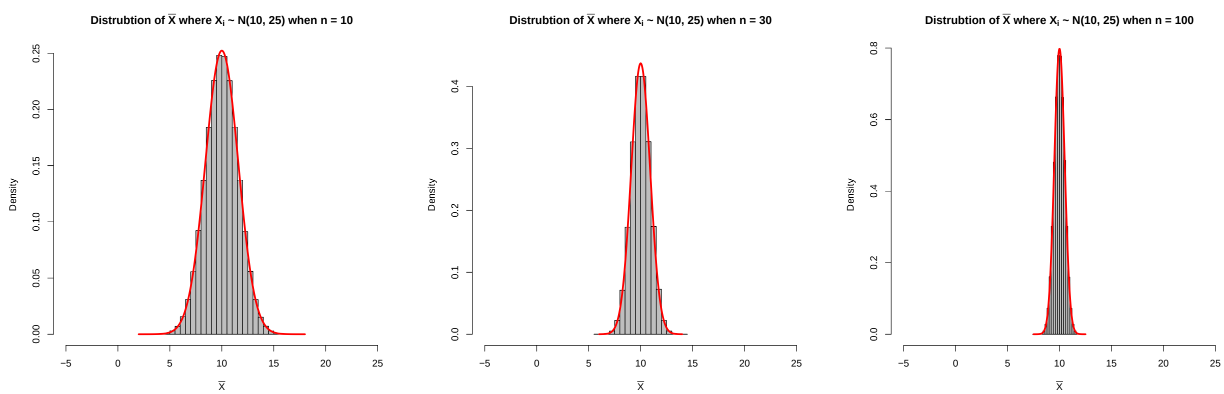
The Central Limit Theorem (Section 5.6)

Remember that the sample mean is $\overline{X} = \frac{X_1+X_2+\cdots+X_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Let's suppose we have random variables each independently drawn from a normal distribution with mean 10 and variance 25 (so standard deviation is 5). We can say $X_i \sim N(10, 25)$. If we generate samples of this random variable and plot a histogram, we get



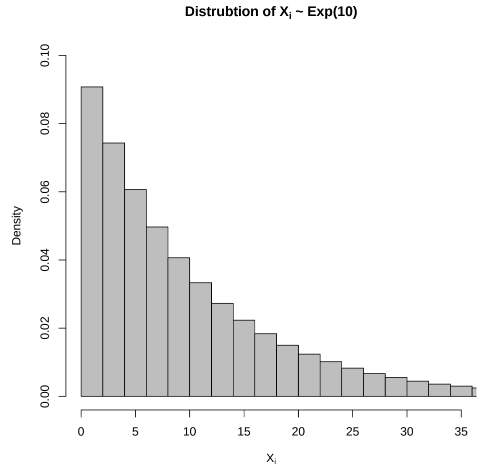
Now, let's take samples of size n for different n values: 10, 30, and 100. If we take many samples of size n and find the sample means for each of those samples, then plot the histogram of the sample means we get the following plots.



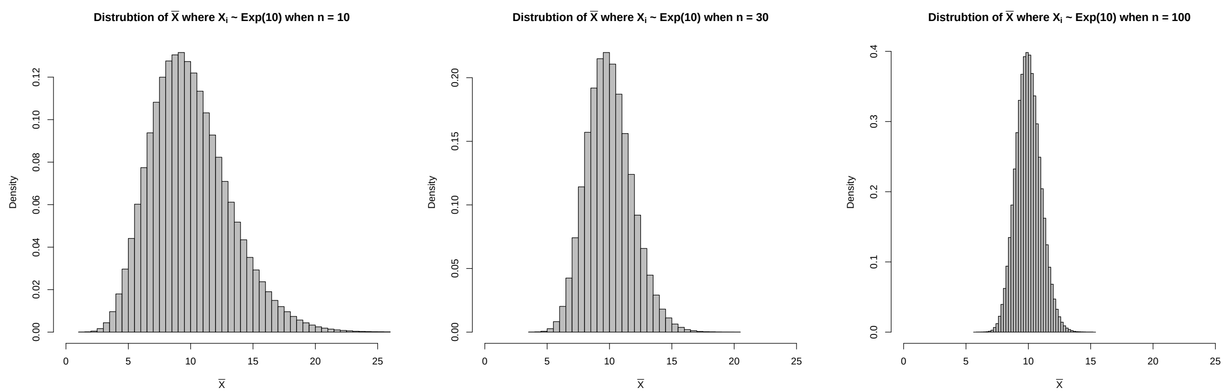
What is happening?

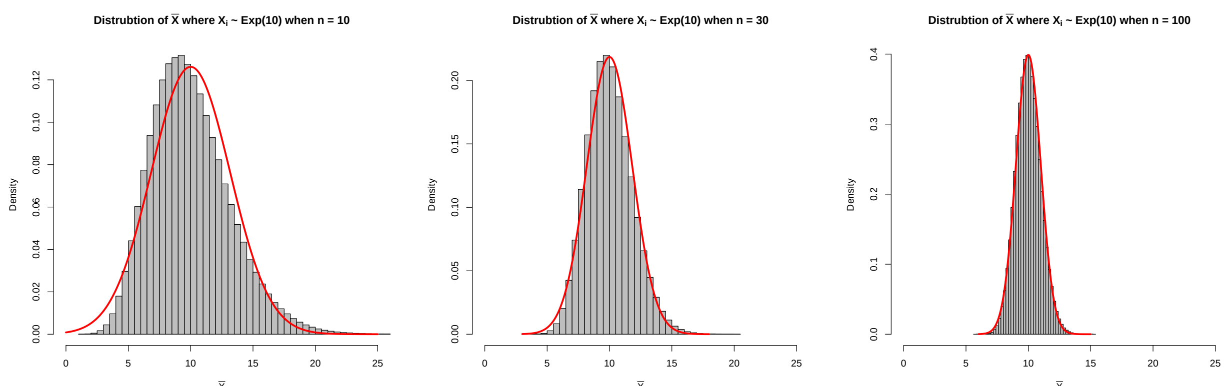
According to Theorem 5.5-1, what should the distribution of $\overline{X}$ be?

What if X_i is not normally distributed? Suppose $X_i \sim \text{Exp}(\theta)$. What are some properties of X_i?



Below are some distributions of $\overline{X}$ when $X_i \sim \text{Exp}(10)$.





Now what is happening?

Theorem 5.6-1: If $\overline{X}$ is the mean of a random sample $X_1, \ldots, X_n$ of size n from *any* distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

is $N(0,1)$ in the limit as $n \to \infty$. Or, put another way, $\overline{X} \sim N(\mu, \sigma^2/n)$ as $n \to \infty$.

But what if n is not infinite?

Example: Suppose that from past experience a professor knows that the test score of a student taking her final examination is a random variable with mean 70 and standard deviation 8. What is the approximate probability that the class average is above 73 if there are 60 students in the class?

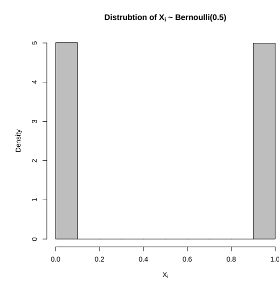
Example: The tips given to a server at a restaurant for any table come from a skewed distribution with a mean of \$12 and a standard deviation of \$4. On a certain long shift, the waiter serves 38 tables. (a) What is the probability that the server will receive a combined total of more than \$500 in tips that night? (b) What total tip amount would be needed for this night to be in the top 1% in terms of tips collected?

Example: Shear strength measurements for spot welds have been found to have standard deviation 10 pounds per square inch (psi). If 100 test welds are to be measured, what is the approximate probability that the sample mean will be within 1 psi of the true population mean?

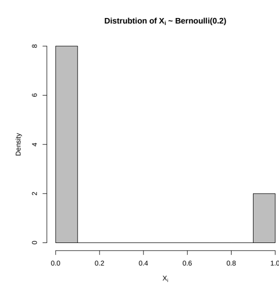
Approximations for Discrete Distributions (Section 5.7)

CLT for Proportions

As stated above, the CLT holds for any distribution. Consider Bernoulli trials. Let X_i be i.i.d. Bernoulli trials with $P(X_i = 1) = p$. Then, in this case,



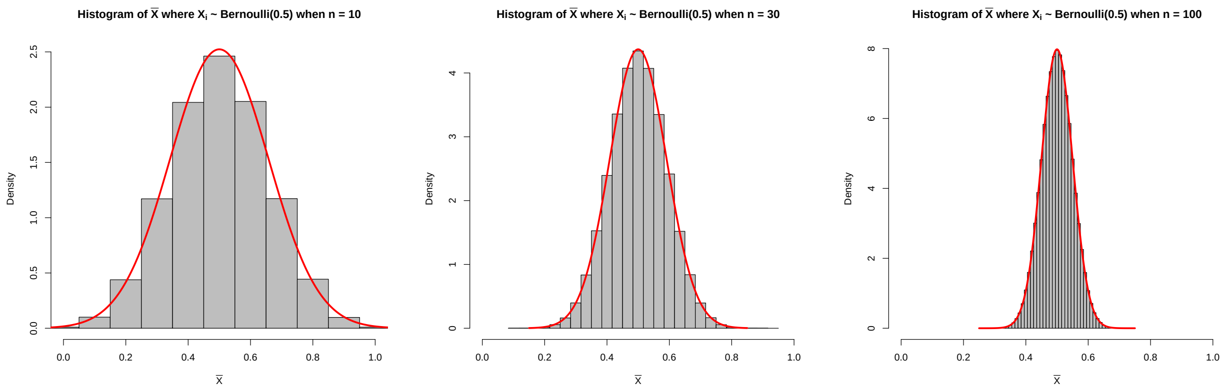
$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} =$$



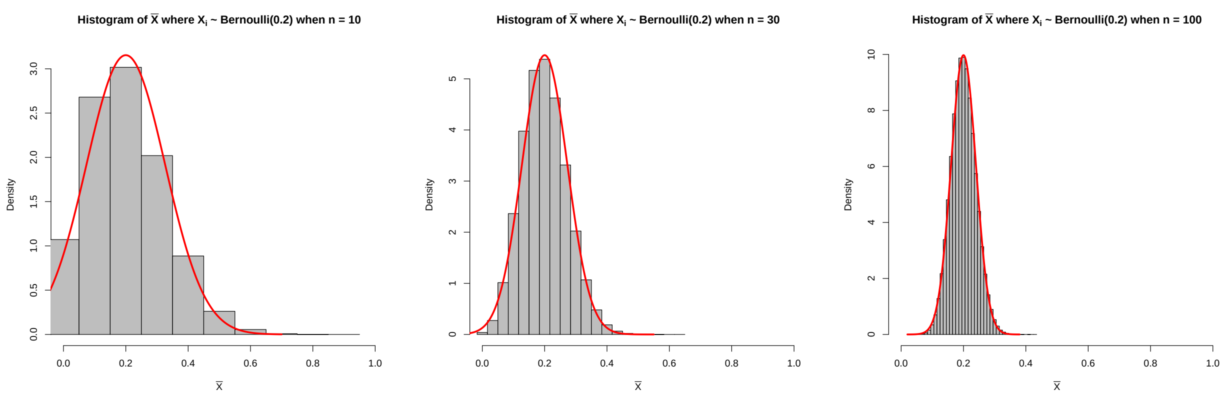
For a sample proportion, $\boxed{\hat{p} \overset{\bullet}{\sim} N\left(p, \frac{p(1-p)}{n}\right)}$.

How large should n be now?

Consider when $p = 0.5$.



Now when $p = 0.2$.

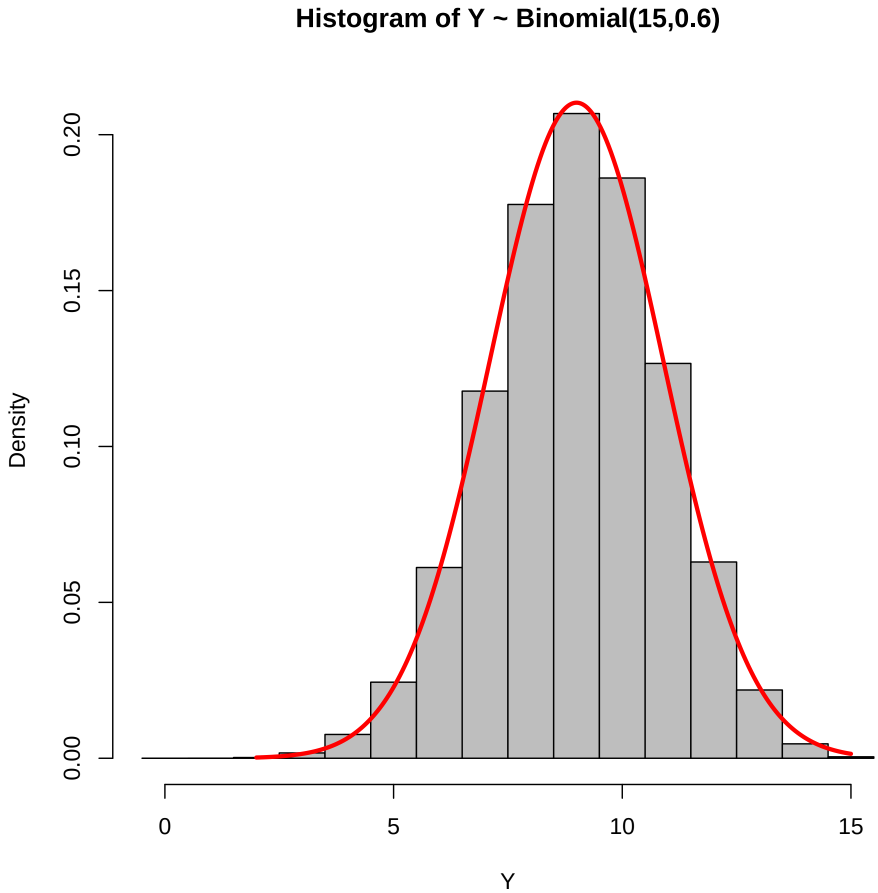


Rule of thumb:

Example: It is known that 83% of the population enjoys the movie *High School Musical*. If you take a random sample of 72 people from that population, what is the approximate probability that fewer than 80% of those in the sample will enjoy *High School Musical*?

Now extending this to the binomial distribution:

Example: Suppose Y has a binomial distribution with $n = 15$ and $p = 0.6$. Find the exact and approximate probabilities that $Y \leq 5$ and $Y = 12$.


Histogram of Y ~ Binomial(15,0.6)
Density
0.00
0.05
0.10
0.15
0.20
0
5
10
15
Y

Continuity Correction

If X is a discrete random variable, then to approximate probabilities related to X with a normal distribution, the following changes must be made:

Discrete	Normal
$P(X \leq a) \approx$	
$P(X \geq a) \approx$	
$P(X = a) \approx$	

Example: Suppose the number of patients in a dermatology office follows a Poisson distribution with a mean of 8 per day. Estimate the probability more than 35 but at most 40 patients will visit the office in a 5-day work week.

Example: Suppose the annual snowfall in Cedar City is normally distributed with mean $\mu = 44$ inches and standard deviation $\sigma = 4$ inches. What is the approximate probability that more than 10 of the next 80 years will have a snowfall over 49 inches? What is the exact probability?

Chapter 5: Important Theorems

__Theorem 5.2-1__: If $U \sim \chi^2(r_1)$ and $V \sim \chi^2(r_2)$, then $X = \dfrac{U/r_1}{V/r_2}$ follows an F distribution. That is, and F distribution is the ratio of χ^2 random variables divided by their degrees of freedom. The pdf of an F is

$$f(x) = \frac{\Gamma\left(\frac{r_1+r_2}{2}\right)(r_1/r_2)^{r_1/2}x^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)[1+(r_1/r_2)x]^{(r_1+r_2)/2}} \text{ for } x > 0$$

We write $X \sim F(r_1, r_2)$. So every F distribution has two different degrees of freedom: one for the <u>numerator</u> and one for the <u>denominator</u>. We will simply use our calculators and Excel to find probabilities regarding F distributions. It is worth knowing that $E[X] = \dfrac{r_2}{r_2 - 2}$.

__Theorem 5.3-1__: Say $X_1, X_2, \ldots, X_n$ are n independent random variables and $Y = u_1(X_1)u_2(X_2) \cdot \ldots \cdot u_n(X_n)$ (the product of functions of each variable), then $E[Y] = E\big[u_1(X_1)u_2(X_2)\cdot\ldots\cdot u_n(X_n)\big] = E\big[u_1(X_1)\big]E\big[u_2(X_2)\big]\cdot\ldots\cdot E\big[u_n(X_n)\big] = \prod_{i=1}^{n} E\big[u_i(X_i)\big]$ if each expected value exists.

__Theorem 5.3-2__: If $X_1, X_2, \ldots, X_n$ are n random variables with respective means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, then the mean of $Y = \sum_{i=1}^{n} a_i X_i = a_1 X_2 + \cdots + a_n X_n$, where $a_1, \ldots, a_n$ are real constants, is

$$\mu_Y = \sum_{i=1}^{n} a_i \mu_i = a_1\mu_1 + \cdots + a_n\mu_2.$$

If $X_1, X_2, \ldots, X_n$ are also independent, then

$$\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2 = a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2.$$

__Theorem 5.4-1__: If $X_1, X_2, \ldots, X_n$ are independent random variables with respective moment-generating functions $M_{X_i}(t)$ for $i = 1, \ldots, n$, then the moment-generating function of $Y = \sum_{i=1}^{n} a_i X_i$ is

$$M_Y(t) = \prod_{i=1}^{n} M_{X_i}(a_i t).$$

__Theorem 5.4-2__: Let $X_1, \ldots, X_n$ be independent chi-square random variables with $r_1, \ldots, r_n$ degrees of freedom, respectively. Then $Y = X_1 + X_2 + \cdots + X_n$ is $\chi^2(r_1 + r_2 + \cdots + r_n)$

__Theorem 5.5-1__: If $X_1, X_2, \ldots, X_n$ are n independent normal variables with means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, respectively, then the linear function $Y = \sum_{i=1}^{n} c_i X_i$ has the normal distribution $N\left(\sum_{i=1}^{n} c_i\mu_i, \sum_{i=1}^{n} c_i^2\sigma_i^2\right)$.

Theorem 5.5-2: Let $X_1, X_2, \ldots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu,\sigma)^2$. That is, $X_i \overset{\text{iid}}{\sim} N(\mu,\sigma^2)$. Then the sample mean $\overline{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$ and the sample variance $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ are independent and $\dfrac{(n-1)S^2}{\sigma^2} = \dfrac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2}$ is $\chi^2(n-1)$.

Theorem 5.5-3: Let $T = \dfrac{Z}{\sqrt{U/r}}$ where Z is a standard normal variable, U is a random variable that is $\chi^2(r)$, and Z and U are independent. Then T has a t distribution with r degrees of freedom and the pdf is

$$f(t) = \frac{\Gamma((r+1)/2)}{\sqrt{\pi r}\,\Gamma(r/2)} \frac{1}{(1+t^2/r)^{(r+1)/2}} \text{ for } -\infty < t < \infty$$

We write $T \sim t(r)$.

Theorem 5.6-1 (Central Limit Theorem): If $\overline{X}$ is the mean of a random sample $X_1, \ldots, X_n$ of size n from *any* distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$$

is $N(0,1)$ in the limit as $n \to \infty$. Or, put another way, $\overline{X} \sim N(\mu, \sigma^2/n)$ as $n \to \infty$.

Practically, if $n \geq 30$ then $\overline{X} \overset{\bullet}{\sim} N(\mu,\sigma^2/n)$, so $\mu_{\overline{X}} = \mu$ and $\sigma_{\overline{X}} = \sigma/\sqrt{n}$.

For a sample proportion, if $np \geq 5$ and $n(1-p) \geq 5$, then $\hat{p} \overset{\bullet}{\sim} N\left(p, \dfrac{p(1-p)}{n}\right)$. So $\mu_{\hat{p}} = p$ and $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

When approximating the probability of a discrete variable with a normal distribution,

Exact (Discrete)	Normal Approximation
$P(X \leq a)$	$P(X < a + 0.5)$
$P(X \geq a)$	$P(X > a - 0.5)$
$P(X = a)$	$P(a - 0.5 < X < a + 0.5)$
$P(a \leq X \leq b)$	$P(a - 0.5 < X < b + 0.5)$

Use the mean and standard deviation of the appropriate distribution (i.e. $\mu = np$ and $\sigma = \sqrt{np(1-p)}$ for binomial).

Chapter 6: Point Estimation

In Chapter 0, we discussed the overview of statistics and descriptive statistical techniques so we would have some background for why probability is important. Therefore, we have already covered most of the Section 6.1 and 6.2 topics. Here are some reminders and highlights of material we should know from these sections.

Descriptive Statistics and Exploratory Data Analysis (Sections 6.1-6.2)

Chapter 0 highlights:

- Variables describe the information we collect from the observational units in our study and come in two primary types: categorical (or qualitative) and quantitative.
- We use statistics that describe a sample to estimate parameters that describe a population. Since a statistic is a single number, it is known as a **point estimate** for the parameter.
- When describing a quantitative data description, it is important to mention center, spread, shape, number of peaks, unusual features, and to write in context.
- The most common measures of center are the mean and the median.
- The most common measures of spread are the standard deviation, the IQR, and the range.
- The mean and standard deviation both use all the values in the data set and are sensitive to outliers, so they are best used when the distribution is approximately symmetric.
- The median and IQR both use percentiles, so they are best used when the distribution is skewed.
- A common collection of statistics is the five-number summary, which includes
- Bar plots are good for displaying categorical variables.
- Histograms and box plots are good for displaying quantitative (numeric) variables.

This list is not exhaustive, so be sure to go back and review the material from Chapter 0.

Maximum Likelihood and Method of Moments Estimation (Section 6.4)

Up until this point when working for probability distributions, we have been studying distributions that have known parameters. For example, $X_i \sim N(10, 25)$ or $X_i \sim \text{Exp}(4)$ or $X_i \sim \text{Gam}(5, 1)$. But what if we don't know the parameters?

Example: Suppose we have a random sample of size 10 from a distribution that we know is exponential, so $f(x) = \dfrac{1}{\theta}e^{-x/\theta}$, but we are not sure what θ is. How can we estimate θ? Here is the sample: 0.2886, 0.0017, 12.70, 12.02, 2.78, 3.23, 21.39, 7.55, 1.28, 2.44.

There are two primary ways to estimate a parameter of a distribution:

1. Maximum likelihood
2. Method of moments

Maximum likelihood estimation

The idea behind maximum likelihood estimation (MLE) is to use knowledge of the pdf (or pmf) of a distribution to find a "good" estimator for the parameter(s). We are trying to maximize something, so calculus will come in handy here! How do we find a maximum of a function?

Here are the steps for MLE:

1. Write down the pdf (or pmf) for the distribution your data came from.
2. Find the likelihood function by multiplying the pdfs (or pmfs) together.
3. Take the natural log of the joint pdf (or pmf).
4. Take the derivative with respect to the parameter you want to estimate.
5. Set the derivative equal to 0, solve for the parameter, and put a hat ($\hat{}$) over it.
6. Repeat steps 4 and 5 if there are more parameters that need estimating.

Note that the **likelihood function** is identical to the joint pdf (or pmf) except we think of the likelihood as a function of the parameter rather than a function of the x values.

For example, if the joint pdf is $f(x_1, x_2, \ldots, x_n; \theta)$ then the likelihood is $L(\theta; x_1, x_2, \ldots, x_n)$.

Why do we take the log of the pdf or pmf?

Example: Find the maximum likelihood estimator for θ if we know our random sample came from a distribution with pdf $f(x) = \dfrac{1}{\theta}e^{-x/\theta}$. The sample is 0.2886, 0.0017, 12.70, 12.02, 2.78, 3.23, 21.39, 7.55, 1.28, 2.44.

Theorem 6.4-1: If $\hat{\theta}$ is the maximum likelihood estimator of θ based on a random sample from the distribution with pdf or pmf $f(x;\theta)$, and g is a one-to-one function, then $g(\hat{\theta})$ is the maximum likelihood estimator of $g(\theta)$.

Example: In the previous example, what is the MLE for e^θ? How about of θ^2?

A reasonable question when finding an estimator for a parameter is "How good is it?" While there are many measures for the "goodness" of an estimate, one desirable property is for it to be **unbiased**.

Definition 6.4-1: If $E[u(X_1, X_2, \ldots, X_n)] = \theta$ for all $\theta \in \Omega$, then the statistic $u(X_1, X_2, \ldots, X_n)$ is called an **unbiased estimator** for θ. Otherwise, it is said to be **biased**.

Example: Is $\hat{\theta} = \overline{x}$ an unbiased estimator for θ for an exponential distribution?

Example: Find the MLEs for μ and σ^2 if $X_i \sim N(\mu, \sigma^2)$ for $i = 1, \ldots, n$. Are $\hat{\mu}$ and $\hat{\sigma}^2$ unbiased?

Method of moments estimation

The idea for the method of moments (MoM) is quite simple: just equate the theoretical first moment to the sample first moment. Next, if needed, equate the theoretical second moment to the sample second moment, etc. Recall the following:

Moment Number	Theoretical Moment	Sample Moment
1	$E[X] = \mu$	$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
2	$E[X^2] = \sigma^2 + \mu^2$	$\frac{1}{n}\sum_{i=1}^{n} x_i^2$
r	$E[X^r]$	$\frac{1}{n}\sum_{i=1}^{n} x_i^r$

The estimator for the MoM typically is denoted with a tilde over the parameter: $\tilde{\theta}$.

Example: Find the method of moments estimators of α and θ for a random sample of size n from a gamma distribution.

Example: Find the ML and MoM estimators for a random sample of size n from the pdf $f(x;\alpha) = \alpha x^{\alpha-1}$ for $0 < x < 1$.

Chapter 7: Interval Estimation

Now that we have learned about probability and descriptive statistics, we can return to **inferential statistics**. Recall the notation we have established previously:

Symbol	Description
n	Sample size
x or x_i	Observation
$\overline{x}$	Sample mean
μ	Population mean
s	Sample standard deviation
σ	Population standard deviation
s^2	Sample variance
σ^2	Population variance
$\widetilde{m} = \tilde{q}_2 = \tilde{\pi}_{0.50}$	Median
$\tilde{q}_1 = \tilde{\pi}_{0.25}$	Lower quartile
$\tilde{q}_3 = \tilde{\pi}_{0.75}$	Upper quartile
IQR	Interquartile range
df	Degrees of freedom
z	Standard score or z-score
x	Number of "successes"
$\widehat{p}$	Sample proportion
p	Population proportion or probability of success
q	$1 - p$ or probability of failure

Remember also that capital letters are typically variables and lowercase letters are values.

Inferential Statistics Overview Review

Throughout these notes, it will help to reference the "Inference Summary Sheet" handout on Canvas.

A **point estimate** is a single value estimate for a population parameter.

- A point estimate for the population mean μ is

- A point estimate for the population standard deviation σ is

- A point estimate for the population proportion p is

One issue we face is that our estimates are nearly always wrong! They are wrong not because we make mistakes (though this is possible), but because they are based on limited, random samples. A sample mean, for instance, is random due to the sample itself being random. If a new sample is taken, a different mean will be calculated. This is known as **sampling variability**.

We deal with the fact that we are wrong by determining how wrong we may be. We quantify this by calculating a **margin of error**, which we add and subtract from our point estimate in order to form a **confidence interval** (CI).

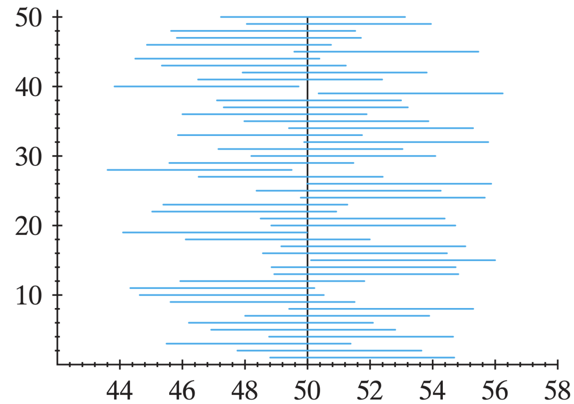
This way, we have not only our "best guess" for the value of the unknown parameter, but we have an idea of the extent to which our best guess may be in error.

Confidence Interval for Means (Sections 7.1)

More formally, we just showed that for a $100(1-\alpha)\%$ confidence interval,

$$P\left(-z_{\alpha/2} < \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = P\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right] = 1-\alpha$$

90% CIs When $\mu = 50$



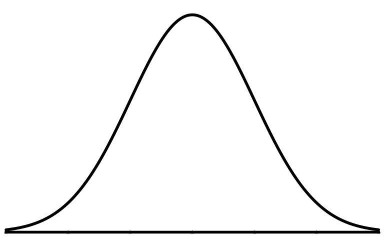
That gives us the following formula for a confidence interval for population mean (μ):

$$\boxed{\overline{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}}$$

where $z_{\alpha/2}$ is known as the **critical value** that corresponds to a certain **confidence level**. The **margin of error** is the part of the formula that follows the "$\pm$". This confidence interval is only valid if our sample size is large enough.

Must check that the sample is random and $n \geq 30$ or data come from a normal distribution.

How to determine $z_{\alpha/2}$:



Confidence Level	$z_{\alpha/2}$ value
90%	1.645
95%	1.96
98%	2.326
99%	2.576

Example: After having to get these parts replaced yourself, you are interested in knowing what the average cost to replace the brake pads and rotors for a 2018 Kia Soul is in 2022. You do some research and get a few cost estimates (including labor) of \$502, \$473, \$538, \$561, and \$488. Assuming this sample is random and the population is normal with a standard deviation of \$34, find a 99% confidence interval for the true mean cost.

Writing a confidence interval

The way I would like the confidence intervals written in this class is in this double inequality form:

$$\boxed{\text{Lower Bound of CI} < \text{Parameter of Interest} < \text{Upper Bound of CI}}$$

This not only is easy to read, it also lets me know that you know the parameter for which you a finding a confidence interval.

Interpreting a confidence interval

Once the CI is constructed, we need to give an English interpretation. The general forms of some acceptable interpretations are:

1. "We are (confidence level)% confident that the true (parameter) lies between (lower bound) and (upper bound)."

2. "Under repeated sampling, (confidence level)% of similarly constructed intervals will contain the true (parameter)."

Note that we do **not** say, "There is a (confidence level)% chance that the true (parameter) lies between (lower) and (upper)."

This is because the value of the parameter is fixed (but unknown). Once we have constructed a CI, the parameter either is or is not contained in this interval. We just don't get to know whether it is or is not.

The key here is that the CI is the thing that is random, not the parameter. We can talk about the probability that a random CI will contain the parameter, not the probability that a specific CI contains the parameter; it either does or does not.

The correct interpretations above are in very basic form. We will always need to include context in our interpretations as well.

Interpret the confidence interval from the example on the previous page:

The case where $\boldsymbol{\sigma}$ is unknown

It is usually not very realistic that you would want to calculate a confidence interval for μ when you know σ. If one is unknown, they are almost always both unknown. Since the interval is based on the fact that $Z = \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, what happens when we don't know σ?

We discussed in Section 5.5 that the statistic $T = \dfrac{Z}{\sqrt{U/r}}$ where Z is $N(0,1)$ and U is $\chi^2(r)$ follows a t distribution with r degrees of freedom. We also mentioned that $\dfrac{(n-1)S^2}{\sigma^2}$ is $\chi^2(n-1)$. Putting these together gives

$$T = \frac{Z}{\sqrt{U/r}} =$$

That means that

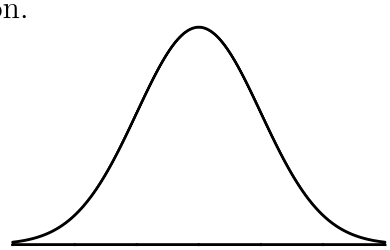
$$P\left(-t_{\alpha/2} < \frac{\overline{X} - \mu}{S/\sqrt{n}} < t_{\alpha/2}\right) = P\left[\overline{X} - t_{\alpha/2}\left(\frac{S}{\sqrt{n}}\right) < \mu < \overline{X} + t_{\alpha/2}\left(\frac{S}{\sqrt{n}}\right)\right] = 1 - \alpha$$

The t-distribution allows us to account for the fact that we also have to estimate the standard deviation. That gives us the following formula for a $100(1-\alpha)\%$ confidence interval for population mean (μ) when σ is unknown:

$$\boxed{\overline{x} \pm t_{\alpha/2} \cdot s/\sqrt{n};\ \ df = n - 1}$$

where $t_{\alpha/2}$ is known as the **critical value** that corresponds to a certain **confidence level**. Each $t_{\alpha/2}$ value will have associate degrees of freedom, $df = n - 1$ and will be larger than a $z_{\alpha/2}$ value since it is accounting for the fact that we are estimating σ. The **margin of error** is the part of the formula that follows the "$\pm$". This confidence interval is only valid if our sample size is large enough. **Must check** that the sample is random and $n \geq 30$ or data come from a normal distribution.

How to determine $t_{\alpha/2}$:



Example: Star Wars: The Rise of Skywalker was a divisive end to the Skywalker saga. To determine the average rating of the film (from 1 to 10 with 10 being the best), a random sample of 71 people was taken. The average rating for the sample was 6.2 with a variance of 12.25. Find and interpret a 98% confidence interval for the true mean rating for the film.

Confidence Interval for the Difference of Two Means (Sections 7.2)

Transitioning from one sample to two, the ideas for confidence intervals remain the same except now the interval is constructed to capture the difference in means: $\mu_1 - \mu_2$. There are four cases we will consider when we have two means:

1. Independent samples when the population variances are known.
2. Independent samples when the population variances are unknown, but are equal.
3. Independent samples when the population variances are unknown and are not equal.
4. Dependent samples when the population variance is unknown.

Independent samples when the variances are known

In this case, we use the fact that $Z = \dfrac{\overline{X}_1 - \overline{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$. This gives us the $100(1-\alpha)\%$ confidence interval formula of

$$\boxed{\overline{x}_1 - \overline{x}_2 \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

This is valid when both samples are random, the samples are independent (the subjects are different in each sample), and the data are normal or each sample is at least 30.

When constructing intervals for the difference in parameters, it is important to pay attention to the signs of the bounds.

- If both bounds are positive, then we have evidence that parameter 1 is larger than parameter 2.
- If both bounds are negative, then we have evidence that parameter 1 is smaller than parameter 2.
- If zero is in the confidence interval (one bound is negative and the other is positive), we do not have evidence that the parameters differ.

Why?

Example: Suppose Amile wants to go to Disneyland, but he doesn't care for crowds. He wants to go on the day that is the least crowded. He was told that would typically be a Tuesday or a Wednesday. Amile randomly samples 14 Tuesdays and 21 Wednesdays throughout the year and find that the sample average number of visitors were 48,500 and 51,800, respectively. Suppose it is known that the standard deviations in the number of visitors to Disneyland is 4,500 on all Tuesdays and 5,200 on all Wednesdays. Assuming the data follow normal distributions, find a 95% confidence interval for the true mean difference in the number of Disneyland attenders on Tuesdays and Wednesdays.

Independent samples when the variances are unknown, but are equal

In this case, we are assuming $\sigma_1^2 = \sigma_2^2$, but we don't know what value they are equal to. So, our best guess is a weighted average of the sample variances. This is known as **pooling** the variances and the formula is

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Then we use the fact that $T = \dfrac{\overline{X}_1 - \overline{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t(n_1+n_2-2)$ to get the $100(1-\alpha)\%$ confidence interval formula

$$\boxed{\overline{x}_1 - \overline{x}_2 \pm t_{\alpha/2}\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}; \quad df = n_1 + n_2 - 2}$$

This is valid when both samples are random, the samples are independent (the subjects are different in each sample), the data are normal or each sample is at least 30, and the population variances are equal.

Example: Suppose Amile wants to go to Disneyland, but he doesn't care for crowds. He wants to go on the day that is the least crowded. He was told that would typically be a Tuesday or a Wednesday. Amile randomly samples 14 Tuesdays and 21 Wednesdays throughout the year and find that the sample average number of visitors were 48,500 with a SD of 4,500 and 51,800 with a SD of 5,200, respectively. Suppose it is known that the standard deviations in the number of all visitors to Disneyland on Tuesdays and on Wednesdays are equal. Assuming the data follow normal distributions, find a 95% confidence interval for the true mean difference in the number of Disneyland attenders on Tuesdays and Wednesdays.

Independent samples when the variances are unknown and are not equal

In this case, we are assuming $\sigma_1^2 \neq \sigma_2^2$, which is the more common assumption. Then we use the fact that $W = \dfrac{\overline{X}_1 - \overline{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \stackrel{\bullet}{\sim} t(r)$ where $r = \dfrac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$. Obviously, the formula for the degrees of freedom is ugly, so it is common to use a conservative estimate if doing these by hand. It will be okay if you use $df = \min(n_1 - 1, n_2 - 1)$. The $100(1-\alpha)\%$ confidence interval formula is

$$\boxed{\overline{x}_1 - \overline{x}_2 \pm t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}; \quad df = \min(n_1 - 1, n_2 - 1)}$$

This is valid when both samples are random, the samples are independent (the subjects are different in each sample), the data are normal or each sample is at least 30, and the population variances are not equal.

Example: Suppose Amile wants to go to Disneyland, but he doesn't care for crowds. He wants to go on the day that is the least crowded. He was told that would typically be a Tuesday or a Wednesday. Amile randomly samples 14 Tuesdays and 21 Wednesdays throughout the year and find that the sample average number of visitors were 48,500 with a SD of 4,500 and 51,800 with a SD of 5,200, respectively. Suppose it is not known that the standard deviations in the number of all visitors to Disneyland on Tuesdays and on Wednesdays are equal. Assuming the data follow normal distributions, find a 95% confidence interval for the true mean difference in the number of Disneyland attenders on Tuesdays and Wednesdays.

Dependent samples when the variance is unknown

The final case we will consider with the difference in means is if the samples are dependent. That is, if the same subjects are being measured twice. In this case, we define $\mu_D = \mu_1 - \mu_2$ and find a confidence interval for μ_D. μ_D is the population average of the differences between the two samples. We define $D_i = X_{1i} - X_{2i}$ for $i = 1, \dots, n$. Then we use the fact that $T = \dfrac{\overline{D} - \mu_D}{S_d/\sqrt{n}} \sim t(n-1)$ where $\overline{D}$ is the sample mean of the differences and S_D is the sample standard deviation of the differences. The $100(1-\alpha)\%$ confidence interval formula is

$$\boxed{\overline{d} \pm t_{\alpha/2} \cdot s_D/\sqrt{n}; \quad df = n - 1}$$

This is valid when both subjects are selected randomly, the samples are dependent (the subjects are the same in each sample), and the data are normal or each sample is at least 30. Notice that this formula is exactly the same as the formula in the one-sample case, but we use the differences instead of the raw data.

Example: A physical fitness program is designed to increase a person's upper body strength. To determine the effectiveness of this program a simple random sample of 9 members of a health club was selected and each member was asked to do as many push-ups as possible in 1 minute. After 1 month on the program the participants were once again asked to do as many push-ups as possible in 1 minute. These values were recorded and the difference (After - Before) was computed. Construct a 90% confidence interval for the difference in the number if push-ups in one minute after vs before the program.

Subject	Before	After	Difference
1	28	32	
2	34	32	
3	28	42	
4	60	64	
5	20	41	
6	25	33	
7	32	49	
8	19	32	
9	29	50	

Confidence Interval for Proportions (Sections 7.3)

The theory for confidence intervals for means also applies to proportions. We will consider a confidence intervals for a single proportion and for the difference in proportions when the samples are independent.

Single proportion

In Section 5.7, we stated that the CLT can be applied to *any* distribution and so $\hat{p} \overset{\bullet}{\sim} N(p, p(1-p)/n)$, where $\hat{p}$ is the number of "successes" over the sample size: x/n. That means

$$P\left(-z_{\alpha/2} < \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} < z_{\alpha/2}\right) = P\left(\hat{p} - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) \approx 1-\alpha.$$

This gives the interval formula of $\hat{p} \pm z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$. There is a problem with this, though!

Due to this issue, we need to use an estimate and therefore, the (approximate) $100(1-\alpha)\%$ confidence interval for a single proportion is

$$\boxed{\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

This interval is valid when the sample is random, $n\hat{p} \geq 5$, and $n(1-\hat{p}) \geq 5$.

<u>Example</u>: Suppose a political polling firm would like to determine whether or not a majority of registered voters in a certain congressional district plan to vote for the incumbent congressional representative. The firm conducts a poll in which 300 registered voters are surveyed. Of these 300 voters, 159 state that they are planning on voting for the incumbent candidate. Construct a 95% CI for the proportion of voters in this district who plan to vote for the incumbent representative.

Difference in proportion

In this case, we are assuming that the samples are independent (the subjects are different in each sample). We then use the fact that $Z = \dfrac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \overset{\bullet}{\sim} N(0,1)$ to get the formula for the (approximate) $100(1-\alpha)\%$ confidence interval:

$$\boxed{\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}}$$

This interval is valid when the samples are random and independent and $n_1\hat{p}_1 \geq 5$, $n_1(1-\hat{p}_1) \geq 5$, $n_2\hat{p}_2 \geq 5$, and $n_2(1-\hat{p}_2) \geq 5$.

Example: Suppose the Acme Drug Company develops a new drug, designed to prevent colds. The company states that the drug is equally effective for men and women. To test this claim, they choose a a simple random sample of 100 women and 200 men from a population of 100,000 volunteers. At the end of the study, 38% of the women caught a cold; and 51% of the men caught a cold. Based on these findings, can we reject the company's claim that the drug is equally effective for men and women? Construct a 98% confidence interval for the difference in the proportions to answer the question.

Sample Size (Sections 7.4)

A common problem in statistics is finding the sample size that is required in order to reach a given margin of error. This is common when polling. The researcher will often want a margin of error of less than 3% so that the results are more likely to be more accurate. We usually don't want a larger sample than we need, however, since more time and money must be spent to get a larger sample.

Consider the margin of error for a population proportion: $\varepsilon = z_{\alpha/2}\sqrt{\dfrac{\widehat{p}(1-\widehat{p})}{n}}$.
Using the wonders of algebra:

$$\boxed{n = \qquad\qquad\qquad}$$

Problems with the formula?

Example: At 95% confidence, calculate the sample size required to obtain a margin of error of 0.03 for the estimation of a population proportion.

The process is the same when dealing with a mean when <u>σ **is known**</u>. Setting $\varepsilon = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ gives us a required minimum sample size of

$$\boxed{n = \frac{z_{\alpha/2}^2\sigma^2}{\varepsilon^2}}$$

If <u>σ **is not known**</u>, we would typically use the t distribution. In this case, $\varepsilon = t_{\alpha/2}\frac{s}{\sqrt{n}}$. So, the sample size required is $n = \dfrac{t_{\alpha/2}^2 s^2}{\varepsilon^2}$. Problems?

In the case where σ is not known, it is typical to either use an estimate for σ and plug that into the boxed formula above, or take a small sample and then plug in the sample variance from that small sample and not worry about the t critical value. Use the z instead.

<u>Example</u>: If you estimate that a population has a mean of 38 and a standard deviation of 9.7 based on last year's survey, but want to be 96% sure that you get a sample average for a new survey that is within 5% of the true mean, what sample size will you need?

Chapter 8: Tests of Statistical Hypotheses

There are two primary procedures we use in inferential statistics to draw inference on a population:

1. Confidence intervals (introduced in Chapter 7)
2. Hypothesis tests (which we are introducing now!)

With confidence intervals, we set up an interval that contains bounds on what we believe are the plausible values of our parameter based on our sample.

Hypothesis tests are a bit different. Instead of obtaining bounds where we believe the parameter lies, we instead test to see if we have sufficient evidence to reject a null hypothesis (H_0) in favor of an alternative hypothesis (H_A).

Example: I want to know if the proportion of all students who have seen *High School Musical* has increased since Disney+ launched. Let's say that 30% of students had seen the movie before Disney+. I take a random sample of 100 students and calculate the proportion of them who have seen *High School Musical* to help answer my research question.

What is the variable?

What is the parameter of interest?

What value of the sample proportion would it take for you to be confident that the true proportion of all students who have seen the movie has increased?

Hypothesis testing outline

The hypothesis testing procedure can be performed in 6 steps. Note that how these steps are defined is subjective; other instructors and textbooks will define them differently, but the outcome will be the same.

1. State the parameter(s) of interest (that is, define the symbol(s) that will be used in the hypotheses).
2. Set up the null & alternative hypotheses.
3. State the significance level (if we have one) and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p-value (if we are basing the decision on p-values).
5. Make the statistical decision.
6. Interpret the results.

We will go through each of these steps using the following example.

Tests About One Mean (Sections 8.1)

Illustrative Example: An education researcher is interested in determining whether high school students give correct answers to SAT reading comprehension questions that refer to passages they have not read at a rate that differs from what would be expected by random chance. 12 high school students are each given 100 SAT reading comprehension questions to answer. None of the students have read the relevant passage before answering the questions. If the students were just randomly guessing, then they should, on average, answer 20 correctly just by chance. The average number of correct answers given by the 12 students is 33.7 and it is known the standard deviation of all test takers in this situation is 10.2. Is this significantly different from what we would expect from just random guessing alone? Use $\alpha = 0.01$.

Step 1: State the parameter(s) of interest

The parameter of interest is a single value that summarizes what we want to know about the population. It is crucial that we know what this is and its symbol so we can set up our hypotheses correctly. For now, this will be either a population mean (______), proportion (______), variance (______), or standard deviation (______).

Step 2: Set up the null and alternative hypotheses

In hypothesis testing, we usually make the null hypothesis (H_0) the proposal we would like to __________________.

This proposal can be stated as a comparison between some unknown parameter (such as a true population mean or proportion) and some hypothesized value.

If we reject H_0, we do so in favor of the alternative hypothesis, H_A. Thus if you are trying to find evidence in favor of a proposal, that proposal goes in H_A and its opposite goes in H_0.

Two tailed or one tailed test?

When setting up the null and alternative hypotheses, we must determine which of three scenarios makes the most sense for the question at hand: a left-tailed test, a right-tailed test, or a two-tailed test.

In every case, the notion of equality goes in the ____________ hypothesis and the notion of non-equality goes in the __________________ hypothesis.

Two-tailed test

If the null hypothesis that we are trying to reject is that a population parameter is ____________ to some value, and the alternative hypothesis is that this population parameter is ____________ to this value, then the test is a two-tailed test.

In a two tailed test, we are willing to reject the null hypothesis if we find evidence that the unknown parameter is either less than or greater than the hypothesized value, and we do not specify ahead of time which it should be.

So, if the question is whether or not the true unknown parameter is "different from" or "not equal to" some specified value, then the test should be a two-tailed test.

One-tailed test

On the other hand, we sometimes are only interested in testing to see if the true unknown parameter is ____________ than some specified value, or ____________ than some specified value, but not in testing for both possibilities at once.

In this case, the test will be either "____________ tailed" (if our alternative hypothesis is that the parameter is less than the hypothesized value) or "____________ tailed" (greater than the hypothesized value).

Illustrative Example: Steps 1 and 2. We would like to see if the true mean number of correct answers differs from what we would expect to see if the students were simply guessing randomly. When the parameter is a mean, we denote it μ. Also, since we are testing to see if the mean "differs" from some value, this will be a ________________________ test.

We can now write the parameter of interest and the null and alternative hypotheses:

Step 3: State the significance level and the corresponding critical value

The **significance level (α)** is the probability we are willing to accept of rejecting a true null hypothesis. If this is not given to you, the rule of thumb is to use $\alpha = 0.05$.

The **critical value** is the value of the test statistic such that α is the area remaining in the tail of the sampling distribution of the test statistic under H_0.

The critical value is the "cutoff" that distinguishes values of the test statistic that result in rejecting H_0 from values that result in failing to reject H_0.

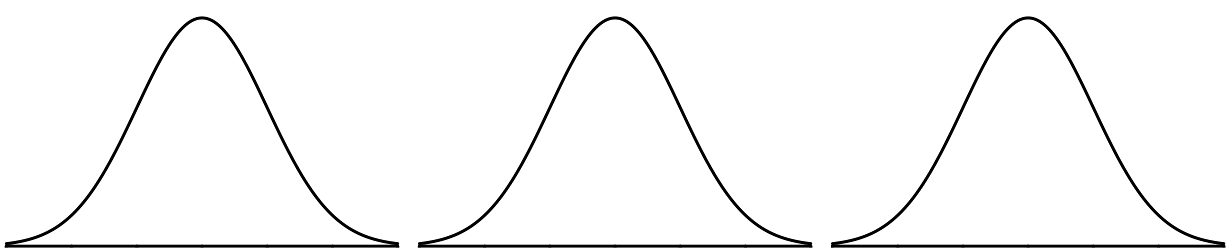
The sampling distribution of the test statistic under H_0 shows the distribution of values of the test statistic that we would expect to obtain if H_0 were true. Values in the middle of the distribution are likely to occur when H_0 is true; values in the tail(s) of the distribution are unlikely to occur when H_0 is true.

For two-tailed tests, we will reject H_0 if we obtain a test statistic in either the left or right tail of this distribution. For one-tailed tests, we will reject H_0 only if we obtain a test statistic in the relevant tail (either the left or the right).

The area in the tail of the distribution next to the critical value will be α.

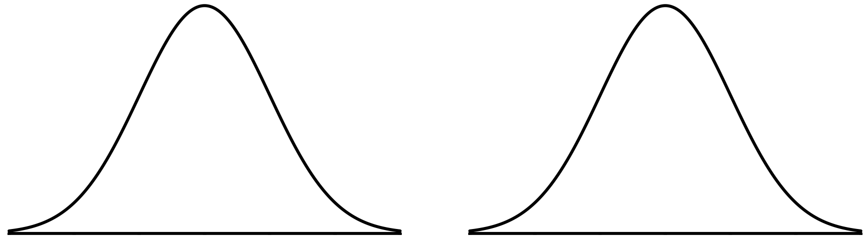
For one-tailed tests, this area is contained only in the relevant tail. For two-tailed tests, this area is split between both tails, so that the area in each tail next to the critical value is $\alpha/2$.

For means, the distribution from which critical value comes depends on if we know σ. If σ is known, the critical value is a z. If σ is unknown, the critical value is a t with $df = n - 1$. We can use the t table, a calculator, or Excel to get critical values for z and t.



In calculator and Excel:

Illustrative Example: Step 3



Step 4 Part 1: Compute the test statistic

The test statistic tells us how much evidence we have against H_0. The bigger the test statistic, the ____________________ the evidence.

The test statistic we use will depend on what kind of parameter we are dealing with. You can refer to your Inference Summary Sheet to see which test statistic is used for which parameter.

For a single mean, the test statistic is a z when σ is known and a t when σ is unknown.

$$z_{\text{stat}} = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} \qquad\qquad t_{\text{stat}} = \frac{\overline{x} - \mu_0}{s/\sqrt{n}}; \quad df = n - 1$$

where $z_{\text{stat}} \sim N(0,1)$ and $t_{\text{stat}} \sim t(n-1)$.

Step 4 Part 2: Compute the -value

The p-value is the area in the tail(s) of the distribution next to the test statistic.

If the test statistic is a z value, you can use `normalcdf`, `NORM.DIST`, or a z table to find it. If the test statistic is a t value, you can use `tcdf` or `=T.DIST`. The bounds are determined by what type of test we have.

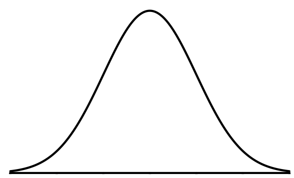
This again gives evidence against the null hypothesis. We can interpret it in one of a couple ways:

<u>For left-tailed tests</u>: The p-value is the probability of obtaining a sample statistic equal to the one we observed or smaller if the null hypothesis is true.

For right-tailed tests: The p-value is the probability of obtaining a sample statistic equal to the one we observed or larger if the null hypothesis is true.

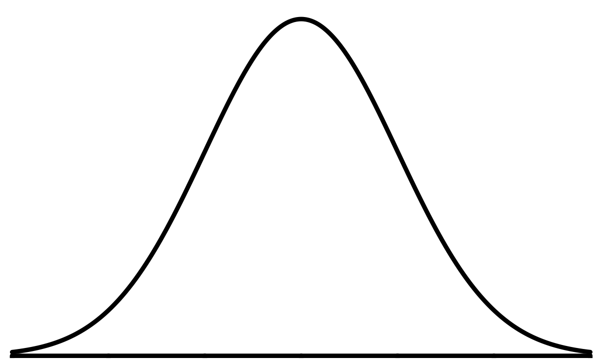
For two-tailed tests: The p-value is the probability of obtaining a sample statistic at least as extreme as the one we observed if the null hypothesis is true.

Important: if we have a two-tailed test, we must multiply the probability we get from our calculator by 2 to get our p-value. Why?



In calculator:

Illustrative Example: Step 4. The parameter we are drawing inference on is μ, which in this case is the true (i.e. population) mean of the population of all students in this situation. The test statistic is:



Step 5: Make the statistical decision

There are two ways to make the statistical decision about whether we want to reject the null hypothesis or not.

1. Compare our p-value to our level of significance (α).

   If the p-value $< \alpha$, we reject H_0.

   If the p-value $> \alpha$, we fail to reject H_0.

2. Compare our test statistic to our critical value. This is known as using rejection regions.

   For left-tailed test:
   If the test statistic $<$ critical value, we reject H_0.

   For right-tailed test:
   If the test statistic $>$ critical value, we reject H_0.

   For two-tailed test:
   If the test statistic is not between the two critical values, we reject H_0.

Step 6: Interpret the results

Once we have made the statistical decision, we should state what our results mean in plain English. In other words, relate the statistical results back to the original question of interest. We generally interpret our results in terms of the alternative hypothesis.

We will simply say whether we do have enough evidence (if we reject H_0) or do not have enough evidence (if we fail to reject H_0) to suggest the alternative hypothesis is true.

In general, results are significant if they are unlikely to come about as the result of chance alone. That is, if the p-value is small.

Illustrative Example: Steps 5 and 6.

This example demonstrates one kind of hypothesis tests. In practice, there are hundreds (maybe thousands) of different kinds of hypothesis tests. We will only look at a handful of them in this class.

What all hypothesis tests have in common is that they identify null and alternative hypotheses, and they utilize observed data to answer the question "how likely would it be to obtain results like this if the null hypothesis was true?"

We can think of hypothesis tests similar to the creed of "innocent until proven guilty". That is, we will always go with the null hypothesis (innocent) unless there is enough evidence to reject it (guilty).

Subjective View: When thinking of it in these terms, another way to make the decision is done without a significance level. Here, we don't conclude that H_0 is correct or incorrect; we simply assess the weight of evidence against H_0 and in favor of H_A. As a guideline, we might conclude:

- If P-value $> .10$, we have little or no evidence in favor of H_A.
- If $.05 <$ P-value $< .10$, we have suggestive evidence in favor of H_A.
- If $.01 <$ P-value $< .05$, we have moderate evidence in favor of H_A.
- If $.001 <$ P-value $< .01$, we have strong evidence in favor of H_A.
- If P-value $< .001$, we have very strong evidence in favor of H_A.

Example: Suppose it was reported that the average GRE score in 1997 was 558 out of a possible 800 points. After taking your GRE test, you believe that the average GRE score has changed since 1997. You obtain a random sample of 101 students who have taken the GRE and find the average is 584 with a standard deviation of 120. At the 0.05 significance level, test your hypothesis.

Construct a 95% confidence interval for the true average GRE score. Do the results agree with the hypothesis test?

Relationship between a hypothesis test and a confidence interval

A confidence interval can be thought of as an "inverted" two-tailed hypothesis test. What is meant by this is that a CI contains all the possible null values that would result in failing to reject a null hypothesis.

This should make intuitive sense: if we consider a value for an unknown parameter plausible, then we would not want to reject it. However, if a value is outside the range of what we consider plausible, then we would want to reject it.

Note that this relationship holds for two-tailed hypothesis tests. You can see this on your t-table; the 95% confidence level column corresponds to a two-tailed test at $\alpha = 0.05$; the 99% confidence level column corresponds to a two-tailed test at $\alpha = 0.01$, and so on.

For a one-tailed hypothesis test, it is possible to reject a null value that falls inside a corresponding confidence interval.

Hypothesis Testing Errors

When doing a hypothesis test, there are two types of errors that can come about.

1. The first type, which we call a Type I error, is when we reject the null hypothesis when the null hypothesis is actually true.
2. A Type II error is failing to reject the null hypothesis when it is actually false.

Usually, Type I errors are regarded as worse. If we make a Type II error, then that just means that we say we have no additional information than what we started off with. If we make a Type I error, we say things have changed when they actually haven't. Because a Type I is worse, that is the one we control with our significance level:
α = probability of committing a Type I error. We often denote the probability of committing a Type II error with β.

Continuing with our courtroom metaphor, a Type I error is saying someone is guilty when they are actually innocent and a Type II error is saying someone is innocent when they are actually guilty.

	Null Hypothesis	
Decision	True	False
Reject H_0	Type I Error	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error

Example: A manufacturer of compact fluorescent light (CFL) bulbs claims the average lifetime of its CFL bulbs is greater than the current industry standard of 8000 hours.

What is a Type I error in the context of this problem?

What is a Type II error in the context of this problem?

Tests of the Equality of Two Means (Sections 8.2)

When we have two samples and want to compare means, the alternative hypothesis are usually one of the following:

$$H_A : \mu_1 - \mu_2 > 0$$
$$H_A : \mu_1 - \mu_2 < 0$$
$$H_A : \mu_1 - \mu_2 \neq 0$$

There is no reason why we need to have a zero on the right-hand side. You could replace that with any number, but zero is used most often since if the difference is not equal to zero, then that is evidence that the two means are not equal.

Just like with confidence intervals, there are four cases when doing a hypothesis test for the equality of two means:

1. Independent samples when the variances are known.
2. Independent samples when the variances are unknown, but are equal.
3. Independent samples when the variances are unknown and are not equal.
4. Dependent samples when the variance is are unknown.

Independent samples when the variances are known

The test statistic in this case is

$$\boxed{z_{\text{stat}} = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}} \quad \text{if zero is in the hypotheses.}$$

and $z_{\text{stat}} \sim N(0,1)$. This test is valid when both samples are random, the samples are independent (the subjects are different in each sample), and the data are normal or each sample is at least 30.

Independent samples when the variances are unknown, but are equal

In this case, we are assuming $\sigma_1^2 = \sigma_2^2$, but we don't know what value they are equal to. Like with the confidence intervals, we must <u>**pool**</u> the variances to get the test statistic

$$\boxed{t_{\text{stat}} = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}; \quad df = n_1 + n_2 - 2}$$

where $s_p^2 = \dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ and $t_{\text{stat}} \sim t(n_1+n_2-2)$. This test is valid when both samples are random, the samples are independent (the subjects are different in each sample), the data are normal or each sample is at least 30, and the population variances are equal.

Independent samples when the variances are unknown and are not equal

In this case, we are assuming $\sigma_1^2 \neq \sigma_2^2$. The test statistic is

$$\boxed{t_{\text{stat}} = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}; \quad df = \min(n_1-1, n_2-1)}$$

Then $t_{\text{stat}} \overset{\bullet}{\sim} t\Big(\min(n_1-1, n_2-1)\Big)$. Remember that taking $df = \min(n_1-1, n_2-1)$ is just an easy estimate. The correct degrees of freedom value is $df = \dfrac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$, but you don't have to use that. This test is valid when both samples are random, the samples are independent (the subjects are different in each sample), the data are normal or each sample is at least 30, and the population variances are not equal.

Dependent samples when the variance is unknown

Recall in this case that we are measuring the same subjects twice and then are defining $\mu_D = \mu_1 - \mu_2$. We use the differences between observations and get their summary statistics to find $\overline{d}$ and s_D. The hypotheses will be $H_A : \mu_D > 0$, $H_A : \mu_D < 0$, or $H_A : \mu_D \neq 0$ and the test statistic is

$$\boxed{t_{\text{stat}} = \frac{\overline{d} - \mu_D}{s_D/\sqrt{n}} = \frac{\overline{d}}{s_D/\sqrt{n}}; \quad df = n - 1}$$

where $t_{\text{stat}} \sim t(n-1)$. This test is valid when both subjects are selected randomly, the samples are dependent (the subjects are the same in each sample), and the data are normal or each sample is at least 30.

Example: A veterinarian would like to investigate whether or not dogs who are neutered tend to behave less aggressively after the neutering operation than before. Before embarking on a large study, she decides to conduct a small "pilot" study involving seven dogs. Each dog is observed interacting with other dogs for one hour on both the day before the neutering operation, and three weeks after the operation. During each one hour observation period, an observer records how many "violent acts" the dog engages in. The data are below:

Subject #	1	2	3	4	5	6	7
# of violent acts before neutering	4	9	10	2	3	7	0
# of violent acts after neutering	2	12	3	2	0	2	0

Tests for Variances (Sections 8.3)

Hypothesis Testing for One Variance or Standard Deviation

Just like with confidence intervals, the hypothesis tests for a standard deviation do not use z or t values, they use χ^2 values instead. Therefore, the test statistic is

$$\boxed{\chi^2_{\text{stat}} = \frac{(n-1)s^2}{\sigma_0^2}; \qquad df = n-1}$$

where $\chi^2_{\text{stat}} \sim \chi^2(n-1)$. This test is valid if the sample is random and the data come from a normal distribution. The p-value can be found using χ^2`cdf` on a calculator. Otherwise, the 6 hypothesis testing steps still apply.

Note: hypothesis test about a standard deviation and a variance are equivalent. So, we do not have to test both.

<u>Example:</u> A company claims that the standard deviation of the call wait times is less than 1.8 minutes. A random sample of 25 incoming telephone calls has a standard deviation of 1.5 minutes. At $\alpha = 0.10$, is there enough evidence to support the company's claim? Assume the population is normally distributed.

Confidence Interval for One Variance and Standard Deviation

In Chapter 7, we did not discuss a confidence interval for a variance (σ^2) or a standard deviation (σ), so we now will. We use the fact that $\chi^2 = \dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$, so

$$P\left(\chi^2_{1-\alpha/2} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2}\right) = P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right) = 1-\alpha.$$

So, the confidence intervals are given by:

$$\boxed{\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}; \quad df = n-1}$$

and

$$\boxed{\sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2}}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}}; \quad df = n-1}$$

<u>Example</u>: Find an 90% confidence interval for the standard deviation of call wait times from the example on the previous page.

Hypothesis Testing for Two Variances or Standard Deviations

We turn to performing inference on two variances now. It is typical, however, to not find the difference in variances. Rather, we find the <u>ratio</u> of variances.

When testing for the ratio in variances, the alternative hypothesis are usually one of the following:

$$H_A : \sigma_1^2/\sigma_2^2 > 1$$
$$H_A : \sigma_1^2/\sigma_2^2 < 1$$
$$H_A : \sigma_1^2/\sigma_2^2 \neq 1$$

The reason we focus on ratios and not differences is because of the kind of distribution the ratio of variances follows. Remember that when testing a single variance, we needed to use the χ^2 distribution. The difference in χ^2 values does not follow any nice distribution, but the ratio of χ^2 values does. The ratio of χ^2 variables follows an F distribution.

The test statistic for testing the ratio of variances or standard deviations is

$$\boxed{F_{\text{stat}} = \frac{s_1^2}{s_2^2}; \quad df_1 = r_1 = n_1 - 1; \quad df_2 = r_2 = n_2 - 1}$$

where $F_{\text{stat}} \sim F(n_1 - 1, n_2 - 1)$. In order for this test to be valid, we need random, independent samples from normal populations.

Example: The East High Wildcats and West High Knights are bitter basketball rivals. A question has been posed about which team is more consistent in terms of the number of points they score in basketball games. Samples of 16 East High games and 21 West High games were taken and it was found that the Wildcats score 87 points per game on average with a standard deviation of 6.2 points and the Knights score 82 points per game on average with a standard deviation of 9.3 points. Based on this information, can we conclude that one team is more consistent than the other? Use a significance level of 0.10.

Confidence Intervals for Two Variances or Standard Deviations

For the confidence interval, begin with the fact that $F = \dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$. Therefore,

$$P\left(F_{1-\alpha/2} < \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} < F_{\alpha/2}\right) = P\left(\frac{(S_1^2/S_2^2)}{F_{\alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{(S_1^2/S_2^2)}{F_{1-\alpha/2}}\right) = 1 - \alpha$$

So, the confidence intervals are given by:

$$\boxed{\frac{1}{F_{\alpha/2}} \cdot \frac{s_1^2}{s_2^2} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{1}{F_{1-\alpha/2}} \cdot \frac{s_1^2}{s_2^2}; \quad df_1 = n_1 - 1; \quad df_2 = n_2 - 1}$$

and

$$\boxed{\sqrt{\frac{1}{F_{\alpha/2}} \cdot \frac{s_1^2}{s_2^2}} < \frac{\sigma_1}{\sigma_2} < \sqrt{\frac{1}{F_{1-\alpha/2}} \cdot \frac{s_1^2}{s_2^2}}; \quad df_1 = n_1 - 1; \quad df_2 = n_2 - 1}$$

Example: Now create a 90% confidence interval for the ratio of variances and standard deviations of number of points scored for the two teams in the example on the previous page.

Tests About Proportions (Sections 8.4)

Tests About One Proportion

For testing a single proportion, we use

$$\boxed{z_{\text{stat}} = \frac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}}$$

where p_0 is the hypothesized population proportion. Then $z_{\text{stat}} \sim N(0,1)$ if the null hypothesis is true.

Example: Check out the example on the first page of this chapter.

Tests of the Equality of Two Proportion

We will only consider the case where the samples are independent in these notes. That is, we are assuming the samples are not affected by one another. This is often referred to as a **two-proportion z test**.

When testing for the difference in proportion, the alternative hypothesis are usually one of the following:

$$H_A : p_1 - p_2 > 0$$
$$H_A : p_1 - p_2 < 0$$
$$H_A : p_1 - p_2 \neq 0$$

The formula for the test statistic that compares two sample proportions is:

$$\boxed{z_{\text{stat}} = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}}$$

where $\hat{p} = \frac{x_1+x_2}{n_1+n_2} = \frac{\text{successes}_1+\text{successes}_2}{n_1+n_2}$. Then $z_{\text{stat}} \overset{\bullet}{\sim} N(0,1)$. Of course, this is only valid when we have two random, independent samples and the sample size is large enough in each: $n_1\hat{p}_1 \geq 5$, $n_1(1-\hat{p}_1) \geq 5$, $n_2\hat{p}_2 \geq 5$, and $n_2(1-\hat{p}_2) \geq 5$.

Example: Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:

Non-Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
351 said "yes"	41 said "yes"

Is there sufficient evidence at the $\alpha = 0.05$ level to conclude that the two populations – smokers and non-smokers – differ significantly with respect to their opinions?

Symbol	**Description**
n, n_1, n_2	Sample sizes
df	Degrees of freedom
ε	Margin of error
x, x_1, x_2	Number of "successes"
$\widehat{p}, \widehat{p}_1, \widehat{p}_2$	Sample proportion
p, p_1, p_2	Population proportions
$\widehat{p}$	Proportion of all successes from both samples
z_{stat}	z test statistic
$z_{\alpha/2}$	z critical value. Area to the right is $\alpha/2$.
z_0	z critical value for hypothesis tests
$\overline{x}, \overline{x}_1, \overline{x}_2$	Sample means
μ, μ_1, μ_2	Population means
t_{stat}	t test statistic
$t_{\alpha/2}$	t critical value. Area to the right is $\alpha/2$.
t_0	t critical value for hypothesis tests
α	Significance level
H_0	Null hypothesis
H_A	Alternative hypothesis

Symbol	**Description**
s, s_1, s_2	Sample standard deviations
$\sigma, \sigma_1, \sigma_2$	Population standard deviations
s_p^2	Pooled sample variance
s^2, s_1^2, s_2^2	Sample variances
$\sigma^2, \sigma_1^2, \sigma_2^2$	Population variances
χ^2	Test statistic for variance/standard deviation
$\chi^2_{\alpha/2}$	χ^2 critical value. Area to the right is $\alpha/2$.
$\chi^2_{1-\alpha/2}$	χ^2 critical value. Area to the left is $\alpha/2$.
$\overline{d}$	Sample mean of the differences
μ_D	Population mean of the differences
s_D	Standard deviation of the differences
$F_{\alpha/2}$	F critical value. Area to the right is $\alpha/2$.
$F_{1-\alpha/2}$	F critical value. Area to the left is $\alpha/2$.
μ_0	Hypothesized mean
p_0	Hypothesized proportion
σ_0^2	Hypothesized variance
$q, \widehat{q}, q_0, \bar{q}$	$1-p, 1-\widehat{p}, 1-q_0, 1-\bar{p}$
$\chi^2_{\text{stat}}, F_{\text{stat}}$	χ^2, F test statistic

Chapter 9: More Tests

We will end this class by considering three more common hypothesis tests that are based on the χ^2 and F distributions. Recall the outline of hypothesis tests and how we make our decisions:

Hypothesis testing outline

1. State the parameter(s) of interest (that is, define the symbols being used).
2. Set up the null & alternative hypotheses.
3. State the significance level and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p-value (if we are basing the decision on p-values).
5. Make the statistical decision.
6. Interpret the results.

We reject the null hypothesis in the following situations, otherwise we fail to reject H_0:

- For any test, if the p-value is less than α.
- For a left-tailed test, if the test statistic is less than the critical value.
- For a right-tailed test, if the test statistic is greater than the critical value.
- For a two-tailed test, if the test statistic is not between the critical values.

Chi-Square Goodness-of-Fit Tests (Section 9.1)

The first two tests we will discuss have to do with **categorical variables**. Recall that categorical variables take the form of labels or descriptions. Variables like gender, political affiliation, state of birth, and model of car are all categorical.

____________________ categorical variables can be given in no particular order. All of the variables listed above are nominal

____________________ categorical variables have an order they ought to be listed in. Any variable where categories are ranked (from first to last; from best to worst, from strongest to weakest, etc) is ordinal.

The **χ^2 goodness-of-fit** test is used to test whether a frequency distribution of a categorical variable fits an expected distribution. This test is best shown with an example first.

Example: According to Niche.com, the five most popular majors at SUU fall in the categories of Liberal Arts and Humanities, Education, Nursing, Elementary Education, and Biology. Let's limit our attention to only those majors. Of those majors, Niche.com reports that 71.6% of graduates obtain a Liberal Arts and Humanities degree, 12.4% get an Education degree, 5.9% are nursing graduates, 5.2% get a degree in Elementary Education, and Biology majors make up the other 5.0%. Suppose we want to test Niche.com's distribution of these majors. Sample information is given in the table below:

Major	**Observed Frequency**	**Expected %**	**Expected Frequency**
Liberal Arts	171	71.6%	$285(.716) = 204.06$
Education	51	12.4%	
Nursing	26	5.9%	
El. Ed.	16	5.2%	
Biology	21	5.0%	
Total	285	100%	285

The question we must answer is whether the distribution from our sample is different enough from the expected distribution to suggest the expected distribution ______________________. We can do this with a hypothesis test based on the χ^2 distribution. In general, we can write the hypotheses as

$$H_0\colon\ p_i = p_{i0} \text{ for } i = 1, 2, \dots, k$$
$$H_A\colon\ p_i \neq p_{i0} \text{ for at least one } i.$$

Or, we can think about the hypotheses as

H_0: The distribution of our variable is equal to some expected distribution.

H_A: The distribution of our variable differs from the expected distribution.

Of course, we must add context to these hypotheses. In our case,

The **test statistic** is computed in the following way. Let each of the expected frequencies be denoted by $E_i = np_i$ and each of the observed frequencies will be O_i. Then the test statistic is $\chi^2_{\text{stat}} = \dfrac{(O_1 - E_1)^2}{E_1} + \cdots + \dfrac{(O_k - E_k)^2}{E_k} = \sum_{i=1}^{k} \dfrac{(O_i - E_i)^2}{E_i}$. This is also written as

$$\boxed{\chi^2_{\text{stat}} = \sum \frac{(O - E)^2}{E}; \quad df = k - 1}$$

where k is the number of categories in our distribution. This test is only valid if each of the expected frequencies are at least 5 and we have a random sample.

$\boxed{\textbf{Aside}}$: Why does this follow a χ^2 distribution? Remember that $Z = \dfrac{X-\mu}{\sigma}$ and $Z^2 \sim$

If we have two categories, then $n = O_1 + O_2$ and $1 = p_1 + p_2$. Then we can say that $Z \approx \dfrac{O_1 - np_1}{\sqrt{np_1(1-p_1)}}$. So

$$\begin{aligned} Z^2 &\approx \frac{(O_1-np_1)^2}{np_1(1-p_1)} = \frac{(O_1-np_1)^2(1-p_1+p_1)}{np_1(1-p_1)} = \frac{(O_1-np_1)^2(1-p_1)}{np_1(1-p_1)} + \frac{(O_1-np_1)^2(p_1)}{np_1(1-p_1)} \\ &= \frac{(O_1-np_1)^2}{np_1} + \frac{(O_1-np_1)^2}{n(1-p_1)} = \frac{(O_1-np_1)^2}{np_1} + \frac{(O_1-np_1)^2}{np_2} \end{aligned}$$

Also, we can write

$$(O_1-np_1)^2 = (-O_1+np_1)^2 = (n-O_1-n+np_1)^2 = (n-O_1-n(1-p_1))^2 = (O_2-np_2)^2$$

So, $Z^2 \approx \dfrac{(O_1-np_1)^2}{np_1} + \dfrac{(O_2-np_2)^2}{np_2} \overset{\bullet}{\sim} \chi^2(1)$.

By similar (but more complicated) reasoning, if we had three categories, then $\dfrac{(O_1-np_1)^2}{np_1} + \dfrac{(O_2-np_2)^2}{np_2} + \dfrac{(O_3-np_3)^2}{np_3} \overset{\bullet}{\sim} \chi^2(2)$ and in general, $\displaystyle\sum_{i=1}^{k} \frac{(O_i-np_i)^2}{np_i} \overset{\bullet}{\sim} \chi^2(k-1)$.

Now we can finish the test:

<u>Calculator and R Tips</u>:

Contingency Tables and χ^2 Tests for Independence (Section 9.2)

Continuing with categorical variables, we often want to know whether or not two categorical variables are related. By "are they related?", we mean "If an observation falls into a certain category for one variable, does this make it more likely to fall into a certain category of the other?"

In other words, we want to know if our two variables are independent or dependent? If they are independent, then knowing that a variable falls into a certain category for the row variable does not affect the probability that it will fall into any of the categories for the column variable, and vice versa.

Let's consider two categorical variables we have seen before that are included in the following contingency table:

		Marijuana Usage Level			
		Never	Rarely	Frequently	Total
Political Views	Liberal	479	173	119	771
	Conservative	214	47	15	276
	Other	172	45	85	302
	Total	865	265	219	1349

Are the variables "political views" and "marijuana usage level" related? Do they seem to depend on one another?

There is some uncertainty in answering this question since this table is taken from a random sample, and any apparent relationship we observe between variables from a random sample could be due to ________________. We can answer this question much more definitively by conducting a hypothesis test for independence.

The hypotheses for the χ^2 tests for independence will always be the same:

H_0: The variables are independent.

H_A: The variables are dependent.

Remember that we always begin by assuming that the null hypothesis is true. So, we need to know what our contingency table would look like if the variables were ____________ ________________.

Remember how we determined independence in probability. Two events are independent if: $\mathbf{P(A \cap B) = P(A)P(B)}$.

Consider the political/marijuana table. If event A is "Liberal" and event B is "Never", then

$P(A) =$ $\qquad\qquad\qquad$ $P(B) =$

So, if they were independent, then $P(A \cap B) =$

But we know that $P(A \cap B) = \dfrac{\#\text{ A and B}}{\text{sample size}}$. That means if these variables were independent, we would expect:

Writing this more generally, we have:

$$\text{Expected frequency} = \frac{(\text{Sum of row 1})(\text{Sum of column 1})}{\text{sample size}}$$

So, if the variables were perfectly independent, we would expect 494.38 observations in the liberal row and the never column. We have 479. Is that difference big enough to consider the variables as dependent?

There are lots of other levels to consider: Rarely and Conservative, Frequently and Other, Never and Other, etc. We need to compare the expected counts to the actual counts for every cell in our table to really get a sense of how much how actual table is deviating from the expected one if they were independent.

In general, the expected count is

$$\text{Expected frequency } (E_{ij}) = \frac{(\text{Sum of row } i)(\text{Sum of column } j)}{\text{sample size}}$$

Once we have all of our expected counts, we can calculate our test statistic for the test for independence.

$$\boxed{\chi^2_{\text{stat}} = \sum_{j=1}^{c} \sum_{i=1}^{r} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}; \quad df = (r-1)(c-1)}$$

where E_{ij} is the expected count for the cell in row i and column j, O_{ij} is the observed count for the cell in row i and column j, r is the number of rows and c is the number of columns.

This test is only valid if our sample size is large enough. That is, each of the expected counts must be at least 5. Also, <u>we always treat these independence tests as right-tailed</u>. That is, there will only be one critical value and the p-value is the probability of getting a number larger than our test statistic.

Note that another way to formulate the hypotheses are as follows:

$$H_0\colon\ p_{ij} = p_{i\cdot}p_{\cdot j} \text{ for } i = 1, 2, \ldots, r \text{ and } j = 1, 2, \ldots, c$$
$$H_A\colon\ p_{ij} \neq p_{i\cdot}p_{\cdot j} \text{ for at least one } i \text{ or } j.$$

where $p_{i\cdot}$ is the probability of row i and $p_{\cdot j}$ is the probability of column j.

It is more common to simply state the hypotheses in terms of independence, though.

Example: Conduct a hypothesis test to determine whether we have sufficient evidence to suggest that political affiliation and marijuana usage level are dependent.

		Marijuana Usage Level			
		Never	Rarely	Frequently	Total
Political Views	Liberal	479	173	119	771
	Conservative	214	47	15	276
	Other	172	45	85	302
	Total	865	265	219	1349

	Never	Rarely	Freq	Total
Lib.		$\frac{771*265}{1349}$	$\frac{771*219}{1349}$	771
Cons.		$\frac{276*265}{1349}$	$\frac{276*219}{1349}$	276
Other		$\frac{302*265}{1349}$	$\frac{302*219}{1349}$	302
Total	865	265	219	1349

	Never	Rarely	Freq	Total
Lib.		151.46	125.17	771
Cons.	176.98	54.22	44.81	276
Other	193.65	59.33	49.03	302
Total	865	265	219	1349

$$\chi^2_{\text{stat}} = \qquad + \frac{(173-151.46)^2}{151.46} + \frac{(119-125.17)^2}{125.17} + \frac{(214-176.98)^2}{176.98} + \frac{(47-54.22)^2}{54.22} + \frac{(15-44.81)^2}{44.81}$$

$$+ \frac{(172-193.65)^2}{193.65} + \frac{(45-59.33)^2}{59.33} +$$

$$\chi^2_{\text{stat}} = \qquad +3.063 + 0.304$$

$$+6.430 + 0.961 + 19.831$$

$$+2.420 + 3.461 + 26.398$$

$$=$$

It is important to remember that "dependent" in this sense just means that knowing something about one variable gives you information about the probability of the other. <u>It does not imply that one causes the other!</u> It could be that political affiliation affects marijuana use, or that marijuana use affects political affiliation, or that other, confounding variables affect both of these things.

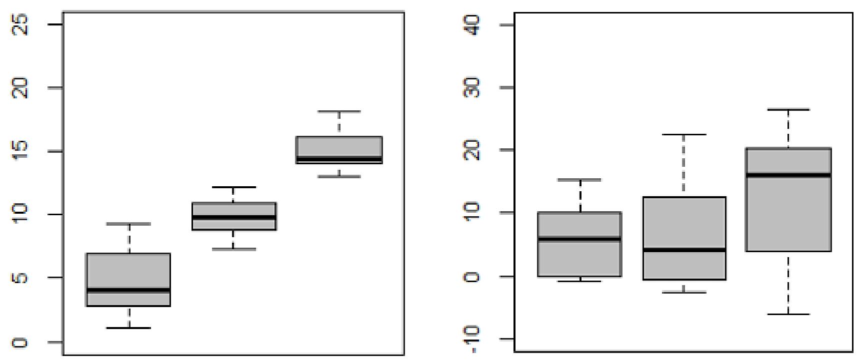
You can see which cells contributed the most to the χ^2 test statistic by looking at which ones have the largest value of $(O-E)^2/E$. In this case, the "conservative and frequently" and "other and frequently" contributed the biggest values (19.831 and 26.389, respectively). This means that these cells had the largest difference between observed and expected counts. Here, the observed count for "conservative and frequently" was much lower than what we would expect under independence (15 vs. 44.81) and the observed count for "other and frequently" was much higher than what we would expect under independence (85 vs. 49.03).

<u>**Calculator Tips**</u>:

One-Way Analysis of Variance (Section 9.3)

In the goodness-of-fit test, we discussed how to deal with more than two proportions. Now, we will discuss how to deal with more than two means. The procedure for comparing multiple population means is known as the **analysis of variance** or ANOVA. Of course, we know that if we are testing for a difference in two independent means, we can use a two-sample t-test. With ANOVA, we test three or more groups to see if any of groups' means differ significantly from the others.

It may seem strange to call the test that compares *means* an analysis of *variance*, but the variance is a crucial part of the process. Consider the following boxplots. In both of these cases, the sample means for the three boxplots are about 5, 10, and 15, respectively. In the first case, it seems clear that the true means must also differ. In the second case, this isn't so clear.



The thing that makes these two cases so different is that in the first, the boxplots are not very spread out, while in the second they are. More variation reflects greater ________________ regarding the values of the true unknown means.

With ANOVA, we compare the average "between group" variance to the average "within group" variance.

Between group variance refers to how much the means of the groups vary from one another. This can be thought of as how much of an effect the treatment is having.
Within group variance refers to how spread out the data is in each group. This can be thought of as how much error there is in each group.

The basic idea is this: if the average amount of variation ________________ the groups is substantially larger than the average amount of variation ________________ the groups, then the true group means likely differ. If not, then we cannot rule out that they are equal. As you might guess, we will make this decision using a hypothesis test. For ANOVA, we have:

H_0: The group means are all equal or $\mu_1 = \mu_2 = \cdots = \mu_k$
H_A: At least one group mean differs from the others.

ANOVA notation

- k: the # of groups we are dealing with. (The book uses m instead of k.)
- $\mu_1, \dots, \mu_k$: the true means for groups 1 through k.
- $\bar{x}_1, \dots, \bar{x}_k$: the sample means for groups 1 through k.
- $s_1, \dots, s_k$: the sample standard deviations for groups 1 through k.
- $n_1, \dots, n_k$: the sample sizes for groups 1 through k.
- n: the overall sample size for all groups combined ($n = n_1 + n_2 + \cdots + n_k$).
- $\bar{\bar{x}}$: the grand mean of all data from all groups $\left(\frac{\sum x}{n} = \frac{n_1\bar{x}_1 + \cdots + n_k\bar{x}_k}{n}\right)$.

Note that if the sample sizes for each group are equal, then the grand mean is just the average of all of sample means: $\bar{\bar{x}} = \frac{\sum \bar{x}_i}{k}$.

Sum of Squares

The way we go about testing this is by dealing with the sum of squares. "Sums of squares" refers to sums of squared deviations from some mean. ANOVA compares "between" to "within" group variation, and these types of variation are quantified using sums of squares. The three important sums of squares we will look at are:

SS_{Tot}: ________________________ sum of squares.

SS_T: Sum of squares ____________________ the samples. Also known as the __________________ sum of squares.

SS_E: Sum of squares __________________ the samples. Also known as the __________________ sum of squares.

Total Sum of Squares

Total sum of squares is calculated by summing up the squared deviations of every point in the data from the grand mean:

$$SS_{Tot} = \sum_{i=1}^{n} (x_i - \bar{\bar{x}})^2$$

This quantifies the amount of overall variation in the data, for all of the groups put together. The degrees of freedom associated with SS_{Tot} are $df = n - 1$.

Fact: $\dfrac{SS_{Tot}}{\sigma^2} \sim \chi^2(n-1)$.

Proof:

Recall that if we have a sample from a normal distribution, then $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$.

If H_0 is true, then $\mu_1 = \cdots = \mu_k$ and therefore, if the variances for all samples are equal, all X_i come from the same distribution.

Therefore, $S^2 = \dfrac{SS_{Tot}}{n-1}$ and $\dfrac{(n-1)S^2}{\sigma^2} = \dfrac{SS_{Tot}}{\sigma^2} \sim \chi^2(n-1)$.

Treatment Sum of Squares

The sum of squares between samples, also known as the treatment sum of squares, is calculated by summing the weighted squared deviations of each sample mean from the grand mean:

$$SS_T = n_1(\bar{x}_1 - \bar{\bar{x}})^2 + \cdots + n_k(\bar{x}_k - \bar{\bar{x}})^2 = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{\bar{x}})^2$$

This quantifies how much the sample means vary. It is what we are referring to when we say "between group variation". The degrees of freedom associated with SS_T are $df_1 = k - 1$.

Fact: $\dfrac{SS_T}{\sigma^2} \sim \chi^2(k-1)$.

Proof:

Not given, but here is some intuition:

Since $SS_{Tot} = SS_T + SS_E$ and we know $SS_{Tot} \sim \chi^2(n-1)$ and $SS_E \sim \chi^2(n-k)$, as we show below, it makes sense that $SS_T \sim \chi^2\Big(n - 1 - (n-k)\Big) = \chi^2(k-1)$.

Error Sum of Squares

The sum of squares within samples, also known as the error sum of squares, is calculated by adding up the squared deviations of each data point around its group mean:

$$SS_E = (n_1 - 1)s_1^2 + \cdots + (n_k - 1)s_k^2 = \sum_{i=1}^{k}(n_i - 1)s_i^2$$

This quantifies how much the data varies around its group sample means. It is what we are referring to when we say "within group variation". The degrees of freedom associated with SS_E are $df_2 = n - k$.

Fact: $\dfrac{SS_E}{\sigma^2} \sim \chi^2(n-k)$.

Proof:

Recall that if we have a sample from a normal distribution, then $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$.

Remember also that if $X_i \overset{\text{indep}}{\sim} \chi^2(r_i)$, then $\displaystyle\sum_{i=1}^{k} X_i \sim \chi^2\left(\sum_{i=1}^{k} r_i\right)$.

So, that means $\dfrac{(n_i - 1)s_i^2}{\sigma^2} \sim \chi^2(n_i - 1)$ and thus

$\displaystyle\sum_{i=1}^{k} \frac{(n_i - 1)s_i^2}{\sigma^2} = \frac{SS_E}{\sigma^2} \sim \chi^2(n_1 - 1 + n_2 - 1 + \cdots + n_k - 1) = \chi^2(n-k)$.

Mean Squares
Finally, to get an idea of how much variation we have on average, we will divide SS_T and SS_E by their degrees of freedom to obtain **mean squares**:

$$\text{Mean Square Treatment} = MS_T = \frac{SS_T}{df_1} = \frac{\sum_{i=1}^{k} n_i(\bar{x}_i - \bar{\bar{x}})^2}{k-1}$$

$$\text{Mean Square Error} = MS_E = \frac{SS_E}{df_2} = \frac{\sum_{i=1}^{k}(n_i - 1)s_i^2}{n-k}$$

Test Statistic
Using Theorem 5.2-1, when we divide these mean squares, we obtain a value that follows an F distribution. So, we can then compute our test statistic by calculating:

$$\boxed{F_{\text{stat}} = \frac{\frac{SS_T}{\sigma^2}/(k-1)}{\frac{SS_E}{\sigma^2}/(n-k)} = \frac{MS_T}{MS_E}; \quad df_1 = k-1; \quad df_2 = n-k}$$

where $F_{\text{stat}} \sim F(k-1, n-k)$. If the F test statistic is large, that means there is much more variation ________________________ samples than there is ________________________ them. If that is the case, we should ________________________ the null hypothesis that all of the means are equal.

There are some necessary conditions to use ANOVA:

1. The population of each group is normal.
2. Each population has the same variance (or standard deviation).
3. Each sample is random and independent.

The ANOVA Table
All of the information regarding this ANOVA procedure can be summarized in an ANOVA table:

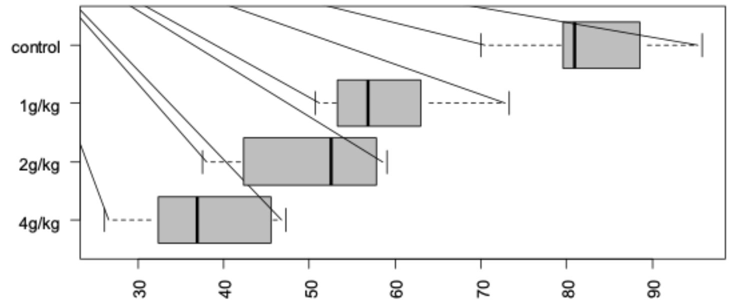
Source of Variation	**Sum of Squares**	**df**	**Mean Square**	**Statistic**
Treatment	SS_T	$k-1$	$MS_T = \frac{SS_T}{k-1}$	$F = \frac{MS_T}{MS_E}$
Error	SS_E	$n-k$	$MS_E = \frac{SS_E}{n-k}$	
Total	SS_{Tot}	$n-1$		

or often written with the sum of squares and degrees of freedom columns switched.

Calculator Tips:

Example: A researcher who studies sleep is interested in the effects of ethanol on sleep time. She gets a sample of 20 rats and gives each an injection having a particular concentration of ethanol per body weight. There are 4 treatment groups, with 5 rats per treatment. She records REM sleep time for each rat over a 24-hour period. The results are given in the table and plotted below:

Treatment	**Mean**	**SD**
1) 0 (control)	83.0	9.7
2) 1 g/kg	59.4	9.0
3) 2 g/kg	49.8	9.66
4) 4 g/kg	37.6	8.8



Source of Variation	**Sum of Squares**	**df**	**Mean Square**	**Statistic**
Between				
Within				
Total				

Note that this overall ANOVA test does not identify which means differ from which other means or how many differ. It only tells us that at least one is significantly different from at least one other. There are other statistical methods that can be used to deduce which means differ, but we will not consider them here.

Standard Normal Probabilities

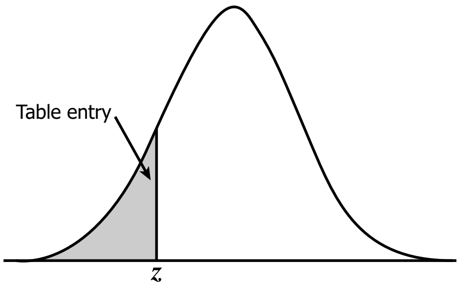


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Standard Normal Probabilities

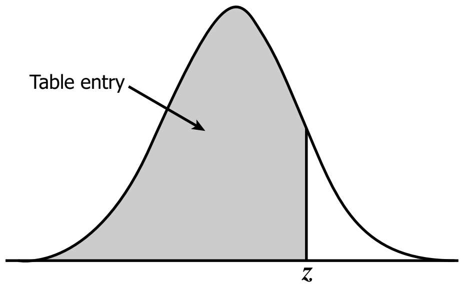


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Inference Summary Sheet

Confidence Intervals

Parameter(s)	TI Calculator Function	CI Form or Test Statistic	Necessary Conditions
Proportion p (single)	1-PropZint	$\hat{p} \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$	1. Random sample. 2. $n\hat{p} \geq 5$ & $n\hat{q} \geq 5$.
Proportions $p_1 - p_2$ (independent)	2-PropZint	$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}_1\hat{q}_1}{n_1} + \dfrac{\hat{p}_2\hat{q}_2}{n_2}}$	1. Independent samples. 2. Random samples. 3. $n\hat{p}_1 \geq 5$ & $n\hat{q}_1 \geq 5$. 4. $n\hat{p}_2 \geq 5$ & $n\hat{q}_2 \geq 5$.
Mean μ (single σ known)	ZInterval	$\overline{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$	1. Random sample. 2. Normal data or $n \geq 30$.
Mean μ (single σ unknown)	TInterval	$\overline{x} \pm t_{\alpha/2} \cdot s/\sqrt{n}$ $df = n - 1$	1. Random sample. 2. Normal data or $n \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with σ_1, σ_2 known)	2-SampZInt	$\overline{x}_1 - \overline{x}_2 \pm z_{\alpha/2} \cdot \sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with $\sigma_1 = \sigma_2$ unknown)	2-SampTInt Pooled: Yes	$\overline{x}_1 - \overline{x}_2 \pm t_{\alpha/2} \cdot \sqrt{s_p^2\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}$ $df = n_1 + n_2 - 2$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with $\sigma_1 \neq \sigma_2$ unknown)	2-SampTInt Pooled: No	$\overline{x}_1 - \overline{x}_2 \pm t_{\alpha/2} \cdot \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$ $df = \min(n_1 - 1, n_2 - 1)$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means μ_d (dependent)	TInterval	$\overline{d} \pm t_{\alpha/2} \cdot s_D/\sqrt{n}$ $df = n - 1$	1. Dependent samples. 2. Random samples. 3. Normal data or $n \geq 30$.
Variance or SD σ^2 or σ (single)	None	Left: $\frac{(n-1)s^2}{\chi^2_{\alpha/2}}$, Right: $\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$ (Take square roots for SD.) $df = n - 1$	1. Random sample. 2. Data are normal.
Variances or SDs σ_1^2/σ_2^2 or σ_1/σ_2 (ratio)	None	Left: $\frac{1}{F_{\alpha/2}} \cdot \frac{s_1^2}{s_2^2}$, Right: $\frac{1}{F_{1-\alpha/2}} \cdot \frac{s_1^2}{s_2^2}$ (Take square roots for SDs.) $df_1 = n_1 - 1$, $df_2 = n_2 - 1$	1. Independent samples. 2. Random sample. 3. Both populations normal.

ANOVA formulas for ANOVA test on other side of this sheet:

$$\overline{\overline{x}} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \cdots + n_k\bar{x}_k}{n} = \frac{\sum_{i=1}^{k} n_i\bar{x}_i}{n}$$

$$SS_T = n_1(\bar{x}_1 - \overline{\overline{x}})^2 + \cdots + n_k(\bar{x}_k - \overline{\overline{x}})^2 = \sum_{i=1}^{k} n_i(\bar{x}_i - \overline{\overline{x}})^2$$

$$SS_E = (n_1 - 1)s_1^2 + \cdots + (n_k - 1)s_k^2 = \sum_{i=1}^{k}(n_i - 1)s_i^2$$

$$MS_T = \frac{SS_T}{k-1} \qquad MS_E = \frac{SS_E}{n-k}$$

The ANOVA Table

Source of Variation	Sum of Squares	df	Mean Square	*F* Statistic
Treatment	SS_T	$k - 1$	$MS_T = \frac{SS_T}{k-1}$	$F = \frac{MS_T}{MS_E}$
Error	SS_E	$n - k$	$MS_E = \frac{SS_E}{n-k}$	
Total	SS_{Tot}	$n - 1$		

Hypothesis Tests

Parameter(s)	TI Calculator Function	CI Form or Test Statistic	Necessary Conditions
Proportion p (single)	1-PropZTest	$z_{\text{stat}} = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$	1. Random sample. 2. $np_0 \geq 5$ & $nq_0 \geq 5$.
Proportions $p_1 - p_2$ (independent)	2-PropZTest	$z_{\text{stat}} = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\widehat{pq}}{n_1} + \frac{\widehat{pq}}{n_2}}}$ $\widehat{p} = \frac{\text{successes}_1 + \text{successes}_2}{n_1 + n_2}$	1. Independent samples. 2. Random samples. 3. $n_1 \widehat{p} \geq 5$ & $n_1 \widehat{q} \geq 5$. 4. $n_2 \widehat{p} \geq 5$ & $n_2 \widehat{q} \geq 5$.
Mean μ (single σ known)	ZTest	$z_{\text{stat}} = \dfrac{\overline{x} - \mu_0}{\sigma / \sqrt{n}}$	1. Random sample. 2. Normal data or $n \geq 30$.
Mean μ (single σ unknown)	TTest	$t_{\text{stat}} = \dfrac{\overline{x} - \mu_0}{s / \sqrt{n}}$ $df = n - 1$	1. Random sample. 2. Normal data or $n \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with σ_1, σ_2 known)	2-SampZTest	$z_{\text{stat}} = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with $\sigma_1 = \sigma_2$ unknown)	2-SampTTest Pooled: Yes	$t_{\text{stat}} = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ $df = n_1 + n_2 - 2$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means $\mu_1 - \mu_2$ (independent, with $\sigma_1 \neq \sigma_2$ unknown)	2-SampTTest Pooled: No	$t_{\text{stat}} = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ $df = \min(n_1 - 1, n_2 - 1)$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Means μ_d (dependent)	TTest	$t_{\text{stat}} = \dfrac{\overline{d}}{s_D / \sqrt{n}}$ $df = n - 1$	1. Dependent samples. 2. Random samples. 3. Normal data or $n \geq 30$.
Variance or SD σ^2 or σ (single)	None	$\chi^2_{\text{stat}} = \dfrac{(n-1)s^2}{\sigma_0^2}$ $df = n - 1$	1. Random sample. 2. Data are normal.
Variances or SDs σ_1^2 / σ_2^2 or σ_1 / σ_2 (independent)	2-SampFTest	$F_{\text{stat}} = \dfrac{s_1^2}{s_2^2}$ $df_1 = n_1 - 1$ $df_2 = n_2 - 1$	1. Independent samples. 2. Random samples. 3. Both populations normal.
Distribution of proportions (goodness of fit)	χ^2GOF-Test	$\chi^2_{\text{stat}} = \sum \dfrac{(O - E)^2}{E}$ $df = k - 1$	1. Random sample. 2. $E_i = np_i \geq 5; i = 1, \ldots, k$.
Independence of variables	χ^2-Test	$\chi^2_{\text{stat}} = \sum \dfrac{(O - E)^2}{E}$ $df = (r-1)(c-1)$	1. Random samples. 3. Expected frequencies each at least 5.
Means $\mu_1, \ldots, \mu_k$ (ANOVA)	ANOVA	$F_{\text{stat}} = \dfrac{MS_T}{MS_E}$ $df_1 = k - 1$ $df_2 = n - k$	1. Independent samples. 2. Random samples. 3. Normal populations. 4. All populations have same variance.

Note that the goodness-of-fit test, the test for independence, and the ANOVA test are always right-tailed tests. See other side of the sheet for the ANOVA formulas and table.

Common Critical Z-Values

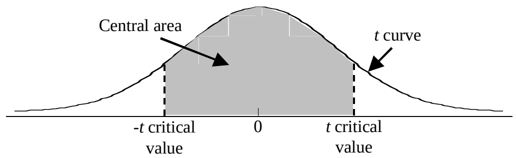
Confidence Level	90%	95%	98%	99%
Critical Value ($z_{\alpha/2}$)	1.645	1.96	2.326	2.576

<u>Central Limit Theorem Statements:</u>

$\hat{p}$ is approximately normal with mean p and standard deviation $\sqrt{pq/n}$.

$\overline{x}$ is approximately normal with mean μ and standard deviation $\sigma / \sqrt{n}$.

t critical values



Confidence area captured: Confidence level:	**0.90** **90%**	**0.95** **95%**	**0.98** **98%**	**0.99** **99%**
Degrees of Freedom **1**	6.314	12.706	31.821	63.657
2	2.920	4.303	6.965	9.925
3	2.353	3.182	4.541	5.841
4	2.132	2.776	3.747	4.604
5	2.015	2.571	3.365	4.032
6	1.943	2.447	3.143	3.707
7	1.895	2.365	2.998	3.499
8	1.860	2.306	2.896	3.355
9	1.833	2.262	2.821	3.250
10	1.812	2.228	2.764	3.169
11	1.796	2.201	2.718	3.106
12	1.782	2.179	2.681	3.055
13	1.771	2.160	2.650	3.012
14	1.761	2.145	2.624	2.977
15	1.753	2.131	2.602	2.947
16	1.746	2.120	2.583	2.921
17	1.740	2.110	2.567	2.898
18	1.734	2.101	2.552	2.878
19	1.729	2.093	2.539	2.861
20	1.725	2.086	2.528	2.845
21	1.721	2.080	2.518	2.831
22	1.717	2.074	2.508	2.819
23	1.714	2.069	2.500	2.807
24	1.711	2.064	2.492	2.797
25	1.708	2.060	2.485	2.787
26	1.706	2.056	2.479	2.779
27	1.703	2.052	2.473	2.771
28	1.701	2.048	2.467	2.763
29	1.699	2.045	2.462	2.756
30	1.697	2.042	2.457	2.750
40	1.684	2.021	2.423	2.704
60	1.671	2.000	2.390	2.660
70	1.667	1.994	2.381	2.648
80	1.664	1.990	2.374	2.639
90	1.662	1.987	2.368	2.632
100	1.660	1.984	2.364	2.626
1000	1.646	1.962	2.330	2.581
z critical values **∞**	1.645	1.960	2.326	2.576
α for 2-tailed tests	0.10	0.05	0.02	0.01
α for 1-tailed tests	0.05	0.025	0.01	0.005